Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Global Production Centre
GLO_ANALYSISFORECAST_PHY_001_024

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table. The third column specifies which sections, or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	April 2022	All	Creation of the document: major release of 001_024	Marie Drevillon	Marie Drevillon
1.1	October 2023	II.3	Creation of a new section for the dataset SST uncertainty	Charly Regnier, Jean-Michel Lellouche	Marie Drevillon
1.1	November 2023	All	Correcting/rephrasing of the entire text	Jean-Michel Lellouche	Marie Drevillon
1.1	March 2024		Minor corrections following postponement of system updates to June 2024 Release.	Renaud Dussurget	Olivier Le Galloudec
1.2	June 2024		High frequency seal level dataset SST error maps	Stéphane Law Chune, Charly Regnier, Jean-Michel Lellouche	Olivier Le Galloudec
1.2	December 2024	All	Correcting/rephrasing/reviewing of the entire document	Jean-Michel Lellouche	Olivier Le Galloudec
1.3	May 2025	All	Impact of correction in November 2024	Jean-Michel Lellouche	Olivier Le Galloudec

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

The products assessed in this document are referenced as:

GLOBAL_ANALYSISFORECAST_PHY_001_024 for the physical 14-day hindcast (analyses, updated weekly), and for the physical 10-day forecast (updated daily). These products, hereafter referred to as GLO_HR products, contain global nominal fields on global standard grid in 1/12° (0.0833° lat x 0.0833° lon) with 50 geopotential levels from 0 to 5000 m, for the following variables:

* sea_ice_area_fraction

* sea_ice_thickness

* eastward_sea_ice_velocity

* northward_sea_ice_velocity

* sea_surface_height_above_geoid

* sea_water_potential_temperature_at_sea_floor

* ocean_mixed_layer_thickness_defined_by_sigma_theta

* sea_water_potential_temperature

* sea_water_salinity

* eastward_sea_water_velocity

* northward_sea_water_velocity

* sea_water_potential_salinity_at_sea_floor

* sea_water_pressure_at_sea_floor

* surface_snow_thickness

* sea_ice_velocity

* sea_ice_surface_temperature

* sea_ice_albedo

* age_of_sea_ice

* upward_sea_water_velocity

Note that seventeen different datasets are available, corresponding to ocean parameters available at various temporal frequency:

Instantaneous fields:

- cmems_mod_glo_phy-so_anfc_0.083deg_PT6H-i containing salinity **6 hourly** outputs
- cmems_mod_glo_phy-thetao_anfc_0.083deg_PT6H-i containing temperature **6 hourly** outputs
- cmems_mod_glo_phy-cur_anfc_0.083deg_PT6H-i containing horizontal currents **6 hourly** outputs

- cmems_mod_glo_phy_anfc_merged-uv_PT1H-i containing **hourly** surface current components from multi model sources. This dataset is called SMOC for "Surface Merged Ocean Currents" product
- cmems_mod_glo_phy_anfc_merged-sl_PT1H-i containing hourly sea levels from multi model sources. This dataset is called MOWL for "Merged Ocean Water Levels" product. t It should be noted that only tide data are instantaneous whereas other sea level contributions are from hourly averaged data

Time averaged fields:

- cmems_mod_glo_phy_anfc_0.083deg_P1D-m containing surface **daily** outputs
- cmems_mod_glo_phy-so_anfc_0.083deg_P1D-m containing salinity **daily** outputs
- cmems_mod_glo_phy-thetao_anfc_0.083deg_P1D-m containing temperature **daily** outputs
- cmems_mod_glo_phy-cur_anfc_0.083deg_P1D-m containing horizontal current **daily** outputs
- cmems_mod_glo_phy-wcur_anfc_0.083deg_P1D-m containing upward current **daily** outputs
- cmems_mod_glo_phy_anfc_0.083deg_PT1H-m containing **surface hourly** outputs
- cmems_mod_glo_phy_anfc_0.083deg_P1M-m containing surface **monthly** outputs
- cmems_mod_glo_phy-so_anfc_0.083deg_P1M-mcontaining salinity **monthly** outputs
- cmems_mod_glo_phy-thetao_anfc_0.083deg_P1M-m containing temperature **monthly** outputs
- cmems_mod_glo_phy-cur_anfc_0.083deg_P1M-m containing horizontal currents **monthly** outputs
- cmems_mod_glo_phy-wcur_anfc_0.083deg_P1M-m containing upward current **monthly** outputs
- cmems_mod_glo_phy_anfc_0.083deg-climatology-uncertainty_P1M-m containing climatological monthly **uncertainty statistics for SST** drifting buoys

More details about the different datasets are contained in the product user manual (PUM) (https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-GLO-PUM-001-024.pdf).

I.2 Summary of the results

In the following you will find a summary of the scientific quality assessment performed for most ocean parameters described by GLOBAL_ANALYSIS_FORECAST_001_024, hereafter referred to as GLO12, for which we have reference observations or knowledge. The quality of the global high-resolution products has been assessed in delayed mode using years 2019 and 2020, for which analysed atmospheric fields and delayed time quality-controlled observations (delayed time reprocessed observations when available) were used. Note that when the analyses and forecast are produced in near real time (as opposed to delayed mode), this quality summary still applies but less observations are available and the uncertainty related to the atmospheric fields and assimilated observations (or lack of observations) is higher. Thus, **the estimated accuracy numbers, mostly derived from root mean square differences**

(RMSDs) between available observations and GLO12 analysis counterparts are regularly updated and, recent near real time values (including forecast) can be found on https://pqd.mercator-ocean.fr.

I.2.1 Water masses: temperature and salinity

The systems description of the ocean water masses is very accurate on average with very weak systematic biases (less than 0.05 °C and 0.02 psu (practical salinity unit) at the surface). There is a warm bias in the subsurface layer, with a maximum value of 1°C at 100 m in the Atlantic areas of Zapiola and of the Gulf Stream. Departures from in situ observations rarely exceed 0.5 °C and 0.3 psu Root-Mean-Square Deviation (RMSD) at the surface and, in the thermocline, RMSDs reach 1 °C and 0.2 psu. In high variability regions like the Gulf Stream, the Agulhas Current, or the Eastern Tropical Pacific, RMSDs exceed 2 °C and 0.5 psu locally. The thermocline is usually too deep in the tropics and in upwelling areas resulting in a warm bias in those areas.

I.2.2 Sea Surface Temperature (SST)

The GLO12 sea-surface-temperature analysis is very close to in situ observations. It displays a weak global average bias with respect to Operational Sea Surface Temperature and Ice Analysis (OSTIA) SST product (warm bias of 0.1 °C), as well as a comparable bias with Level 3 along Swath (L3S) ODYSSEA SST product (https://doi.org/10.48670/moi-00164) in terms of amplitude but of opposite sign (cold bias of 0.1 °C). Warm biases up to 1 °C can be observed in upwelling areas, where the thermocline is too deep in GLO12 with respect to observations. The estimated uncertainty is close to 0.6 °C; which is in the range of satellite observations' uncertainty.

I.2.3 Sea Level Anomalies (SLA)

Sea level anomalies are generally very close to altimetric observations (the global average RMSD is around 5 cm, and lower than 5 cm in most ocean basins). GLO12 is lower than the altimetric observations by about 1 to 2 mm on average.
The GLO12 system assimilates along track observations from all altimetric missions available from the Copernicus Marine Sea Level Thematic Assembly Centre (TAC). Correlations between GLO12 hourly Sea Surface Height (SSH) products and tide gauges are high and significant at all frequencies; however, many SSH high frequency fluctuations may not be captured by the GLO12 products (note that tides or pressure effects are not modelled).

I.2.4 Surface currents

Surface currents of the mid latitudes are underestimated on average with respect to in situ measurements of drifting buoys. The underestimation ranges from 20% in strong currents to 60% in weak currents. The SMOC products add the contribution due to the tides and the Stokes drift. They can improve the model velocity where there are strong tidal currents and where there is a big contribution due to the Stokes (usually during storms). Therefore, the use of the SMOC dataset (cmems_mod_glo_phy_anfc_merged-uv_PT1H-i: GLO12 surface currents + tides + Stokes drift) is recommended for drift and Lagrangian applications as it is shown that results can improve locally by around 10 %.

I.2.5 Sea Ice Cover

The large-scale features of the sea ice cover are well captured in the Arctic and the Antarctic. The sea ice concentrations are underestimated in the Arctic mainly during summer (due to atmospheric forcing errors). Sea ice concentration is underestimated in the Antarctic during austral summer (also due to atmospheric forcing errors). Consistently, the temperature at the surface of the ice is overestimated in GLO12.

I.3 Estimated Accuracy Numbers

The following 3D temperature and salinity (T/S), SLA, and SST **accuracy numbers** are estimated using statistical metrics performed with the model counterpart in time and space of the available observations. In all cases these numbers are an estimate of the mean error and not a maximum error on the domain. Here we present only the **accuracy numbers for the global area.** They are also computed for each region but are not shown in this document. Accuracy values are represented by RMSD and Mean bias for SLA, SST, and surface currents. They are given also for 3D temperature and salinity in the following vertical layers: 0-5 m, 5-100 m, 100-300 m, 300-800 m, 800-2000 m, and 2000-5000 m.

In order to calculate these accuracy numbers, we defined a one-year **qualification (or calibration) period** in 2019. Results are presented for each variable in the set of tables below.

I.3.1 Temperature

Table 1. Globally averaged RMSD and mean temperature bias in °C (observation-model) with respect to in situ observations (INS TAC), in contiguous layers from 0 to 5000 m and computed in 2019.

3D T (K) vs in situ	**Hindcast**		**Forecast day 3**	
Observations	**Mean bias**	**RMSD**	**Mean bias**	**RMSD**
0-5 m	0.009	0.51	0.01	0.59
5-100 m	-0.018	0.79	-0.019	0.94
100-300 m	-0.007	0.72	-0.009	0.86
300-800 m	0.024	0.39	0.025	0.48
800-2000 m	0.035	0.18	0.035	0.2
2000-5000 m	0.045	0.28	0.04	0.28
0-5000 m	0.02	0.37	0.03	0.45

I.3.2 Salinity

Table 2. Globally averaged RMSD and mean salinity bias in PSU (observation-model) with respect to in situ observations (INS TAC), in contiguous layers from 0 to 5000 m.

3D S (PSU) vs in situ Observations	Hindcast		Forecast day 3	
	Mean bias	RMSD	Mean bias	RMSD
0-5 m	0.004	0.36	0.007	0.39
5-100 m	0.009	0.21	0.01	0.24
100-300 m	0.002	0.12	0.002	0.15
300-800 m	0.001	0.064	0.001	0.077
800-2000 m	-0.0009	0.033	-0.001	0.037
2000-5000 m	-0.001	0.009	-0.001	0.009
0-5000 m	0.0004	0.073	0.0003	0.085

I.3.3 SST

Table 3. Globally averaged RMSD and mean SST misfits in °C (observation-model) with respect to observations (SST TAC) and computed in 2019.

SST (K) vs Satellite data	Hindcast		Forecast day 3	
	Mean bias	RMSD	Mean bias	RMSD
GLO	-0.06	0.4	0.1	0.6

I.3.4 SLA

Table 4. Globally averaged RMSD and mean SLA misfits in cm (observation-model) with respect to altimetric along tracks observations (Sea Level TAC) and computed in 2019.

Seal Level Anomaly (cm) vs Satellite Along track data	Hindcast		Forecast day 3	
	Mean bias	RMSD	Mean bias	RMSD
GLO	-0.1	5.0	-0.02	7.0

I.3.5 Sea Ice Cover

Table 5. ARC and ACC regions Mean misfits and RMSD for Sea Ice Concentration (%) with respect to observations (Sea Ice TAC) and computed in 2019.

Sea Ice Cover (%)	Hindcast		Forecast day 3	
	Mean bias	RMSD	Mean bias	RMSD
ARC	5	10	-5	18
ACC	2	8	-0.7	10

Remark: First estimates of 3D temperature and salinity Estimated Accuracy Numbers (EANs) for forecast day 3 have been computed using background model first trajectories of the hindcast simulation. This should not be confused with an operational forecast that corresponds to a model trajectory in the future without data assimilation. Even though the quality of real-time forecast is not properly evaluated here (different atmospheric forcing), these statistics give an estimate of the skill of the optimal model forecast. EANs for forecast day 3 for SST, SLA, and sea ice cover have been completed in 2023, not using background model first trajectory of the hindcast simulation but the real-time forecasts.

I.3.6 Surface currents

*Table 6. Averaged separation distance (km/3d) for the Lagrangian simulations as a function of zone of application (with respect to PhOD hourly database (**https://www.aoml.noaa.gov/phod/gdp/index.php**), undrogued drifter observations and simulations computed with the PARCELS Lagrangian tool). Values in light grey correspond to areas where the merged product (SMOC) presents a degradation compared to the initial product.*

	Global	Pacific south	Mid latitudes	Pacific Tropics	Equator	Gulf stream	Shelf Seas
cmems_mod_glo_phy_anfc_0.083deg_PT1H-m **(uo)**	42.7	40.5	42.3	44.3	60.2	61.6	36.4
cmems_mod_glo_phy_anfc_merged-uv_PT1H-i **(utotal)**	39.9	32.8	38.1	40.8	63.7	59.7	38.5

II PRODUCTION SYSTEM DESCRIPTION

These products are provided by Mercator Ocean International which leads the Copernicus Marine Service global monitoring and forecasting centre GLO MFC.

II.1 The physical high resolution global system

Since October 2016, and in the framework of Copernicus Marine Service, Mercator Ocean International has delivered real-time daily services with a global 1/12° high resolution system **GLO12v3** (previously called PSY4V3R1) described and assessed in Lellouche et al. (2018).

A major update of this analysis and forecasting system, now called **GLO12v4** (or sometimes only **GLO12**), is available since November 2022.

II.1.1 Model configuration

The GLO12v4 system uses version 3.6 of the NEMO ocean model (Madec et al., 2017), as well as the LIM3 multi-category sea ice model with the Elastic-Viscous-Plastic rheology formulation and 11 sea ice categories. The configuration is based on the tripolar ORCA12 grid type (Madec and Imbard, 1996) with a horizontal resolution of 9 km at the equator, 7 km at Cape Hatteras (mid-latitudes) and 2 km toward the Ross and Weddell seas (Antarctica). Z-coordinates were used on the vertical and the 50-level vertical discretization retained for this simulation had a decreasing resolution from 1 m at the surface to 450 m at the bottom, with 22 levels within the upper 100 m. The bathymetry used in the system is a combination of interpolated ETOPO1 (Amante and Eakins, 2009) and GEBCO8 (Becker et al., 2009) databases. ETOPO1 datasets are used in regions deeper than 300 m and GEBCO8 is used in regions shallower than 200 m with a linear interpolation in the 200 m – 300 m layer. This bathymetry benefits from a specific correction in the Indonesian Sea inherited from the INDESO system (Tranchant et al., 2016).

II.1.2 Parameterizations

A “partial cells” parameterization (Adcroft et al., 1997) is chosen for a better representation of the topographic floor (Barnier et al., 2006) and the momentum advection term is computed with the energy and enstrophy conserving scheme proposed by Arakawa and Lamb (1981). The advection of the tracers (temperature and salinity) is computed with a total variance diminishing (TVD) advection scheme (Lévy et al., 2001; Cravatte et al., 2007). A Laplacian lateral isopycnal diffusion operator is applied on tracers (100 m^2/s). Special attention was paid to favour the representation of fine scales and fast ocean signals. The simulation thus uses an explicit barotropic mode solved by a split-explicit approach (Shchepetkin and McWilliams, 2005) and an Upstream-Biased Scheme (Shchepetkin and McWilliams, 2008) for computing the horizontal momentum advection without addition of an explicit diffusion. A two-equation vertical mixing scheme (k-epsilon; Rodi, 1987) is used and internal-tide driven is parameterized following Lavergne et al. (2016). Note that the system does not include explicit tides. The depth of light extinction is separated into infrared and red-green-blue-UV bands depending on the chlorophyll data distribution derived from a monthly SeaWIFS climatology.

II.1.3 Relaxations to avoid spurious drifts

As the Boussinesq approximation (density variations are neglected except in their contribution to the buoyancy force; https://www.nemo-ocean.eu/doc/node6.html) is applied to the model equations conserving the ocean volume and varying its mass, the simulation does not represent the global steric effect on the sea level correctly (Greatbatch, 1994). For this reason, only the global mass variations are represented in the sea level and the global steric effect has to be diagnosed. In order to avoid any mean sea-surface-height drift due to the large uncertainties in the water budget closure, the following treatments are applied: firstly, the global surface freshwater budget is set to zero at each time step with a superimposed seasonal cycle spread over the ocean (Chandanpurkar et al., 2021). Secondly, a long-term global mass trend (satellite-based monthly estimates of the Global Mean Sea Level) is added to the water budget in order to somehow represent the recent estimate of the global mass addition to the ocean (Dieng et al., 2017).

Temperature and salinity in the Gibraltar and Bab-el-Mandeb straits is relaxed towards the EN4.2.0 monthly gridded temperature and salinity climatology (Good et al., 2013) to avoid spurious deep water mass formation.

II.1.4 River inputs, atmospheric fluxes

The monthly runoff climatology is built with data on coastal runoffs and 100 major rivers from the Dai et al. (2009) database. This database uses data mostly from recent years, streamflow simulated by the Community Land Model version 3 (CLM3) to fill the gaps in all lands areas except Antarctica and Greenland. In addition, the runoff fluxes were built from Greenland and Antarctica ice sheets as well as glaciers melting, using the Altiberg icebergs database project (Tournadre et al., 2016). This complements the estimate of Silva et al. (2006) for Antarctica.

The atmospheric field forcing comes from the European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecast System (IFS) analyses and forecasts with a high spatial and temporal resolution (1/10° - 1 hour). Fluxes are computed using a bulk formulation derived from IFS (Brodeau et al., 2017) and no atmospheric pressure forcing is used. The surface currents are taken into account in the stress computation as in Renault et al. (2017). Large-scale correction of precipitations (no change of synoptic patterns like cyclones and of interannual signal) are performed using satellite data, except at high latitudes (poleward of 65°N and 60°S) where no data are available.

II.1.5 Data assimilation

Oceanic observations are assimilated in the model using a reduced-order Kalman filter method derived from a SEEK filter (Brasseur and Verron, 2006), with a 3-D multivariate modal decomposition of the forecast error and a 7-day assimilation cycle (Lellouche et al., 2018). It includes an adaptive-error estimate and a localization algorithm. Along track Level 3 (L3) altimeter sea level anomaly (SLA), L3 satellite sea surface temperature (ODYSSEA SST) and Level 4 (L4) sea ice concentration (OSI SAF), as well as in situ temperature and salinity (T/S) at all depths are jointly assimilated to estimate the initial conditions for numerical ocean forecasting. Note that L4 gridded products of SLA and SST are also used to adapt the background error variance (see Table 7). These data are different from the L3 products assimilated in the system. In addition to the quality control performed by data producers, the system carries out two internal quality controls on temperature and salinity vertical profiles in order to minimize the risk of erroneous observed profiles being assimilated in the model. The first quality control is based on temperature and salinity innovation statistics (detection of spikes, large biases). The second quality control is based on dynamic height innovation statistics (detection of vertically constant biases).

The forecast error covariance is based on the statistics of a collection of 3-D ocean state anomalies, typically a few hundred which are assumed representative of the error covariances. This approach is similar to the Ensemble Optimal Interpolation developed by Oke et al. (2008). The anomalies used by GLO12v4 have been computed from the Mercator Ocean reanalysis GLORYS12v1 at 1/12° (Lellouche et al., 2021) with respect to a running mean in order to estimate the 7-day scale error on the ocean state at a given period of the year. Moreover, these anomalies have been spatially filtered in order to retain only the effective model resolution and to avoid injecting noise in the increments.

In addition to the multivariate reduced-order Kalman SEEK filter, GLO12v4 employs a 3D-VAR scheme, which takes into account cumulative three-dimensional T/S innovations over the last month in order to estimate large-scale T/S biases when enough T/S vertical profiles are available. The aim of the bias correction is to correct the large scale slowly evolving error of the model which complements the SEEK assimilation scheme's correction of the smaller scales of the model's first guess. Temperature and salinity are treated separately because temperature and salinity biases are not necessarily correlated. The bias correction is fully effective under the thermocline, away from density gradients. Moreover, a constraint towards the EN4.2.0 climatology (Good et al., 2013) on T/S in the deep ocean (below 2000 m) has been included in the 3D-VAR scheme to prevent drifts in temperature and salinity and as a consequence to obtain a better representation of the sea level trend at global scale in the system while allowing actual climate change signals to emerge at depth.

Before the analysis step, the SLA and SST observations are spatially averaged to reduce the so-called representativeness error (Janjic et al., 2018). This averaging method, known as "super-observation", intends to filter out observations noise and scales that the model does not resolve. In the system, all observations of a given dataset lying inside one model grid point transform into one observation with its position given by the average position and its error transformed by the theory of uncertainty propagation (Ku H. H., 1966). According to this theory, a correlation model is set up for the observation errors averaging. An exponential correlation model with scales of 40 km and 80 km is used for SLA and SST respectively. The latter scale values were estimated using the Hollingsworth and Lönnberg (1986) method. Consequently, the analysis uses a reduced number of observations whose associated error is smaller. Three-dimensional seasonal observations errors computed from GLO12v3 are used for the assimilation of in situ temperature and salinity data.

A 4D extension of the SEEK analysis is used, allowing an improved spatial and temporal continuity of mesoscale structures (Benkiran et al., 2021). Contrary to the 3D method, in the 4D method the observation operator applied to the anomalies takes into account the temporal dimension. In that sense, the 4D method uses an ensemble of model anomalies to fit the innovation in the data assimilation window, allowing a better approximation of the anomaly matrix.

The SEEK analysis grid consists of a reduced horizontal grid (1 out of every 2 points in both horizontal directions and all the points along the coast) to limit the storage and the cost during the analysis stage.

After each analysis, the data assimilation produces 7 daily increments of sea ice concentration, sea surface height, temperature, salinity, and zonal and meridional velocity. These increments are progressively applied using the Incremental Analysis Update (IAU) method (Benkiran and Greiner, 2008; Lellouche et al., 2018) which allows avoiding model shocks which would be induced by the imbalance between the analysis increments and the model physics. The first guess at appropriate time (FGAT) method (Huang et al., 2002) is used. It means that for a given observation, the innovation compares this observation with the model first guess counterpart at the location and at the nearest time of the observation.

The concept of "pseudo-observations" or "Observed-No Change" (innovation equal to zero) is used to overcome known deficiencies of the background errors, in particular for extrapolated and/or poorly observed variables. We apply this approach to the barotropic height at large spatial and temporal scales and to the 3-D coastal salinity at river mouths and all along the coasts (run off rivers). Pseudo-observations are also used for the 3-D variables T (temperature), S (salinity), U (zonal velocity) and V (meridional velocity) under the ice on the one hand, and between 6° S and 6° N below a depth of 200 m on the other hand.

Lastly, a Mean Dynamic Topography (MDT) is used as a reference for SLA assimilation. This MDT is based on the "CNES-CLS18" MDT (Mulet et al., 2021) with adjustments made using the GLORYS12v1 reanalysis at 1/12° (Lellouche et al., 2021). The uncertainties in the MDT estimate and the sparsity of the observation networks (both altimetry and in situ profiles) on the 7-day assimilation window do not allow to accurately estimate the observed global mean sea level. Therefore, the global mean increment of the total sea surface height is set to zero.

Table 7. List of input observations assimilated in GLO12v4.

Type of use	**Parameter**	**Copernicus Marine Service product**	**Original name**	**Time of use**
Assimilated in the system	Sea level anomaly (SLA)	SEALEVEL_GLO_PHY_L3_MY_008_062	AVISO along track REP	Oct 2016 → 28/12/2021
		SEALEVEL_GLO_PHY_L3_NRT_OBSERVATIONS_008_044	AVISO along track NRT	29/12/2021 onwards
	Sea-surface-temperature (SST)	SST_GLO_SST_L3S_NRT_OBSERVATIONS_010_010	ODYSSEA	Oct 2016 onwards
	In situ temperature and salinity (T/S)	INSITU_GLO_TS_ASSIM_REP_OBSERVATIONS_013_051	CORA REP	Oct 2016 à 29/12/2020
		INSITU_GLO_TS_ASSIM_NRT_OBSERVATIONS_013_047	Coriolis NRT	30/10/2020 onwards
	Sea ice concentration (SIC)	SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001	OSISAF	Oct 2016 onwards
Only used for adapting the background error variance	SLA	SEALEVEL_GLO_PHY_L4_MY_008_047	AVISO maps REP	Oct 2016 → 28/12/2021
		SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046	AVISO maps NRT	29/12/2021 onwards
	SST	SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001	OSTIA	Oct 2016 onwards

II.1.6 Time period covered

The GLO12v4 system was run over the October 2016 - November 2022 period before its entry into service, initialised in October 2016 with a GLO12v3 restart file. This means that the GLO12v4 system does not need a long spin up period because it is already in balance at the initial state. The "reprocessed" observations available at that time, and the so-called "near real-time" observations otherwise have been assimilated in the system (see Table 7). In an operational mode (from the entry into service), this system produces daily 10-day forecast and weekly 14-day hindcast (7-day of best-analysis and 7-day of NRT-analysis). Details on the production cycle are in the PUM document. Note that no daily forecast was archived for the 10/2016-09/2022 period.

II.2 Surface drift currents: cmems_mod_glo_phy_anfc_merged-uv_PT1H-I dataset (SMOC)

II.2.1 Short description

A special dataset dedicated to drift application, called **SMOC** (Surface and Merged Ocean Currents), is delivered as the **cmems_mod_glo_phy_anfc_merged-uv_PT1H-i** dataset. SMOC is a composite surface current product that combines velocity components from the Copernicus Marine Service modelling systems to reproduce the net velocity felt at the sea surface.

In SMOC, the total surface current is obtained from the simple addition of contributions from the oceanic general circulation, waves, and tides. Published studies have already demonstrated the importance of wave-induced (e.g., Stokes drift) currents for particle advection (Monismith and Fong, 2004; Constantin, 2006), as well as the role of tides in the exchange between the coast and the open sea (Rynne et al., 2016).

The SMOC product covers the global domain, with a horizontal resolution of 1/12° and an hourly frequency output. This high spatio-temporal resolution should allow it to target mesoscale movements and short forecast scales.

All horizontal components and their sum are delivered, so that the user can select and focus on each component individually. More details are in the PUM.

II.2.2 Input datasets

Technically, three independent systems pool their results into SMOC. They are:

i) the GLO12v4 global high resolution physical forecasting system of Copernicus Marine: *uo* and *vo* are extracted from the **cmems_mod_glo_phy_anfc_0.083deg_PT1H-m** dataset (general circulation component). Hourly averaged 1/12° surface currents.

ii) the Copernicus Marine global wave forecasting system (Lefèvre et al. 2009), dataset **global-analysis-forecast-wav-001-027**: the horizontal components of Stokes drift (*ustokes, vstokes*) are extracted from the VSDX and VXDY variables (wave induced component). Hourly instantaneous 1/10° currents (interpolated on the 1/12° final grid).

iii) the FES tidal model (Carrere et al. 2015): barotropic tide currents *(utide, vtide*) are computed from the **FES2014 tide model** on the GLO12v4 system 1/12° grid at an hourly - instantaneous - time step (tide-induced current).

A composite current that consists of the addition of the three components, namely the **total current**, is also delivered as: Total current: (*utotal=uo+ustokes+utide*, *vtotal=vo+vstokes+vtide*).

II.2.3 Method

We have been largely inspired by what is done for surface current products derived from observations, like the MULTIOBS_GLO_PHY_NRT_015_003 derived from the GLOBCURRENT project (Johannessen et al. 2016), the OSCAR (Ocean Surface Current Analysis Real time) product (Johnson et al. 2007) or the GEKCO (Geostrophy and Ekman Current Observatory) product (Sudre et al., 2013).Those products deliver large-scale satellite-derived geostrophic current added with Ekman drift computed from atmospheric forecasts. The GLO12 horizontal surface velocity components (uo, vo) used in SMOC should represent at least those later physical processes, and the essence of SMOC is to complete them with momentum effects linked with tides and waves. The advantage of using models instead of derived products from observations is also the capacity to draw forecasts. For now, SMOC forecasts currently extend to 10 days.

Tides currents of SMOC are obtained from the FES2014 model (Carrere et al., 2015), a hydrodynamic tide model developed by LEGOS/NOVELTIS/CLS. FES2014 is based on a version of the TUGO model that assimilates altimeter data (Topex/Poseidon, Jason-1, Jason-2, TP interleaved –J1 interleaved, ERS-1, ERS-2, and Envisat). SMOC uses the Python wrapped version of FES2014 to directly compute tide currents on its 1/12° regular output grid, using the maximal tidal components available in the software. FES2014 model is freely distributed on the CNES/AVISO data distribution platform: https://www.aviso.altimetry.fr/en/data/products/auxiliary-products/global-tide-fes/description-fes2014.html.

Note that since the velocities are simply added together, the coupling processes between the components are not considered. Ad-hoc methods exist to take care of these shortcomings, but they are not applied in SMOC (e.g., Cunningham et al., 2022). Future Copernicus systems in the next few years will be coupled systems that consider these interdependency processes. One other limitation is that SMOC is designed for drifting at the ocean surface (i.e., at 0 m). Ocean currents (uo,vo) are modelled at the 0.5 m level and there is no extrapolation performed to estimate them at 0 m, although some methods exist in the literature (Tamtare et al., 2021).

The dataset **cmems_mod_glo_phy_anfc_merged-uv_PT1H-I** therefore contains GLO12 (uo, v0), (ustokes, vstokes), (utide, vtide) and the sum (utotal, vtotal) which we usually refer to as SMOC. When references are made throughout the document to SMOC uo (=GLO12), SMOC utotal or SMOC currents, they reference the variables of this dataset.

II.3 Climatological monthly uncertainty maps for SST: dataset cmems_mod_glo_phy_anfc_0.083deg-climatology-uncertainty_P1M-m

A special dataset dedicated to climatological monthly uncertainty maps is delivered to give information about SST uncertainty for the Sea Surface Temperature. Two types of information are used to compute the statistics of this dataset: the observations coming from the USGODAE in situ provision of Sea Surface Temperature and the daily mean ocean analysis fields from the product GLO_ANALYSISFORECAST_PHY_001_024. The dataset contains monthly mean statistics that cover the period 2017-2022. The SST drifting buoys measure the Sea Surface Temperature close to the surface around 1 m depth. A quality control is applied to eliminate possible bad measurements but in the polar regions we can still have unrealistic values with good quality control leading to highest differences.

Using CLASS4 files containing information on observations and model values we calculate statistics on a regular 2°x2° output grid. The statistics are the mean difference (MD uncertainty field) between the model and the observations, the RMSD (uncertainty field) and the number of observations. The output 2°x2° regular grid is interpolated on a 1/12° regular grid which is the same as the GLO12 product (GLOBAL_ANALYSISFORECAST_PHY_001_024 in the Copernicus Marine catalogue). For this dataset some cells are empty meaning that there are no values in certain areas during this period.

II.4 Total water levels: cmems_mod_glo_phy_anfc_merged-sl_PT1H-I dataset

II.4.1 Short description

A special dataset called internally MOWL for Merged Ocean Water Levels is dedicated to water levels and delivered in the "cmems_mod_glo_phy_anfc_merged-sl_PT1H-i" dataset. MOWL's design was inspired by two products:

- SMOC (cmems_mod_glo_phy_anfc_merged-uv_PT1H-i) for technical design and operational production
- Data produced in the scope of the ECFAS project (Le Gal, 2023), a European project aiming to set up a European coastal flood awareness system, where total water levels come from the linear contribution of various physical models.

In short, this dataset distributes additional components not modelled by the GLO12 physical model and add them linearly for a practical total sea level variable. Those components represent high-frequency variations in ocean elevation caused by:

1) The tidal cycle, with ocean tide and load tide treated separately.
2) The deformation of the ocean surface caused by atmospheric pressure (the so-called invert barometer).

This special dataset also provides diagnostics of global ocean rise (global average):

3) Ocean dilatation in relation to global salt and heat content (steric contribution).
4) Oceanic rise in terms of volume variation linked to the global mass budget (mass budget = precipitation - evaporation + ice melt).

The dataset also includes:

5) Water level variations linked to ocean dynamics from the GLO12 physical ocean system (the so called “ssh”: sea surface height. GLO12 doesn’t consider tides and invert barometer).

Finally, a variable accounting for total water levels as the sum of all components (1+2+3+4+5) is also distributed to users for practical use.

All variables are distributed at an hourly frequency and on the regular GLO12 grid.

Remark: For this, global average variables (3 and 4) are remapped on the GLO12 grid. The main reason for this choice is that the Copernicus Marine Service web viewer currently only works with 2D fields.

II.4.2 Method and input data

II.4.2.1 Tide contribution (ocean tide and load tide)

The gravitational forces exerted by the moon and the sun cause the ocean surface to rise and fall in a rhythmic pattern, known as tidal motion. Ocean tide and load tide elevations of MOWL are, as for SMOC, obtained from the FES2014 model (Carrere et al., 2015). Readers are kindly referred to the corresponding documentation of FES2014 (https://www.aviso.altimetry.fr/en/data/products/auxiliary-products/global-tide-fes/description-fes2014.html) for more information.

II.4.2.2 Invert Barometer

The inverted barometer effect describes how changes in atmospheric pressure influence the elevation of the ocean surface. In MOWL, the atmospheric pressure from IFS atmospheric forcing is further transformed into an equivalent inverse barometer sea surface height:

$$\eta_{ib} = \frac{-1}{g\rho_o}(P_{atm} - P_o) \tag{1}$$

where P_{atm} is the atmospheric pressure, P_o the ocean domain average of atmospheric pressure, ρ_o is the water density and g the gravity acceleration.

II.4.2.3 Steric contribution

Variations in steric sea level occur due to fluctuations in the density of the water column, leading to either expansion or contraction. These changes primarily result from surface heating or cooling, with additional contributions from non-linear effects such as cabbeling and thermobaricity. Since the NEMO ocean model operates as a Boussinesq model, conserving volume but not mass, it cannot accurately simulate contraction or expansion. However, as proposed by Greatbatch (1994), the steric impact on volume can be assessed by analysing the global ocean's mass budget:

$$\eta_s = -\frac{1}{\mathrm{A}}\int_{\mathrm{V}} \frac{\rho - \rho_o}{\rho_o}\,\mathrm{d}v \tag{2}$$

where $\int_{\mathrm{V}} \mathrm{d}v$ is a volume integral on V the total volume of the ocean, A is the total surface of the ocean ($A = \int_{\mathrm{S}} \mathrm{d}s$) and ρ_o is the reference density chosen as 1035 $Kg.m^{-3}$.

This steric diagnosis is integrated from an initial state, corresponding to the initial state of the GLO12 simulation starting on 2016/10/05 with $\eta_{s0} = -5.817441$ m.

In MOWL, only steric variations from this initial state are distributed such as:

$$d\eta_s = \eta_s - \eta_{s0} \tag{3}$$

II.4.2.4 Barysteric contribution

In a Boussinesq ocean model like NEMO, the total volume (or equivalently the global mean sea level) is altered only by net volume flux exchanges with other earth media like atmosphere, sea ice and land. The temporal change of total volume writes:

$$\partial_t V = A\frac{\overline{-emp}}{\rho_o} = A\,\eta_{mb} \tag{4}$$

where $\overline{emp} = \int_{\mathrm{S}} emp\ ds$ is the net mass flux through the ocean boundaries (evaporation – precipitation + ice forming budget) causing a variation of the ocean sea level of η_{mb}. In practice, since the global average of all other contribution to SSH are zero in the GLO12 system, global SSH temporal variations are only due to volume variations caused by mass fluxes.

$$\eta_v = \overline{SSH} = \frac{1}{V}\int_{\mathrm{V}} \mathrm{SSH}\,\mathrm{d}v \tag{5}$$

As stated before, the global mass budget is constrained by satellite observations to which we add seasonal variations, so that the η_v should be realistic and can be used as such.

II.4.2.5 Total sea level

Finally, MOWL proposes a practical total sea level variable linearly summing the various components, such as:

$$\eta_{TS} = SSH + \eta_{Otide} + \eta_{ib} + d\eta_s + \eta_v \tag{6}$$

Note that load tides are not included in the total sea level variable. It's only relevant to so when comparing to data observed from space.

II.5 Anomaly difference maps for SST: cmems_mod_glo_phy_anfc_0.083deg-sst-anomaly_P1D-m dataset

II.5.1 Short description

A special dataset dedicated to sea surface temperature anomalies is provided to give information on the differences between SST from the GLO_ANALYSISFORECAST_PHY_001_024 product in comparison with a monthly climatology from the GLOBAL_MULTIYEAR_PHY_001_030 product, covering the period 1993-2016. The Sea Surface Temperature represents the temperature of the ocean near the surface at a depth of around 1m.

II.5.2 Method and input data

The input data concerned are the two products described below, the GLO_ANALYSISFORECAST_PHY_001_024 real time system and the GLOBAL_MULTIYEAR_PHY_001_030 reanalysis (https://doi.org/10.48670/moi-00021).

This dataset adds a new variable for the GLO_ANALYSISFORECAST_PHY_001_024 product. This means that for each daily and weekly run, an SST anomaly is calculated in addition to the SST field already provided in the output files. This anomaly is a simple difference between the SST field for a given day and the corresponding daily climatological SST field of GLOBAL_MULTIYEAR_PHY_001_030 annual cycle. This dataset is evaluated in the section IV.3.

III VALIDATION FRAMEWORK

Most of the scientific assessment framework for the GLO12 global system is summarized in Lellouche et al. (2013, 2018). In the present quality information document, all validation diagnostics are computed using data assimilation scores and CLASS 4 metrics when available.

*8. **Error! Reference source not found.Error! Reference source not found.***

8Oceanic parameter	Area of computation	Type of metric	Reference observational dataset	Ref name of metric
Qualification of temperature and salinity fields				
3D temperature	Global, and regional basins	Residual difference=obs-model Time evolution of mean and RMSD on layers from 0 to 5000 m: 0-5 m; 5-100 m; 100-300 m; 300-800 m; 800-2000 m; 2000-5000 m; 0-5000 m Vertical profile of mean and RMSD	T(z) profiles from CORIOLIS database	CLASS4
3D salinity	Global, and regional basins	Residual difference=obs-model Time evolution of mean and RMSD on layers from 0 to 5000 m: 0-5 m; 5-100 m; 100-300 m; 300-800 m; 800-2000 m; 2000-5000 m; 0-5000 m Vertical profile of mean and RMSD	S(z) profiles from CORIOLIS database	CLASS4
Monthly 3D T and S	Global	Visual inspection of seasonal and interannual signal at several immersions	Levitus 2013 monthly climatology of T/S	CLASS1
Sea Surface Temperature SST	Global, and regional basins	Residual Error=obs-model Time evolution of RMS and mean error	L3S super-collated SST	Data assimilation statistics
Qualification of sea surface height and currents				
Sea surface height	At tide gauges (Global but near coastal regions)	Residual difference=model-obs Time series correlation and RMSD	Tide gauges sea level time series from GLOSS, BODC, Imedea, WOCE, OPPE and SONEL	CLASS2
Sea level anomaly (SLA)	Global, and regional basins	Residual difference=obs-model Time evolution of RMSD and mean residual difference	Along track SLA observations from all available missions	Data assimilation statistics
Surface layer zonal and meridional velocity	Global, and regional basins	Residual difference=obs-model Mean, RMSDs and correlation	SVP drifting buoys or YOMAHA07 Argo floats	CLASS4, Eulerian metric
Surface layer zonal and meridional velocity	Global	Maps of separation distance for 3-day Lagrangian forecasts, same for initial error in velocity and direction	PhoD hourly drifters from NOAA	Lagrangian, user-oriented metric

*8 (continued). **Error! Reference source not found.***

Oceanic parameter	Area of computation	Type of metric	Reference observational dataset	Ref name of metric
Qualification of sea surface height and currents				
Surface layer zonal and meridional velocity	Local and regional	Residual difference=obs-model Mean, RMSDs and correlation	European Current meters and high frequency radars	CLASS2 and CLASS4
Qualification of sea ice concentration				
Sea ice concentration	Antarctic and Arctic regions	Visual inspection of seasonal and interannual signal	CMEMS: SI TAC - Sea ice concentration	CLASS1
Sea ice concentration	Antarctic and Arctic regions	Residual Error=obs-model Time evolution of RMSD and mean error	CMEMS: SI TAC - Sea ice concentration	Data assimilation statistics
Sea ice concentration	Antarctic and Arctic regions	Time evolution of sea ice extent	NSIDC sea ice extent from SSM/I observations	CLASS3
Qualification of ocean processes				
HF fluctuations of T, S, SSH, and atmospheric forcings	Locally, at moorings	Visual inspection of high frequencies, comparisons with observed time series	CMEMS: CORIOLIS	CLASS2
Wave propagation in SLA and SST	Tropical basins	Visual inspection of Hovmöller diagrams and comparisons with satellite observations	CMEMS: SL and SST TACs	CLASS1
Volume and heat transports using 3D U and V	Global	Visual inspection of volume transports through sections	Literature (e.g, Talley et al., 2003; Ganachaud and Wunsch, 2003; Lumpkin and Speer, 2007; Stammer et al., 2004; Haines et al., 2012; Bricaud et al., 2018).	CLASS3

IV VALIDATION RESULTS

The reference calibration years used for this scientific qualification exercise are 2019-2021. Some diagnostics computed over the whole hindcast run (2017-2022) are also included to show that the quality is stable over time. Hindcast statistics are presented by geographical area and depth layer.

Estimated Accuracy Numbers (EANs) for GLO12 analysis were computed for 11 regions corresponding to the main basins of the global ocean: Antarctic Circumpolar (ACC), Arctic (ARC), North Pacific (NPA), North Atlantic (NAT), Mediterranean (MED), Tropical Pacific (TPA), Tropical Atlantic (TAT), South Pacific (SPA), South Atlantic (SAT), Indian (IND) and Global (GLO) (see Figure 70 in Appendix for the geographical location of these regions). The modelled physical variables were compared to assimilated observations and to independent observations when available. When possible, the global ocean estimates and EANs were intercompared with other available global products to ensure the consistency of the results and to quantify the improvements with respect to previous versions of the system (verification of non-degradation).

IV.1 Temperature

The global average of the temperature bias shows that GLO12v4 has a cold bias near the surface and a warm bias at around 100 m (maximum 0.08 °C, see Figure 1). The warm bias is inherited from the initial condition. This warm bias decreases significantly in February 2022. This may correspond to the reduction of the warm bias of the L3S ODYSSEA SST observation, and as our assimilation scheme is multivariate, a change in the assimilated SST observation may produce a better correction for temperature at 100 m. The moderate La Nina event 2021-2022 also contributes to cooling the system.

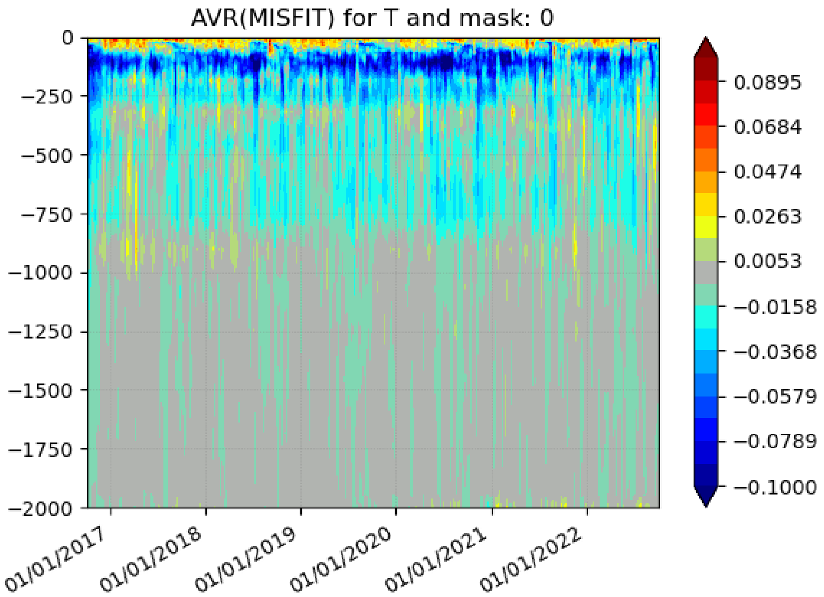


Figure 1. Mean innovation (observation minus GLO12v4 background model first trajectory) as a function of time and depth with respect to the assimilated temperature vertical profiles (including Argo) over the 2017-2022 hindcast period. Units are °C.

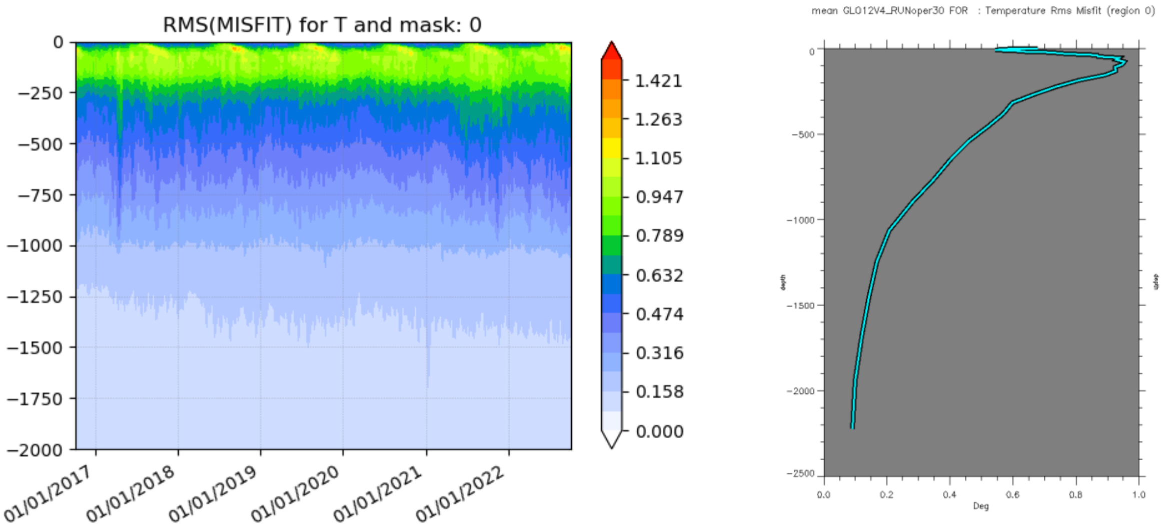


Figure 2. RMSD (observation minus GLO12v4 background model first trajectory) as a function of time and depth with respect to the assimilated temperature vertical profiles (including Argo) over the 2017-2022 hindcast period (on the left). On the right is the average profile over the entire period. Units are °C.

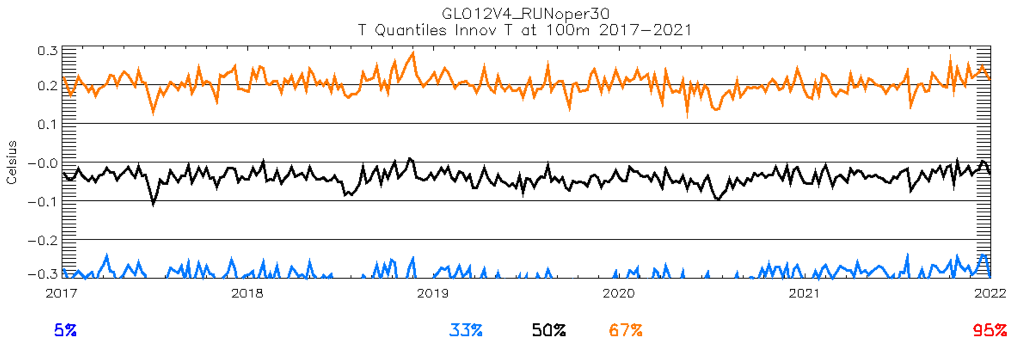
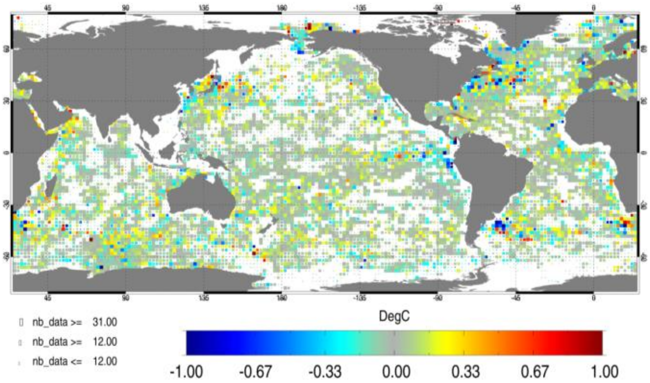


Figure 3. Percentiles (5, 33, 50, 67, 95) of the GLO12v4 temperature innovations as a function of time at 100 m depth over the 2017-2022 hindcast period. Units are °C. Note that the 5 and 95 percentiles are outside the window.

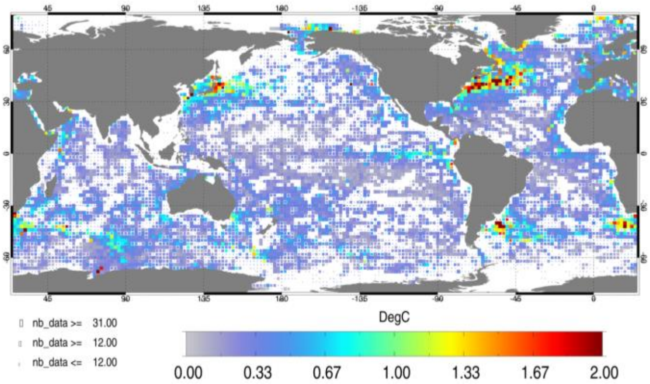
The temperature RMSD is around 0.6 °C at the surface; the error is maximum at 100 m (0.95 °C) and is less than 0.4 °C below 700 m (Figure 2). There is no obvious explanation for the increase in RMSD between 300 m and 400 m in June 2021 except for an increase in the number of XBT (eXpendable Bathy Thermograph) profiles (probe used to measure the temperature throughout the water column), and an increased number of discordant profiles.

The stability of GLO12v4 scores is good as illustrated by the different percentiles of temperature innovations at 100 m depth that remain stable over the entire period (Figure 3).

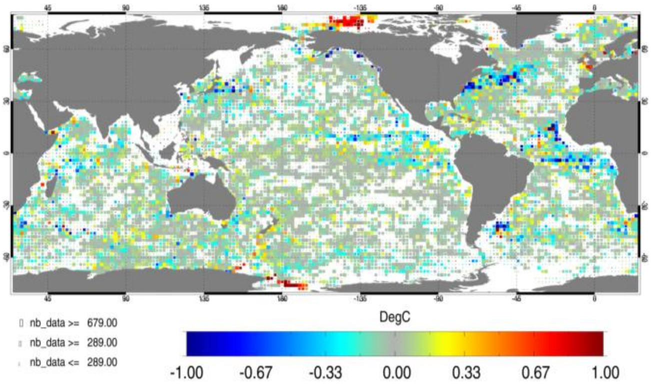
a) TEMP mean OBS-MODEL 0-5 m 2019



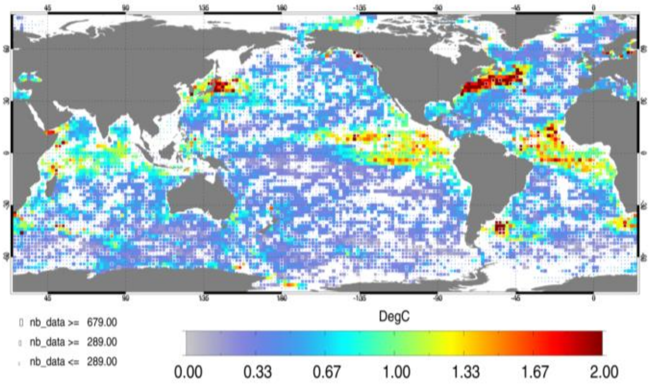
b) TEMP RMS OBS-MODEL 0-5 m 2019



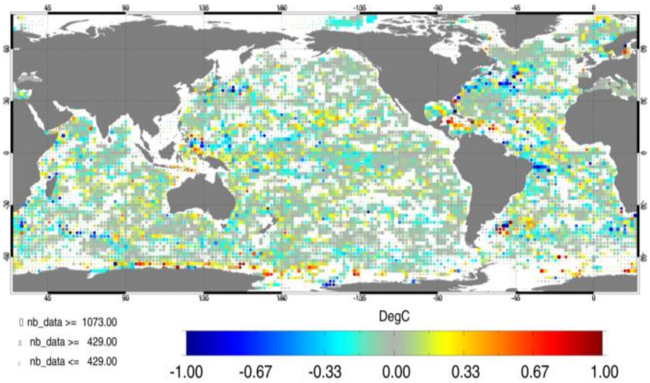
c) TEMP mean OBS-MODEL 5-100 m 2019



d) TEMP RMS OBS-MODEL 5-100 m 2019



e) TEMP mean OBS-MODEL 100-300 m 2019



f) TEMP RMS OBS-MODEL 100-300 m 2019

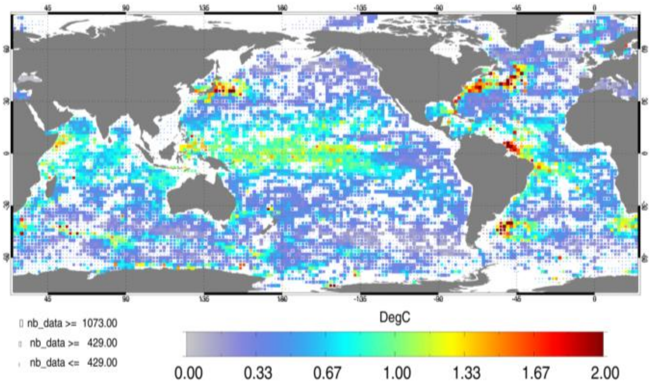


Figure 4. Argo observation - GLO12v4 analysis mean (a,c, and e) and RMSD (b,d, and f) for temperature (°C) during the calibration year 2019. Differences are averaged in the 0-5 m (a and b), 5-100 m (c and d), and 100-300 m (e and f) layers. Model outputs have been co-located in time and space to observations. For each pixel, statistics are computed for 2°x2° bins, and the size of a pixel is proportional to the number of observations within that bin.

Uncertainty maps in Figure 4 can be established from observation-analysis differences computed for each in situ Argo observation at the time and location of the observation. The differences are then averaged in time and in 2°x2° horizontal bins and over layers. Near the surface, the deviations with respect to in situ observations rarely exceed 0.5 °C; except in high spatial and temporal variability areas such as the Western Boundary Currents: Gulf Stream, Kuroshio, Zapiola eddy, and Agulhas current. A majority of yellow pixels at the surface indicates a cold bias, consistently with Figure 1. In the subsurface between 5 and 100 m, differences can reach 1 °C in the tropical basins and in particular in the Peru Upwelling and Mauritanian upwelling while they can reach 2 °C near Zapiola, in the Gulf Stream, in the North Brazil Current. Between 100 and 300 m differences, up to 1 °C are found in the tropical basins' warm pools (western tropical basins) and can still reach 2 °C locally in the strong variability areas. As these maps are established using one year of data, they include the seasonal cycle and the interannual signal. During El Niño or La Niña years, errors can be larger in the tropical pacific as large temperature anomalies occur.

Regionally averaged vertical profiles of temperature differences between 0 and 2000 m, including percentiles, give a complementary description of the average deviation with respect to assimilated in situ temperature profiles in Figure 5, Figure 6, Figure 7, Figure 8, and Figure 9. The strongest discrepancies are found near 100 m in the main thermocline. Temperature deviations with respect to in situ observations rarely reach more than 0.1 °C below 250 m. Wider error distributions can be found at depth in the deep waters of the Mediterranean Sea between 750 and 1000 m in Figure 5, and in the north Atlantic water inflow into the Arctic Ocean between 1000 and 1250 m in Figure 9.

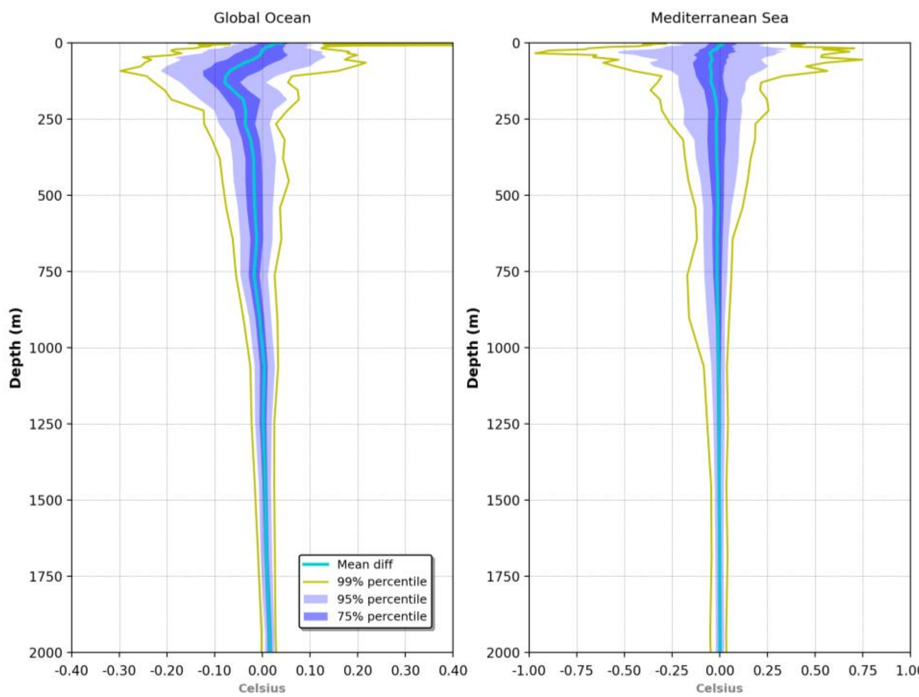


Figure 5. Temperature (°C) observation- GLO12v4 analysis mean and percentiles in 2019 for the Global Ocean (left) and Mediterranean Sea (right).

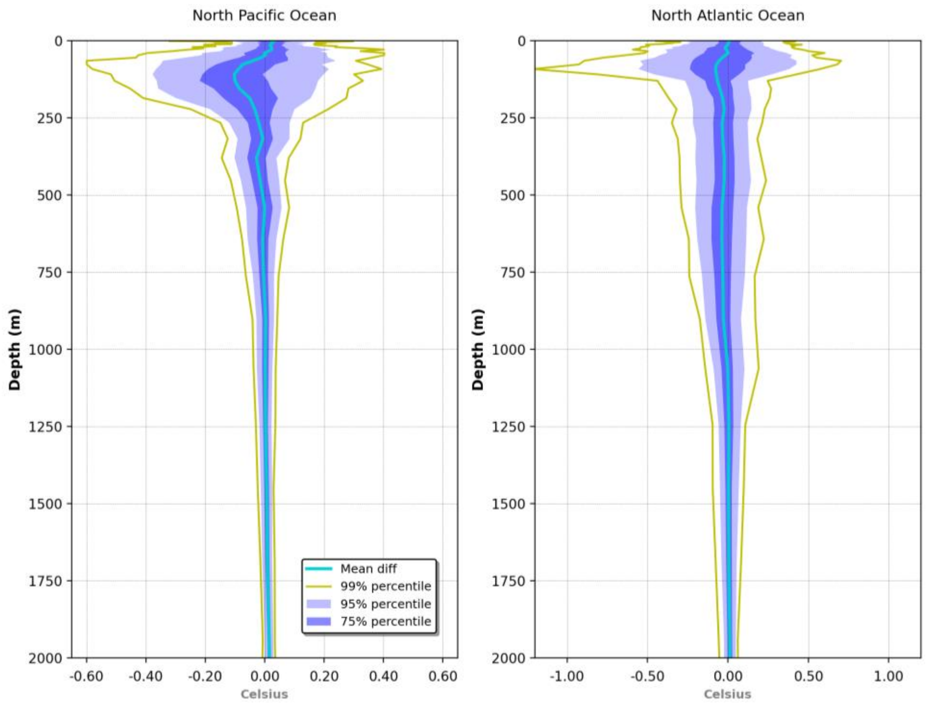


Figure 6. Temperature (°C) observation - GLO12v4 analysis mean and percentiles in 2019 for the North Pacific Ocean (left) and North Atlantic Ocean (right).

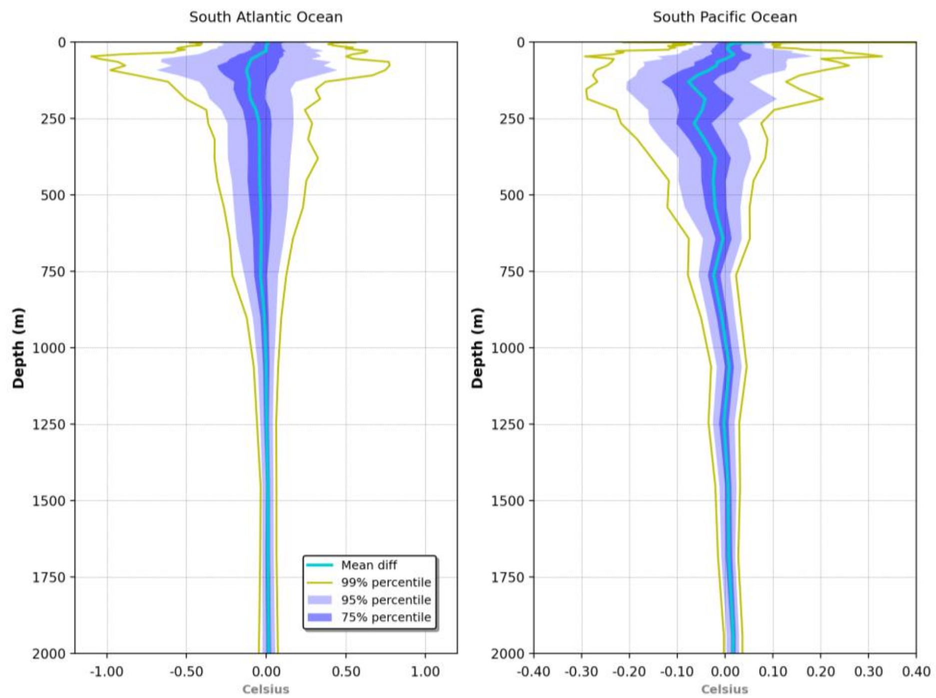


Figure 7. Temperature (°C) observation - GLO12v4 analysis mean and percentiles in 2019 for the South Atlantic (left) and South Pacific Oceans (right).

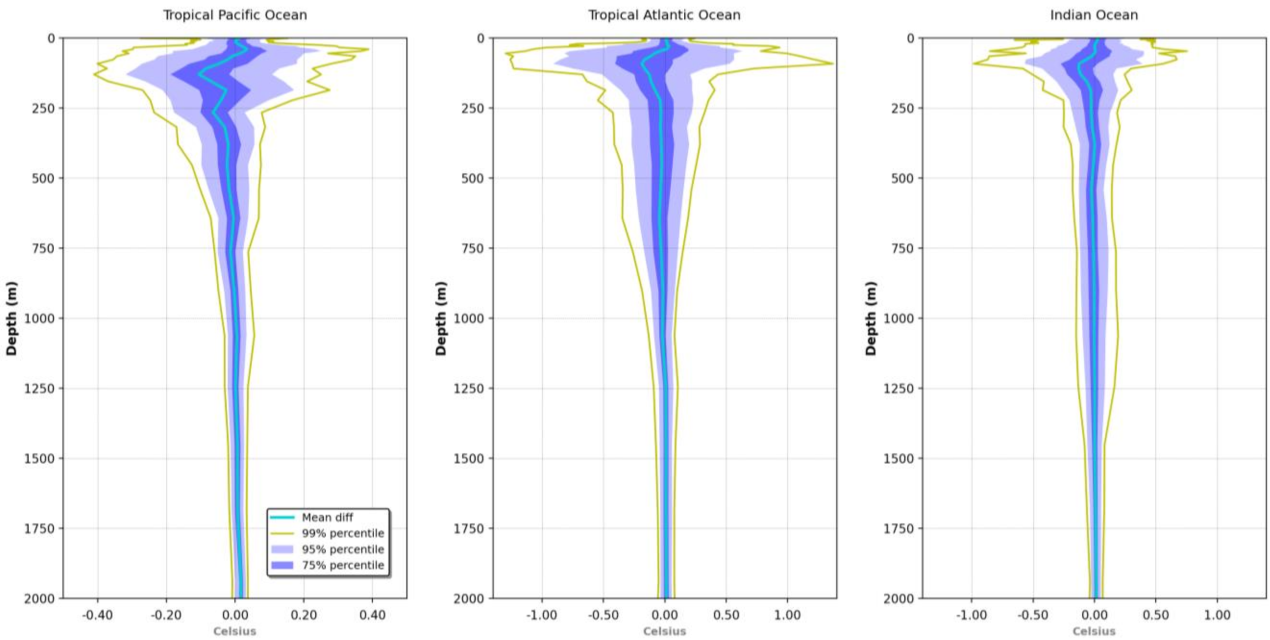


Figure 8. Temperature (°C) observation - GLO12v4 analysis mean and percentiles in 2019 for the Tropical Pacific Ocean (left), the Tropical Atlantic Ocean (middle) and the Indian Ocean (right).

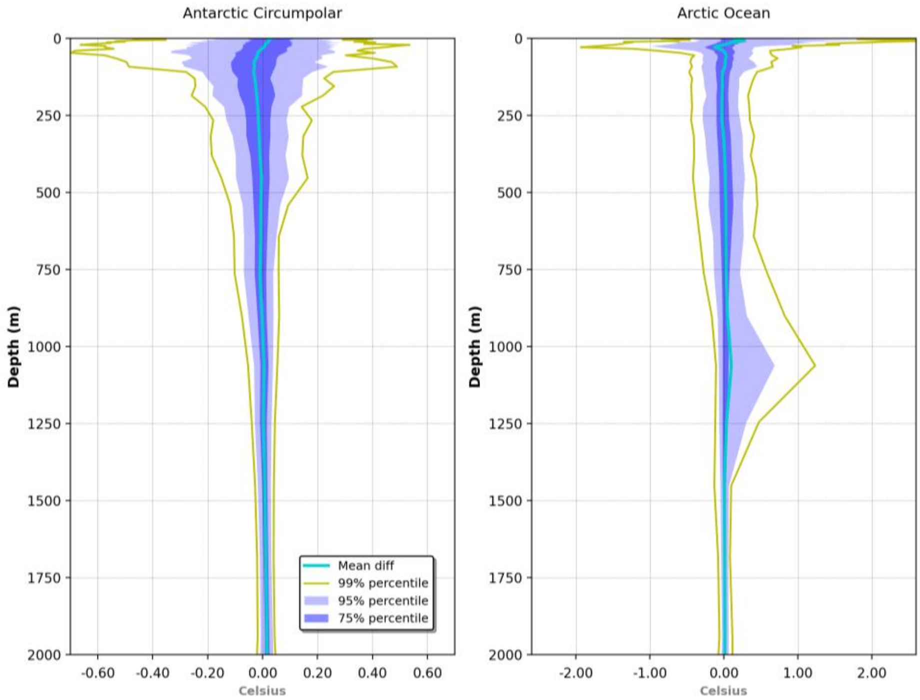


Figure 9. Temperature (°C) observation - GLO12v4 analysis mean and percentiles in 2019 for the Antarctic Circumpolar Ocean (left) and the Artic Ocean (right).

IV.2 Salinity

Figure 10 shows little significant fresh bias in salinity (less than 0.02 psu at the surface) and between 300 m and 400 m. The latter may be related to the salty drift of the Sea-Bird sensors on the profilers between 2015 and 2020 (https://argo.ucsd.edu/argo-salty-drift-salinity-data-issue-notice-2021/).

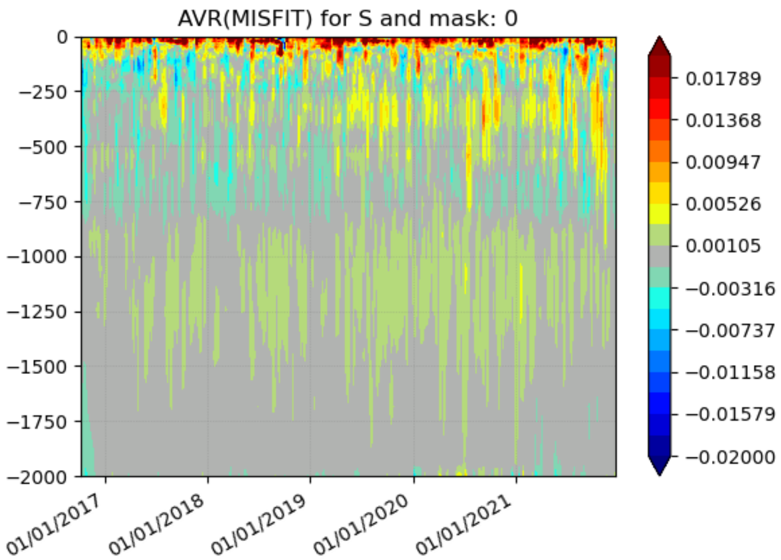


Figure 10. Mean innovation (observation minus GLO12v4 background model first trajectory) as a function of time and depth with respect to the assimilated salinity vertical profiles (including Argo) over the 2017-2022 hindcast period. Units are psu.

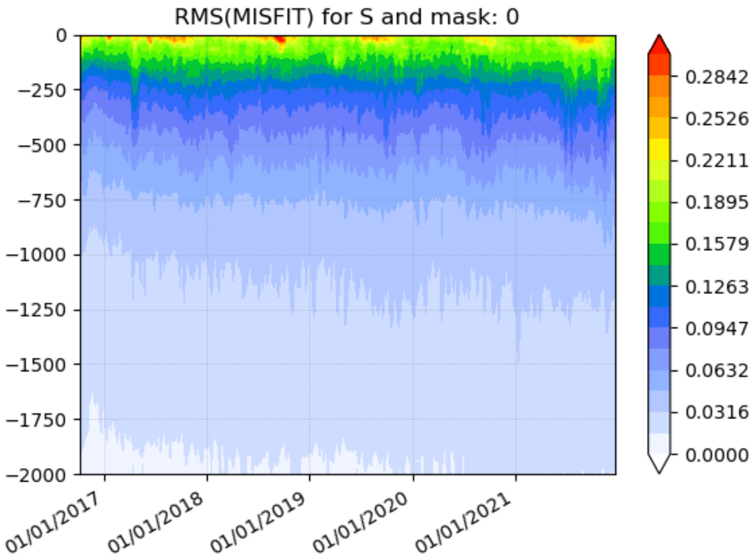


Figure 11. RMSD (observation minus GLO12v4 background model first trajectory) as a function of time and depth with respect to the assimilated salinity vertical profiles (including Argo) over the 2017-2022 hindcast period. Units are psu.

As shown in Figure 11 the RMSD is around 0.25 psu at the surface and decreases steadily below; the error is less than 0.1 psu below 300 m. There is no obvious explanation for the increase in RMSD between 300 m and 400 m in June 2021 except for an increased number of discordant profiles (North Atlantic and around Kerguelen).

The stability of GLO12v4 scores is good as illustrated by the different percentiles of salinity innovations at 100 m depth that remain stable over the entire period (Figure 12).

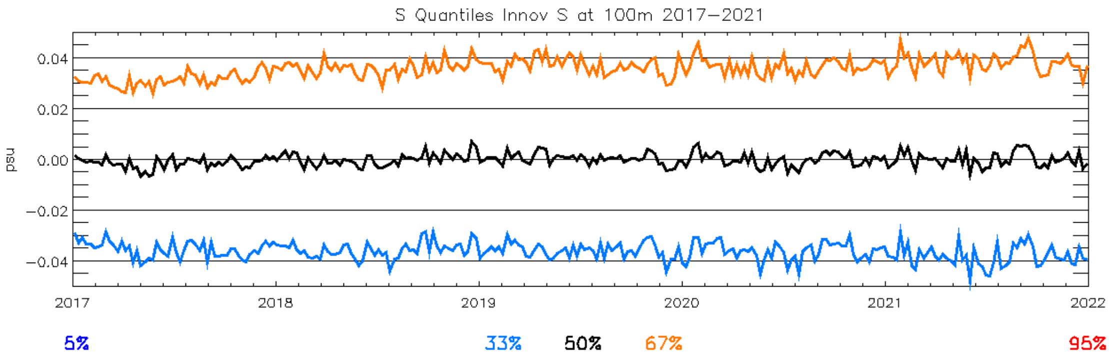
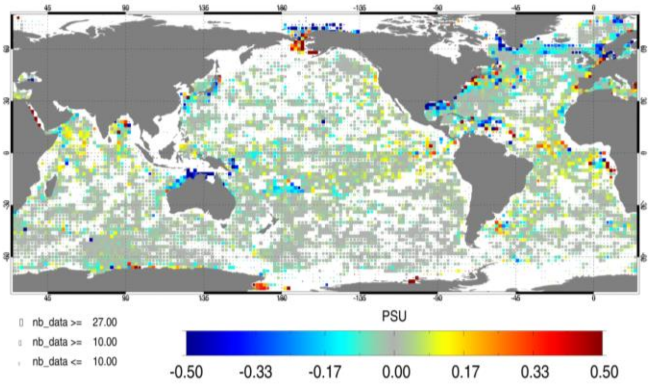


Figure 12. Percentiles (5, 33, 50, 67, 95) of the salinity innovations as a function of time at 100 m depth over the 2017-2022 hindcast period. Units are psu. Note that the 5 and 95 percentiles are outside the window.

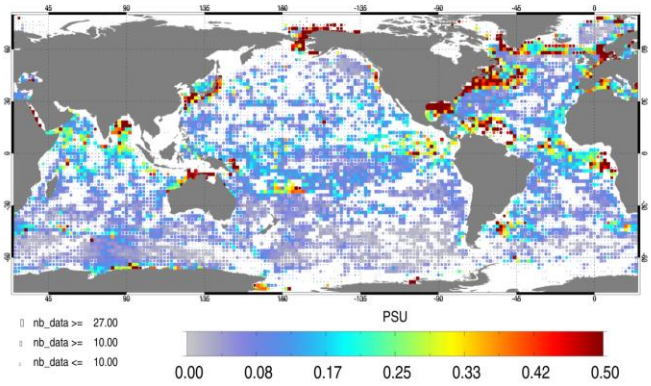
Uncertainty maps in Figure 13 can be established from observation-analysis differences computed for each in situ observation at the time and location of the observation. The differences are then averaged in time and in 2°x2° horizontal bins and over layers. Near the surface the deviations with respect to in situ observations rarely exceed 0.5 psu, except in high spatial and temporal variability areas such as the Western Boundary Currents and in the marginal seas of the Arctic Ocean. A majority of yellow pixels at the surface indicates a fresh bias on average, consistently with Figure 10. In the Gulf Stream and the Arctic marginal seas GLO12v4 is too salty. The dipole at Bering strait may indicate that the transport from the Arctic to the North Pacific could be underestimated in 2019. In subsurface between 5 and 100 m, differences are stronger in the tropical basins reaching 0.3 psu but tend to compensate on average over the year. Between 100 and 300 m, RMSDs up to 0.3 psu are found in the North Atlantic where the salinity variability is highest. As these maps are established using one year of data, they include the seasonal cycle and the interannual signal. During El Niño or La Niña years, errors can be larger in the tropical pacific.

Regionally averaged vertical profiles of salinity differences between 0 and 2000 m, including percentiles, give a complementary description of the average deviation with respect to assimilated in situ salinity profiles in Figure 14, Figure 15, Figure 16, Figure 17 and Figure 18. The strongest discrepancies are found near 100 m in the main thermocline. Salinity deviations with respect to in situ observations rarely reach more than 0.1 psu below 250 m where the widest error distributions can be found in the deep waters of the Mediterranean Sea between 750 and 1000 m in Figure 14, and in the Indian ocean between 1000 and 1250 m in Figure 17.

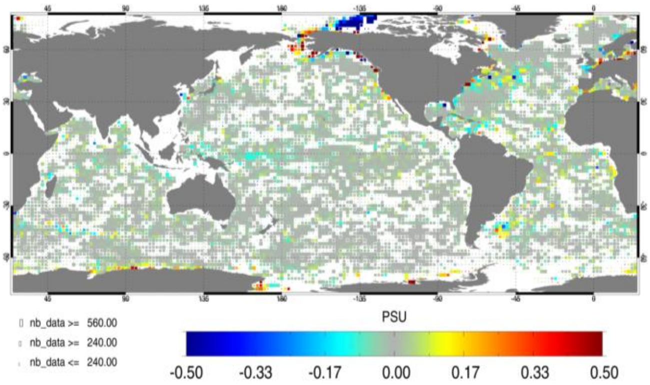
a) SAL mean OBS-MODEL 0-5 m 2019



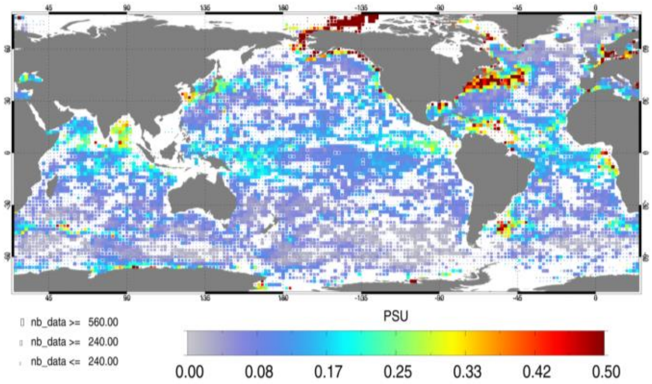
b) SAL RMS OBS-MODEL 0-5 m 2019



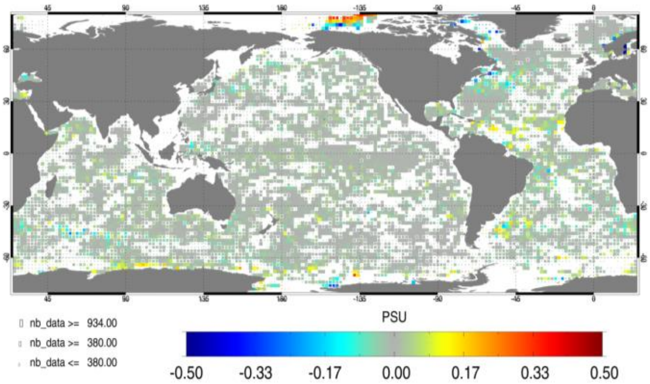
c) SAL mean OBS-MODEL 5-100 m 2019



d) SAL RMS OBS-MODEL 5-100 m 2019



e) SAL mean OBS-MODEL 100-300 m 2019



f) SAL RMS OBS-MODEL 100-300 m 2019

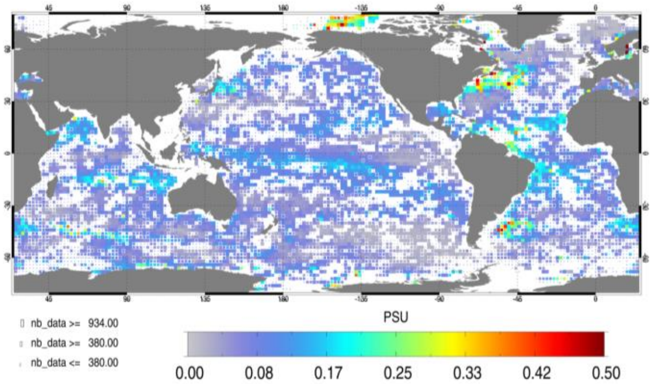


Figure 13. Argo observation – GLO12v4 analysis mean (a,c, and e) and RMSD (b,d, and f) for salinity (psu) during the calibration year 2019. Differences are averaged in the 0-5 m (a and b), 5-100 m (c and d), and 100-300 m (e and f) layers. Model outputs have been co-located in time and space to observations. For each pixel, statistics are computed for 2°x2° bins, and the size of a pixel is proportional to the number of observations within that bin.

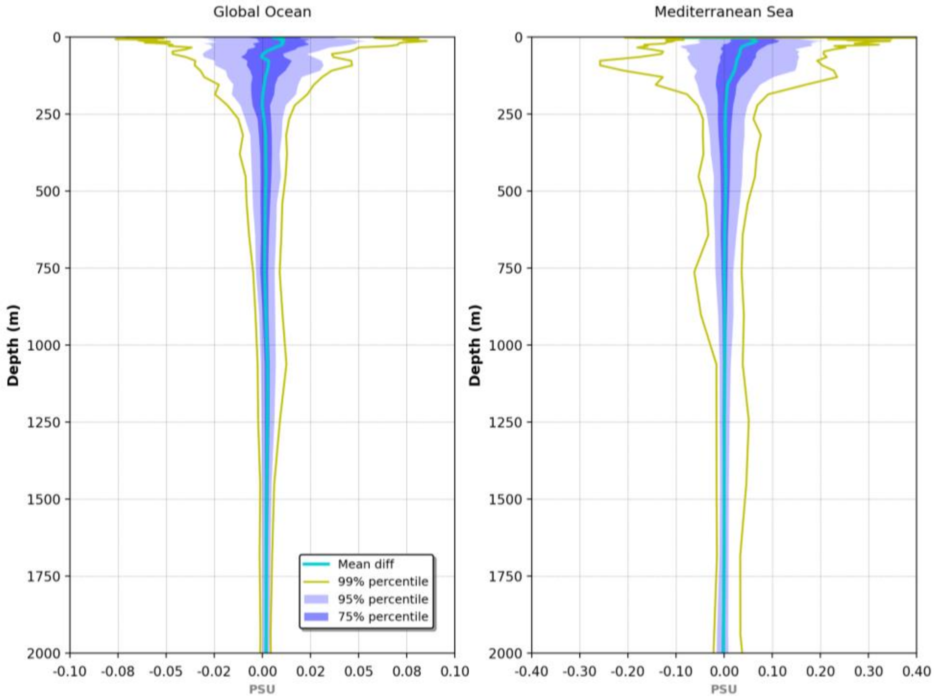


Figure 14. Salinity (psu) observation - GLO12v4 analysis mean and percentiles in 2019 for the Global Ocean (left) and the Mediterranean Sea (right)

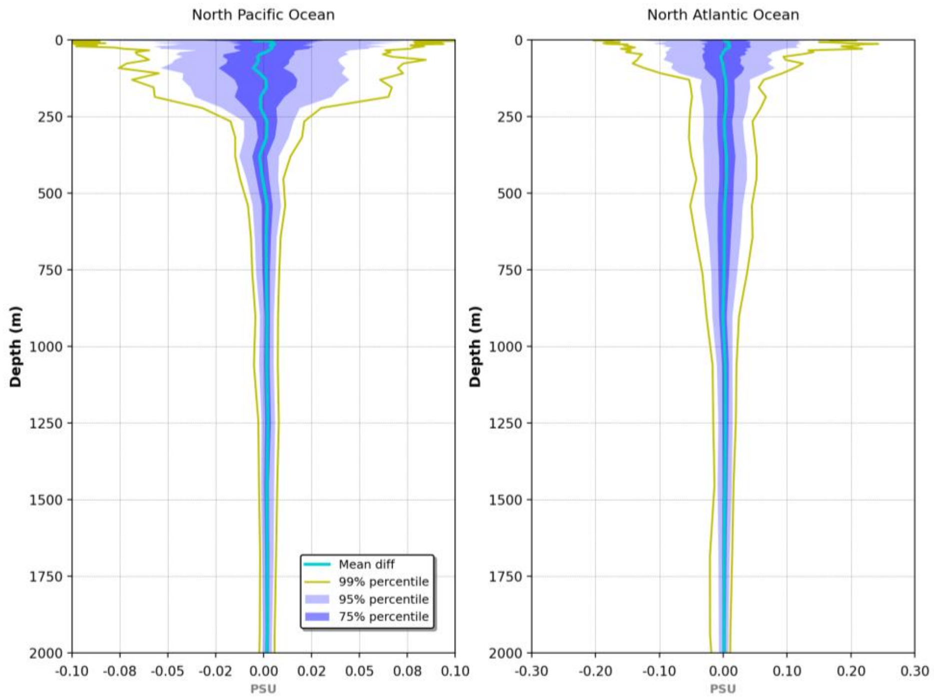


Figure 15. Salinity (psu) observation - GLO12v4 analysis mean and percentiles in 2019 for the North Pacific Ocean (left) and North Atlantic Ocean (right)

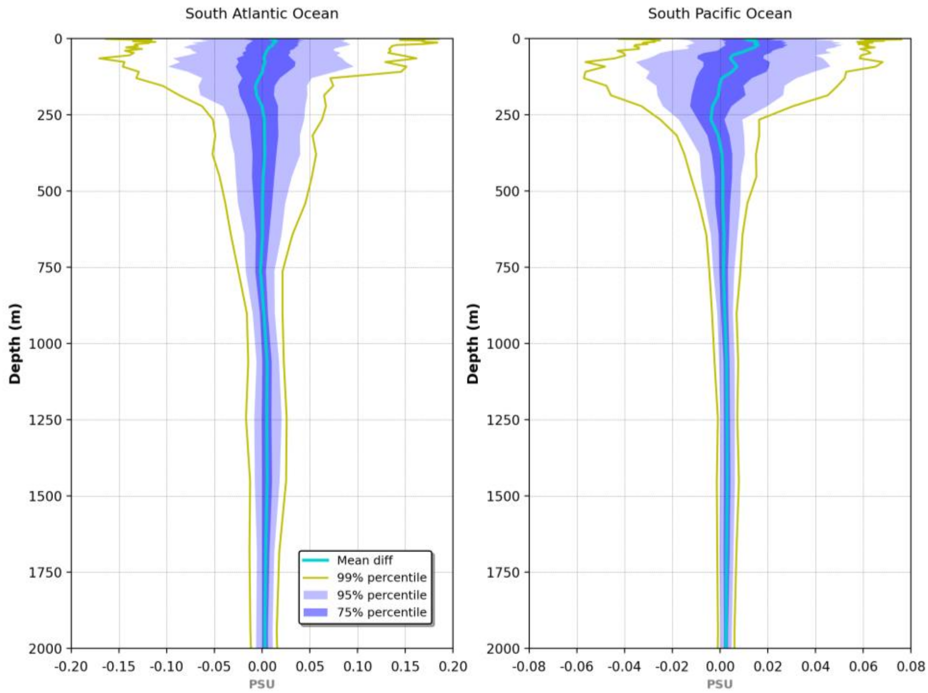


Figure 16. Salinity (psu) observation - GLO12v4 analysis mean and percentiles in 2019 for the South Atlantic (left) and South Pacific Ocean (right)

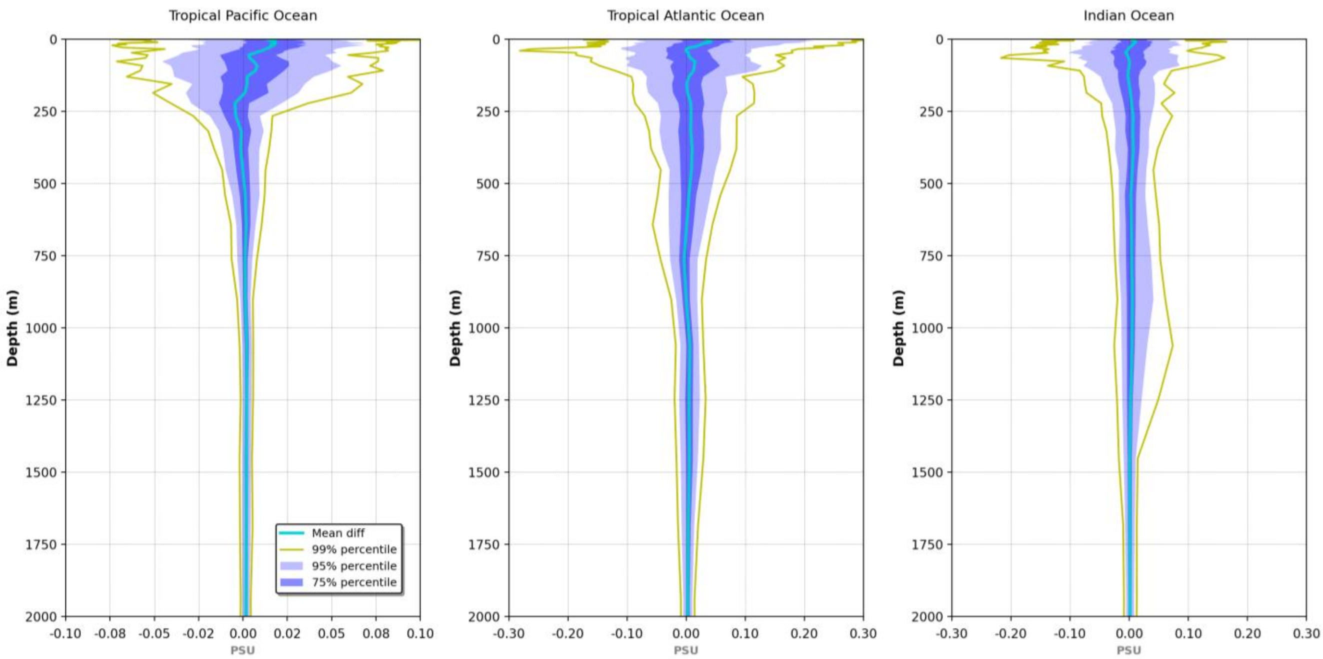


Figure 17. Salinity (psu) observation – GLO12v4 analysis mean and percentiles in 2019 for the Tropical Pacific Ocean, Tropical Atlantic Ocean and Indian

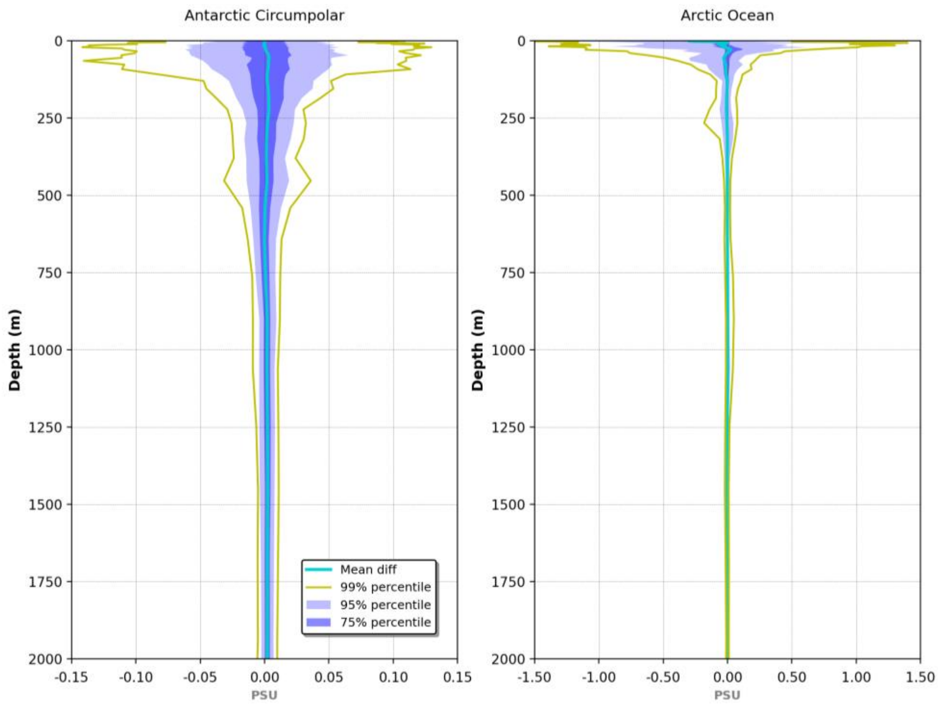


Figure 18. Salinity (psu) observation - GLO12v4 analysis mean and percentiles in 2019 for the ACC and the Artic Ocean

IV.3 Sea-surface-temperature (SST)

Figure 19 shows that the assimilated SST product ODYSSEA is quite different from OSTIA on global average and in terms of variability. The high resolution L3S product ODYSSEA displays a lot of variability which is filtered out by the GLO12v4 system prior to assimilation (super observation); still, GLO12v4 exhibits a warm bias on global average with respect to OSTIA (around 0.1 °C) and a cold bias with respect to ODYSSEA (around 0.1 °C too). On global average, even with the seasonal error cycle the differences between the SST from the in situ data and the model remain very small, less than 0.1°C. The RMSDs stay below 0.5 °C, reaching 0.6 °C when computed with the high resolution L3S or the sparse in situ which is consistent with the errors prescribed to the system.

The maps of mean differences with OSTIA in 2019 (Figure 20) show discrepancies of 1 °C in several upwelling areas (Eastern Tropical Pacific Ocean, Angola and Mauritania) which is consistent with Figure 4 showing mean differences with in situ observed temperatures in the 0-5 m layer. The upwelling areas are too warm as the thermocline is too deep.

The Arctic marginal seas are too cold in GLO12v4, which can be diagnosed from ODYSSEA innovations and from drifter comparisons (not shown) but needs to be investigated further.

The dataset cmems_mod_glo_phy_anfc_0.083deg-climatology-uncertainty_P1M-m helps us to better understand the uncertainties concerning the SST in regions where we have data. If you look at the statistics contained in that dataset, we observe the main regions where the global system exhibits differences with the observations. Figure 21 represents the monthly RMSD for the twelve months. It shows that the model is generally in a good agreement with the observations except for particular regions with high mesoscale activity; in particular the Gulf Stream, the Kuroshio, Agulhas, and the Artic region during the melt season in boreal summer. As described before,it's known that the system is in a better agreement than the previous one even though it's much more energetic. It's statistically coherent to have greatest differences in regions where the Ocean is more turbulent.

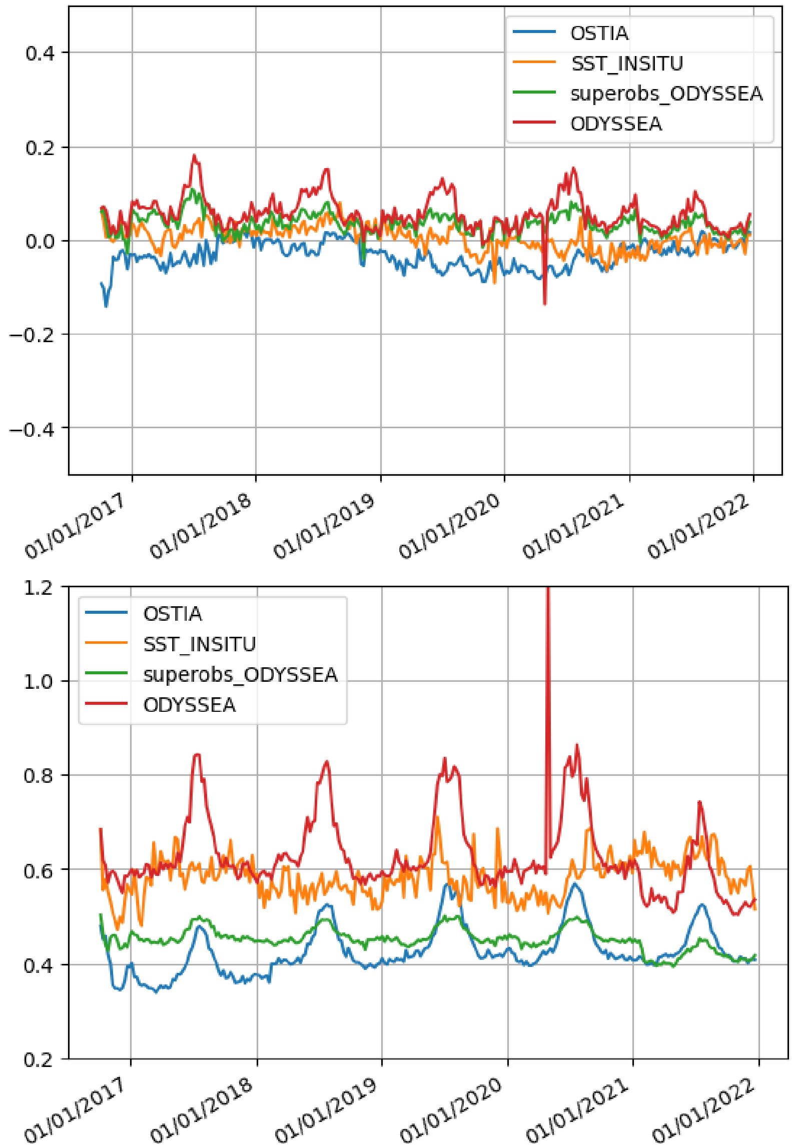


Figure 19. Mean innovation (top) and RSMD (bottom) (observation minus GLO12v4 background model first trajectory) as a function of time with respect to some observations over the 2017-2022 hindcast period: first level of the assimilated temperature vertical profiles including Argo (orange), assimilated ODYSSEA SST (red; super observations are in green), not assimilated OSTIA SST (blue). Units are °C.

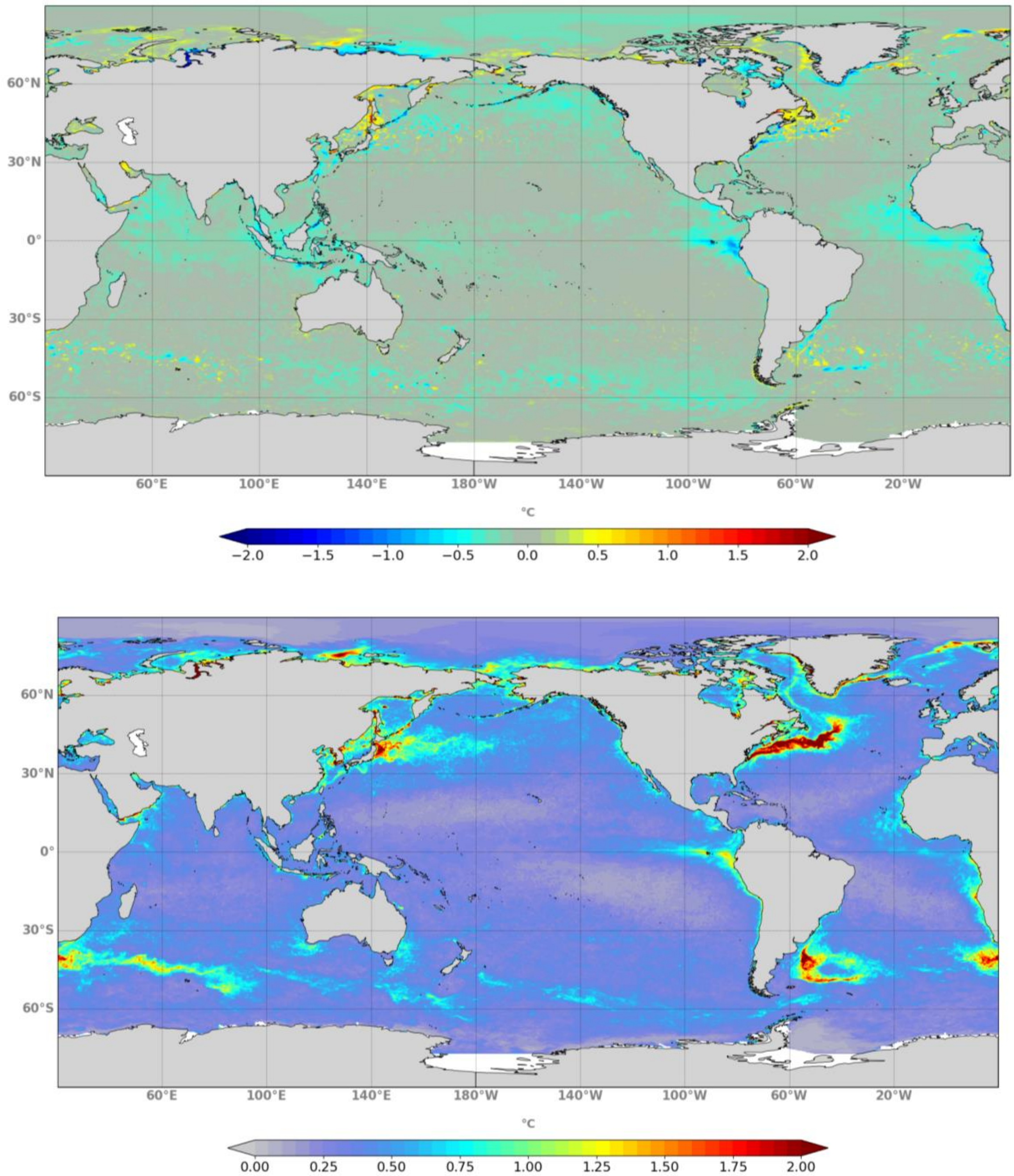


Figure 20. Mean (top panel: contours correspond to 0.1°C) and RMSD (bottom panel: contours correspond to 0.05 RMSD misfit) residual SST errors (°C) OSTIA – GLO12v4, computed for the calibration year 2019.

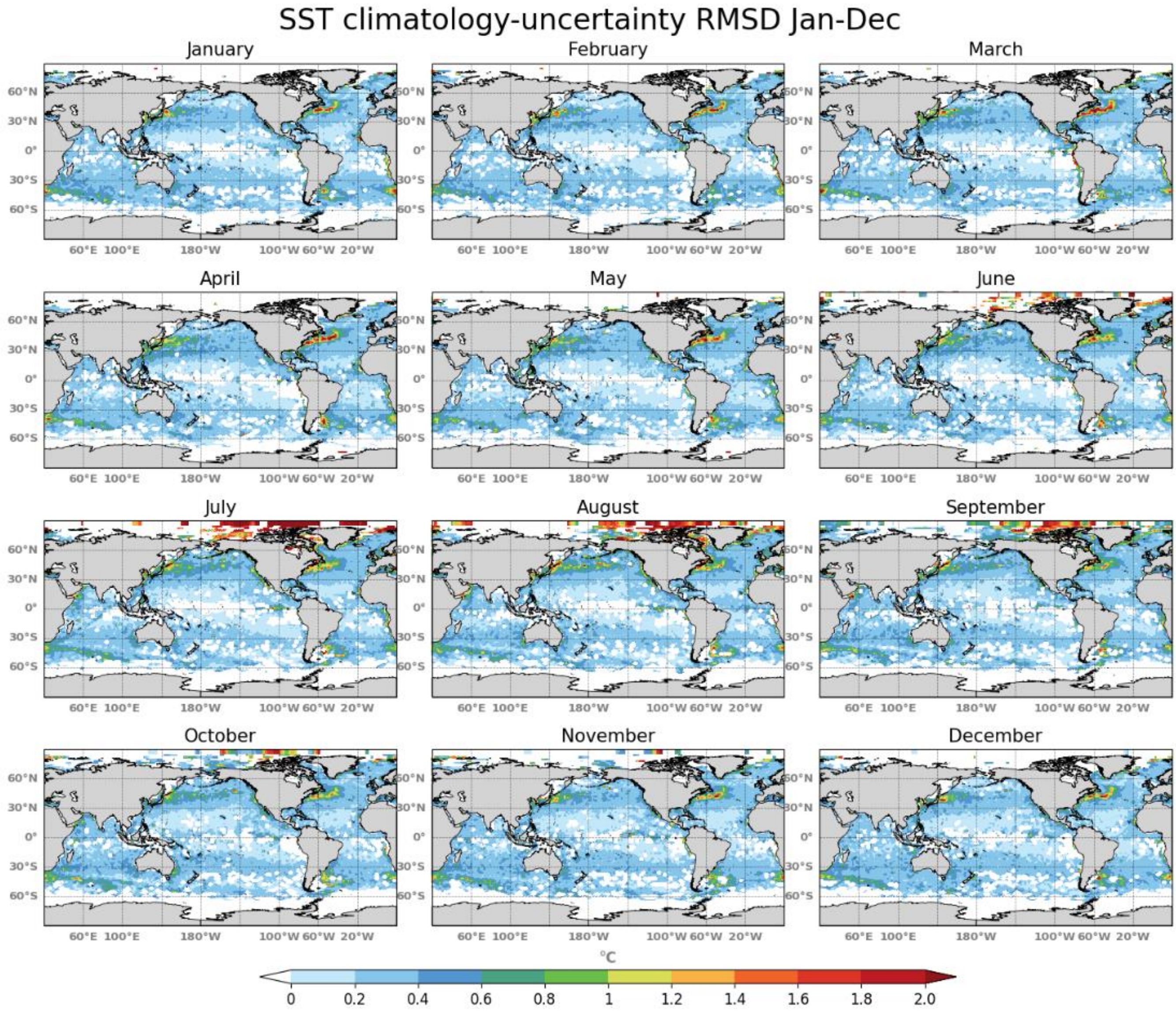


Figure 21. Monthly climatological RMSD SST global uncertainty plots.

For the Arctic region the observations are sparse in space and time, due to the particularity of the polar region with the ice pack during the winter. The statistics show a warming trend during the spring season which can be explained by the fact that the ice melts too early. The waters masses are too warm at the surface, but it's known that the drifters' measurements in the polar regions are not high quality controlled. So, this trend should be interpreted with caution.

The distribution of the observations (Figure 22), shows limited coverage in the tropical band and the polar region. The highest density levels are reached in the North Atlantic and Pacific gyres, and in the South of the three oceans around 30°S. Except for the Artic where observations span five months between June and October, the highest levels of observation density remain high throughout the year with a maximum reached between October and December.

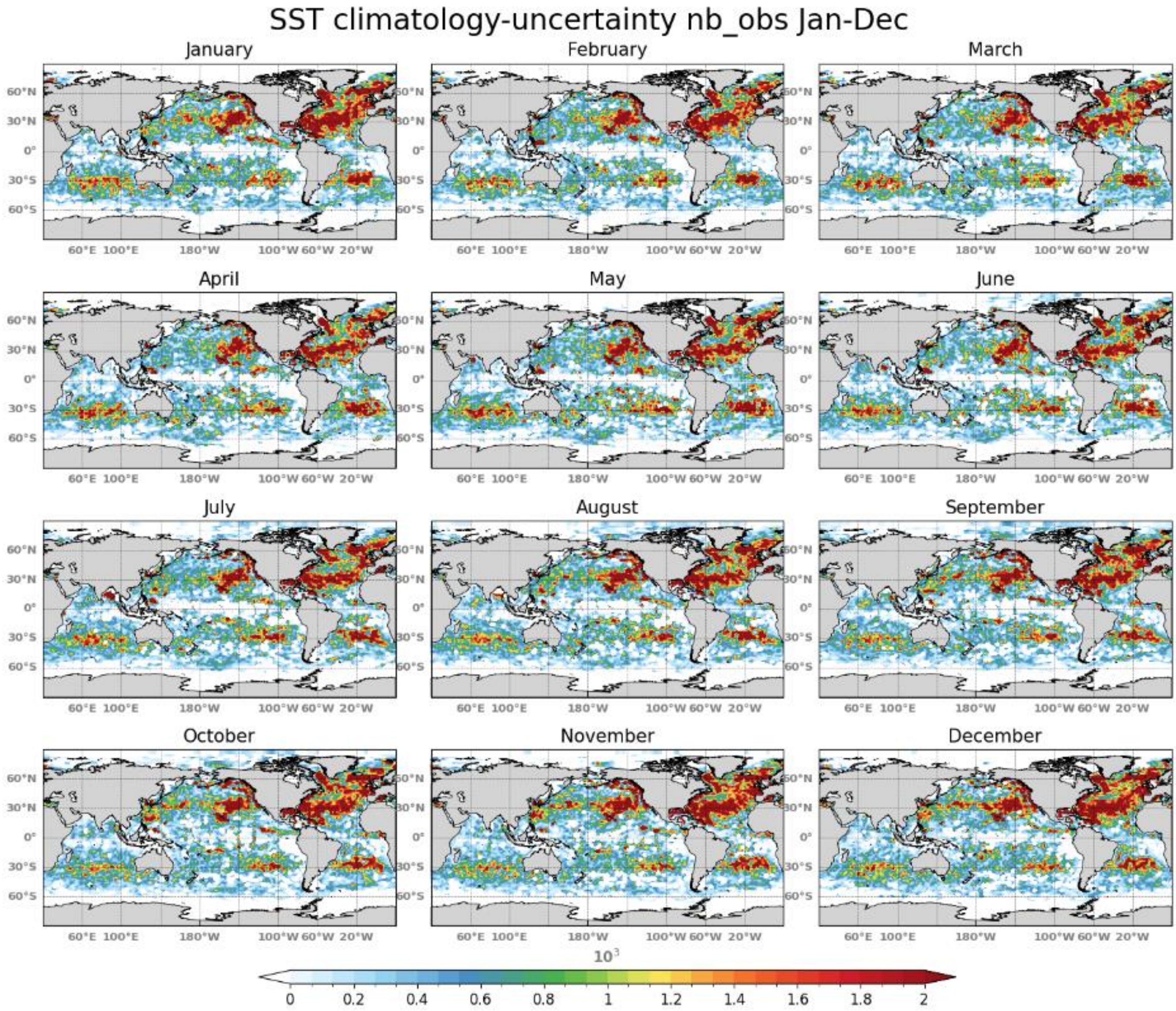
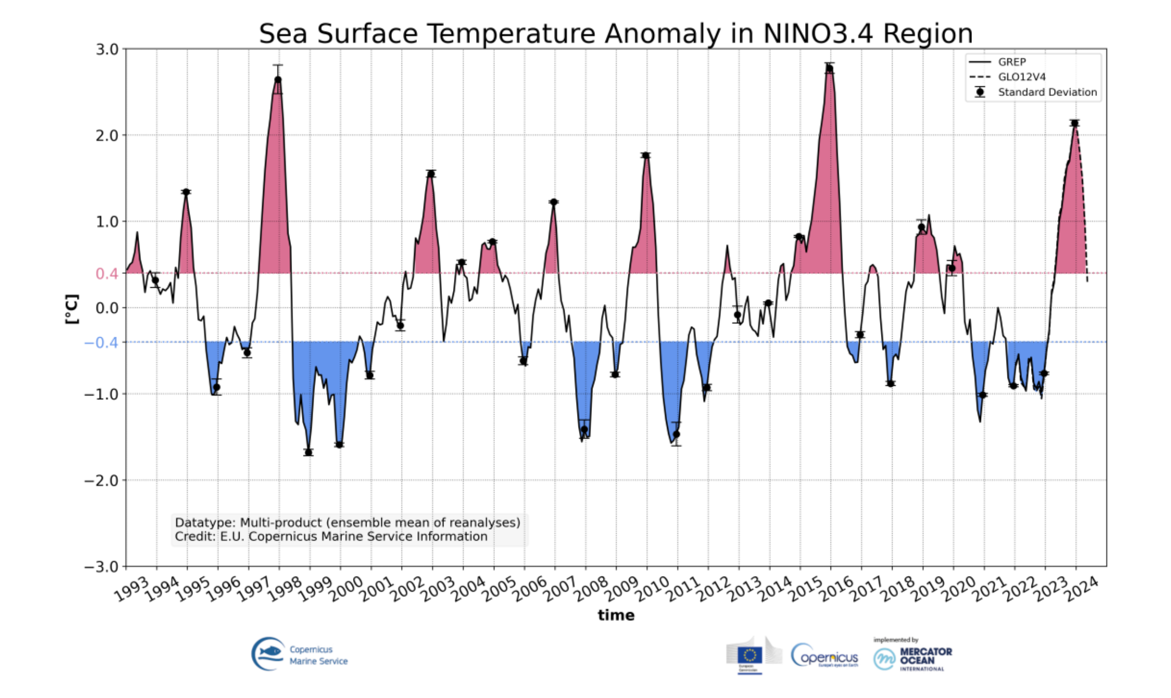


Figure 22. Monthly climatological Number of observation SST uncertainty plots for the global area.

For the dataset cmems_mod_glo_phy_anfc_0.083deg-sst-anomaly_P1D-m the consistency and accuracy of the SST anomaly maps were checked by intercomparing them with SST anomaly maps computed from OSTIA (https://doi.org/10.48670/moi-00165), ESA CCI SST (https://doi.org/10.48670/moi-00169) analysis over the period 2020-2023 (not shown), and with GREP multi-reanalysis anomalies over the years 2020-2023 (see for instance in the nino1+2 box and nino3.4 in Figure 23).



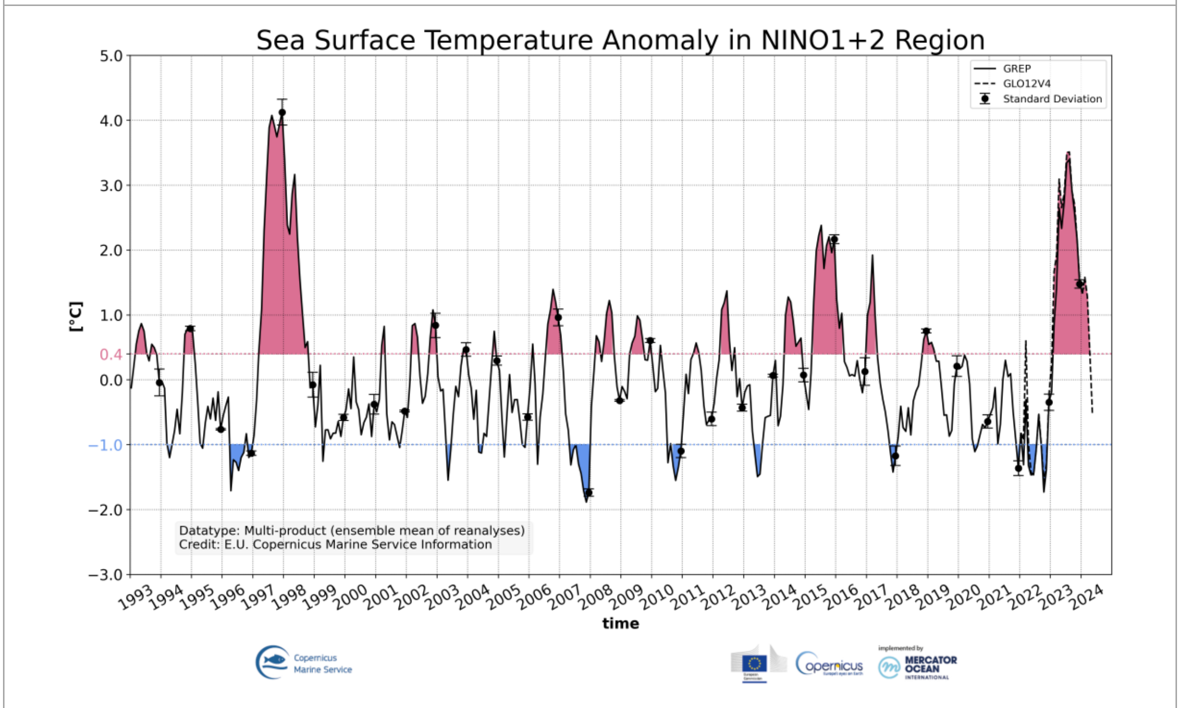


Figure 23. Time evolution of SST anomaly in the nino3.4 (top panel) and nino1+2 (bottom panel) boxes in the tropical pacific, computed from GREP (solid line) and from GLO12v4 (dashed line).

IV.4 Sea level

IV.4.1 Sea level anomalies (SLA)

On global average, GLO12v4 is close to SLA observations (Figure 24). The mean innovation shows that the system is lower than the observations by around 1 to 2 mm on average. The global RMSD rarely exceeds 5.5 cm.

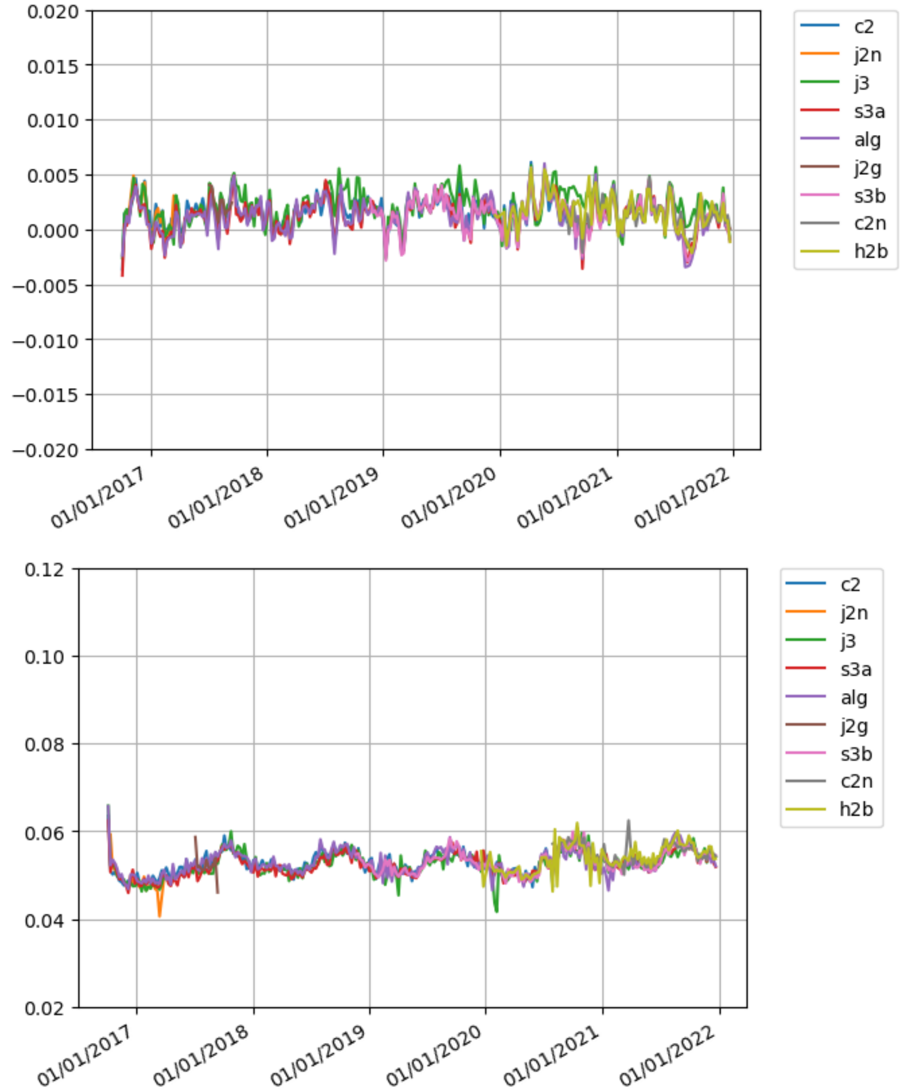


Figure 24. Mean innovation (top) and RSMD (bottom) (observation minus GLO12v4 background model first trajectory) as a function of time with respect to along-track SLA observations over the 2017-2022 hindcast period (blue: cryosat2, orange: Jason 2 new orbit, green: Jason 3, red: sentinel 3a, purple: saral altika geodetic, brown: Jason 2 geodetic, pink: sentinel 3b, grey: cryostat 2 new orbit, yellow: HY2). Units are m.

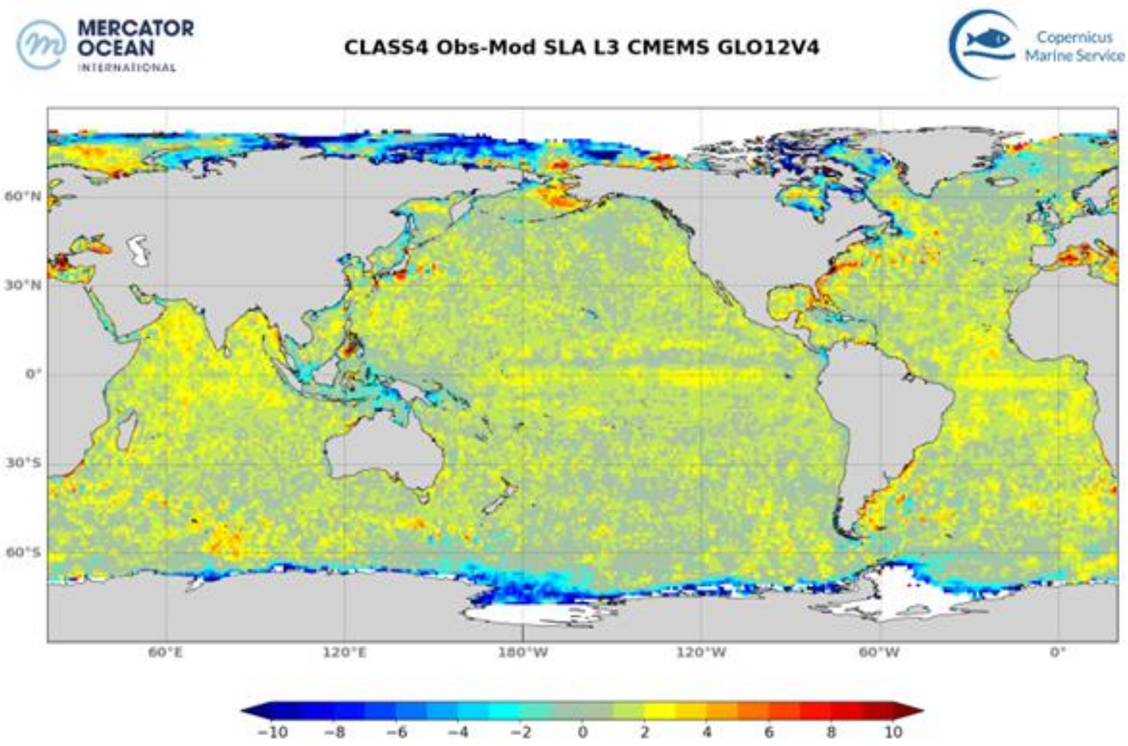


Figure 25. Global mean SLA residual differences (observation - GLO12v4 analysis) in 2019, averaged over 2°x2° bins. Units are mm.

Figure 25 shows the map of SLA residual differences between the observations and the analysis, averaged over 2°x2° bins for the year 2019 confirms that SLA information is well assimilated by the system. The map shows that GLO12v4 SSH is generally lower than the observations (by around 2 mm), except in polar areas. Higher errors are found in the Indonesian area and in the polar oceans, which is consistent with the error prescribed in the systems shown in Figure 27. This map is also showing the areas where there are large uncertainties both for SLA observations and GLO12v4 products: coastal areas, closed seas, polar areas, high variability areas.

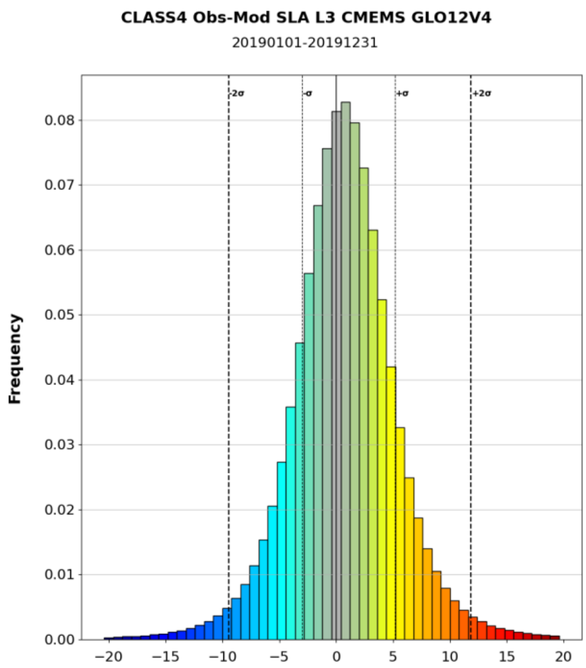


Figure 26. Probability density function of the global mean residual differences (observation - GLO12v4 analysis) in 2019, averaged over 2°x2° bins. Units are mm.

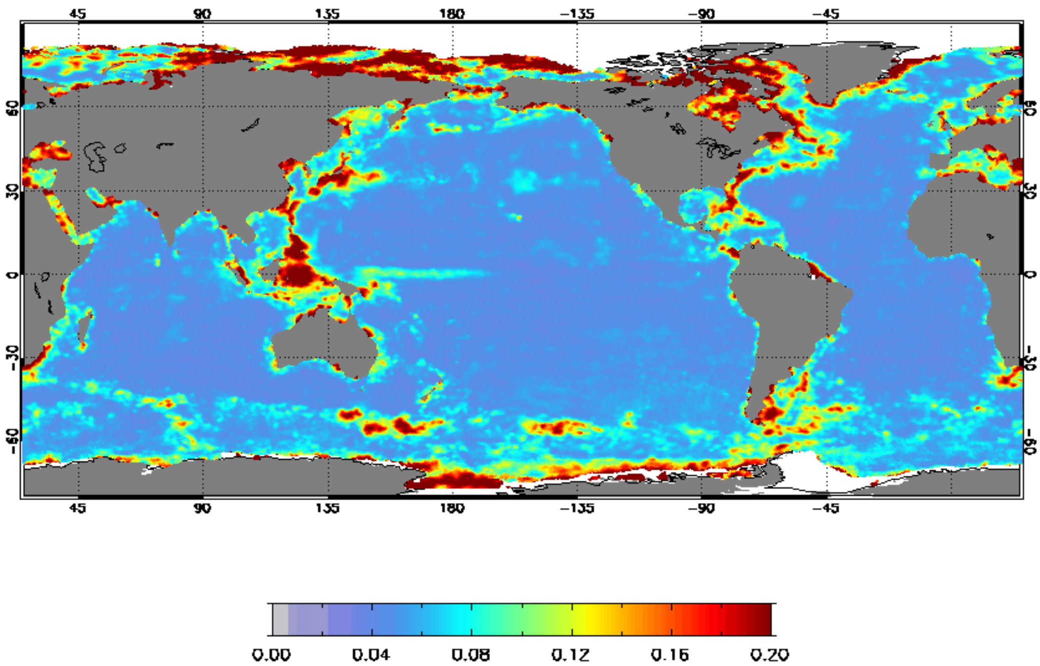


Figure 27. Observation error prescribed for SLA observations (m).

IV.4.2 Total sea levels

In this section, we validate the dataset cmems_mod_glo_phy_anfc_merged-sl_PT1H-I (MOWL: merged ocean water levels). The validation consists in comparison between the sea level from the product to measurements from tide gauges of Copernicus Marine Service database (INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030 product) for the year 2022.

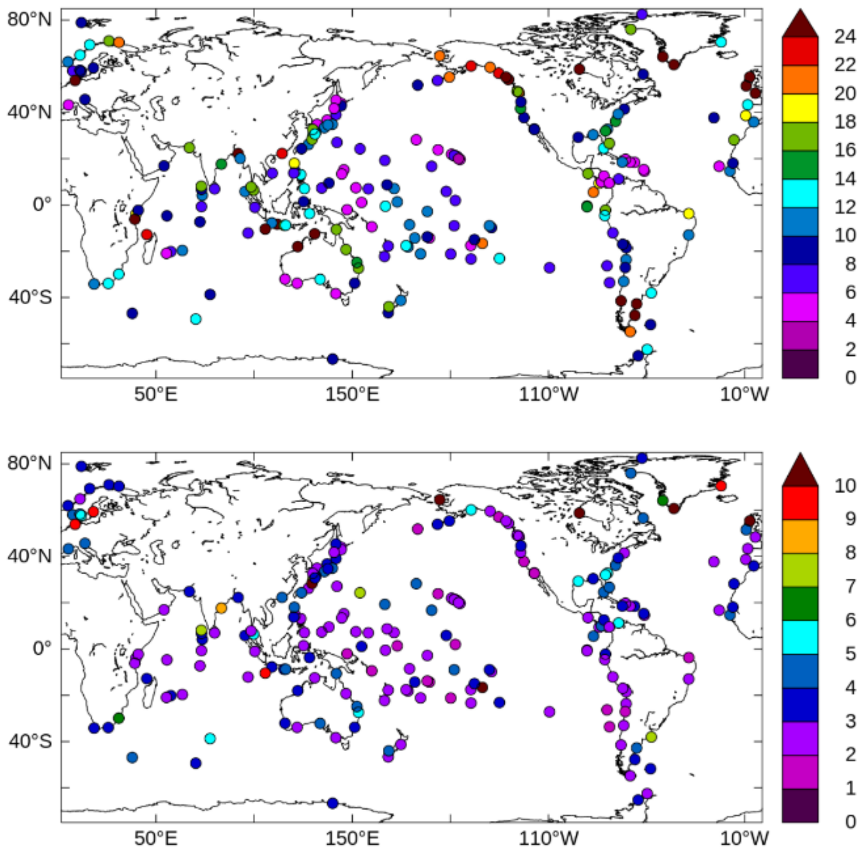


Figure 28. RMSD (cm) of total sea level (top) and residual sea level (bottom) between MOWL data and Copernicus Marine Service tide gauges over the year 2022.

Two components are evaluated: the total sea level corresponding the "total_hs" variable, and a residual component. The residual component is obtained by applying a tide killer (Demerliac filter) to the total sea level time series. Figure 28Figure 29 shows a global map of RMS at tide gauge locations. The areas with the most errors are in the Bay of Biscay, Timor Sea, and Mozambique Channel (RMS over 24 cm) and Alaska (RMS around 20 cm). These errors are primarily driven by errors in the tidal signal since the non-tidal equivalents have much lower RMS at these locations (below 3 cm). In general, total water levels have an RMS error of less than 15 cm. The smallest errors, below 5 cm, are found along the Japanese coast, in Hawaii, in south-eastern Australia, and in the Lion Gulf (Mediterranean, south of France). The non-tidal signal performs fairly well everywhere (RMS less than 2 cm), with the highest errors in the Baltic Sea. Figure 30 is equivalent to Figure 28, but for time correlation. Correlations are high for the total signal, with values generally above 0.94. The MWOL approach works particularly well along the coasts of North America, Japan, Indonesia, and Australia. Regarding the non-tidal signal, we find equivalent results for the above-mentioned areas, but the results are slightly worse for tide gauges in the Central Pacific and the Indian zone, with correlations between 0.8-0.95. Figure 34 shows Taylor plots summarizing the comparison of MWOL's total and non-tidal time series with those of the tide gauges. The additional information compared with the previous figures concerns the comparison of the variability measured by the tide gauges with the variability of MWOL's multi-model approach. For most tide gauges, the multi-model approach for total sea level has a standard deviation within 15% of that of the observations. The Taylor diagram for the non-tidal signal shows more fragmented statistical measurements, due to lower correlation for some measurement points mostly driven by tides, with a greater dispersion. A few geographical outliers can be found for both Taylor diagrams, but the vast majority of statistical comparisons show satisfactory results.

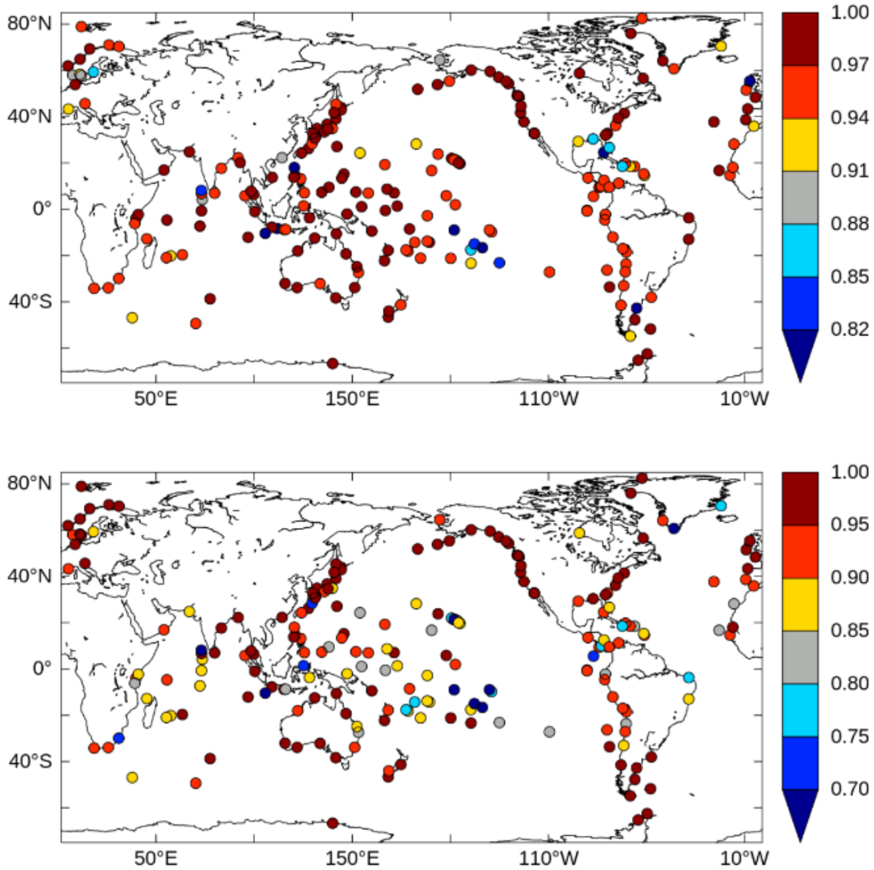


Figure 29. Correlation of total sea level (up) and residual sea level (down) between MOWL data and Copernicus Marine Service tide gauges over the year 2022.

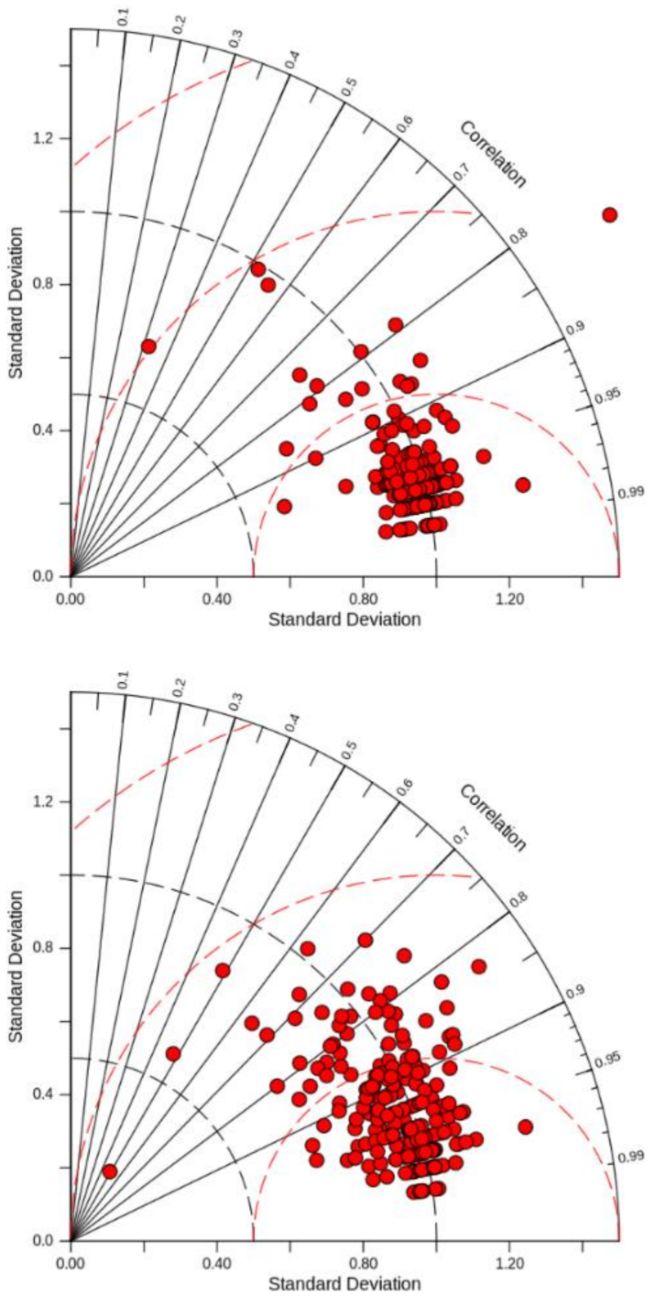


Figure 30. Taylor diagram comparing the total sea level (up) and residual sea level (down) time series between MOWL and Copernicus Marine Service tide gauges over the year 2022. The red arcs represent centred RMS error.

IV.5 Sea Ice

IV.5.1 Validation in the Arctic region

IV.5.1.1 <u>Sea Ice concentration</u>

To represent the discrepancy between the model and the observation, we use the proportion correct total (PCT) following the conclusion of Smith et al. (2016) that, it was more insightful to refer to the PCT rather than other proportions. The PCT quantity is defined as PCT = (hit ice + hit water) / n (see Table 9); where n is the total number of observations with a sea ice concentration greater than 20%. A value of 1 corresponds to a perfect score (perfect fit between the model and the observation).

Table 9. Contingency table entries for sea ice verification of the GLO12v4 system compared to AMSR2 sea ice concentration observations.

	Data ice	**Data water**
Model ice	Hit ice	False hit
Model water	Miss	Hit water

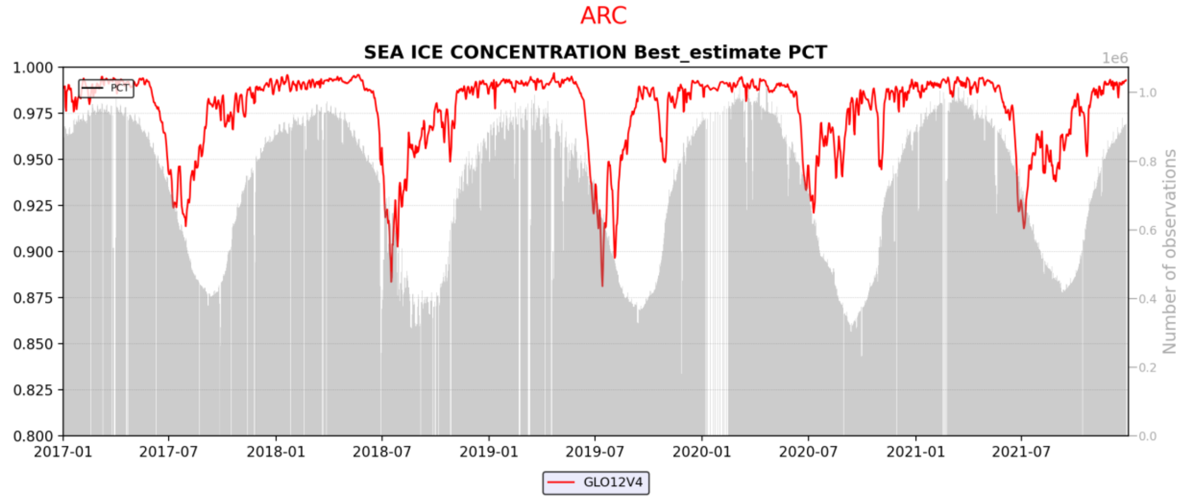
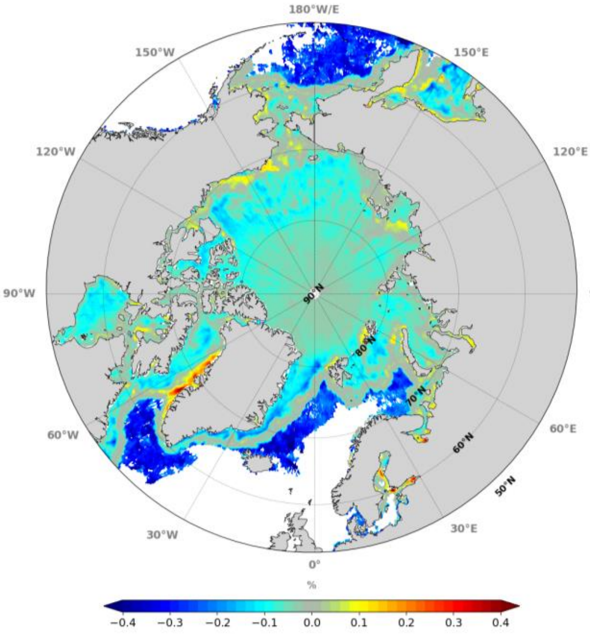


Figure 31. Time evolution of proportion correct total (PCT) scores (in %) using the GLO12v4 best analysis during the 2017-2021 period. The sea ice concentration product is computed with AMSR2 satellite data.

In the Arctic, the time series of the PCT between 2017 and 2021 (Figure 31) exhibits a good agreement between GLO12v4 and observations with very high values close to one during the winter season and also very good value (0.88%) during the melting season in boreal summer. The main differences are in the marginal zones where the model may have a time lag with the observations (too much melting or too much ice formation) in the Bering Sea and in the Eastern part of the Artic where the ice tends to be thinner than in the pack.



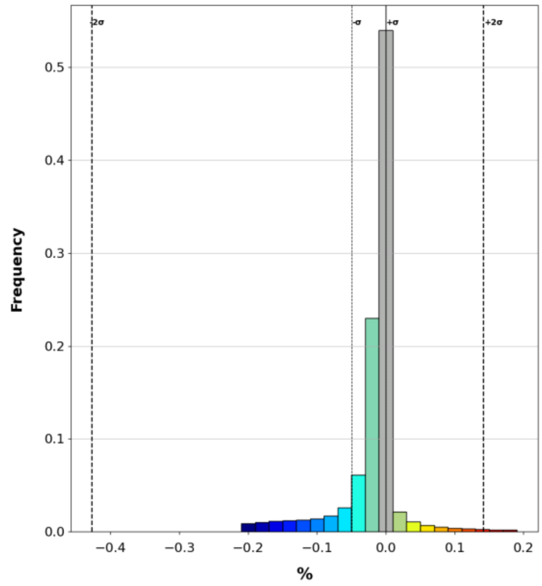


Figure 32. CLASS4 sea ice concentration comparisons for the year 2019. On the left, mean differences in 2°x2° bins (GLO12v4 model minus AMSR2 observations). On the right, pdf of the differences on the whole domain.

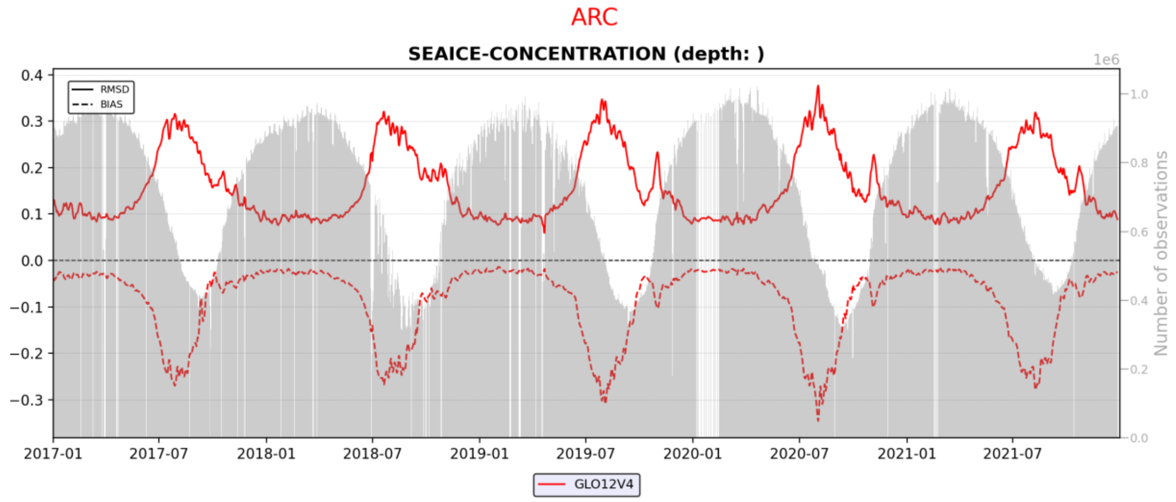


Figure 33. Timeseries of mean bias and RMSD using the GLO12v4 best analysis during the 2017-2021 period. The sea ice concentration model is compared with AMSR2 satellite data.

IV.5.1.2 <u>Sea ice thickness</u>

Comparisons with sea ice thickness measurement using satellite altimetry (Cryostat 2) in March 2019 are shown in Figure 34. The regional distribution of thin and thick ice is well captured by GLO12.

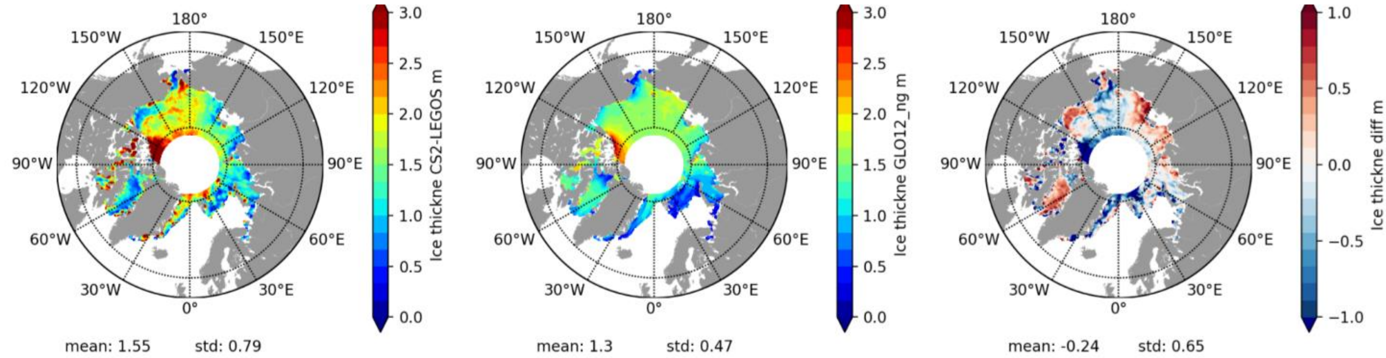


Figure 34. Monthly average for the month of March 2019 of sea ice thickness. Observations (on the left) come from the Cryosat2 Ctoh/Legos (Centre de Topographie des Océans et de l'Hydrosphère) product. GLO12v4 system is in the middle, and the difference between the GLO12v4 system and observations is on the right.

IV.5.1.3 <u>Sea ice volume</u>

Comparisons with sea ice volume measurement using satellite altimetry from altimetry (Cryostat 2) in March 2019 are shown in Figure 35. The spatial features are very similar to those of Figure 34, showing an overall good agreement at large scale between GLO12v4 and these independent observations for this period of sea ice maximum in the Arctic.

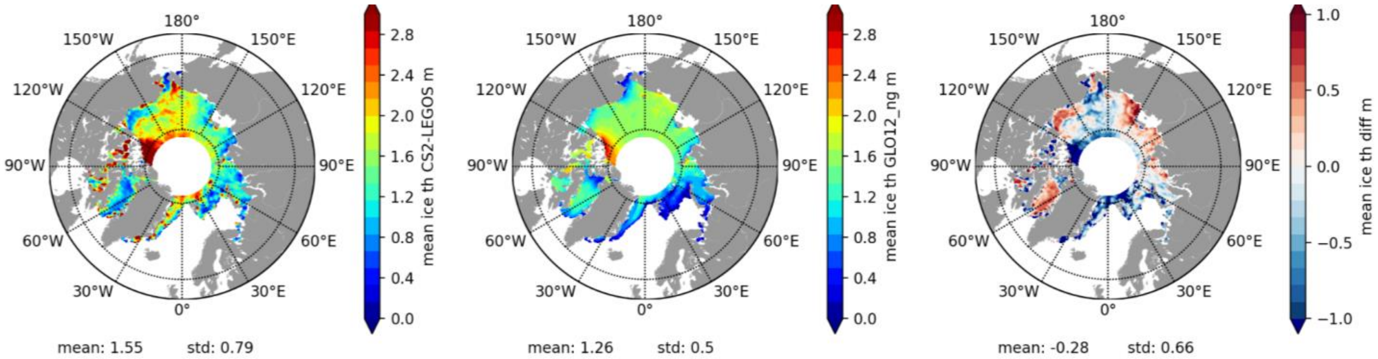


Figure 35. Monthly average for the month of March 2019 of sea ice volume. Observations (on the left) come from the Cryosat2 Ctoh/Legos (Centre de Topographie des Océans et de l'Hydrosphère) product. GLO12v4 system is in the middle, and the difference between the GLO12v4 system and observations is on the right.

IV.5.1.4 Sea ice snow volume

According to comparisons with sea ice snow volume measurement using satellite altimetry from altimetry (Cryostat 2) in March 2019 in Figure 36, the snow volume is overestimated by GLO12v4.

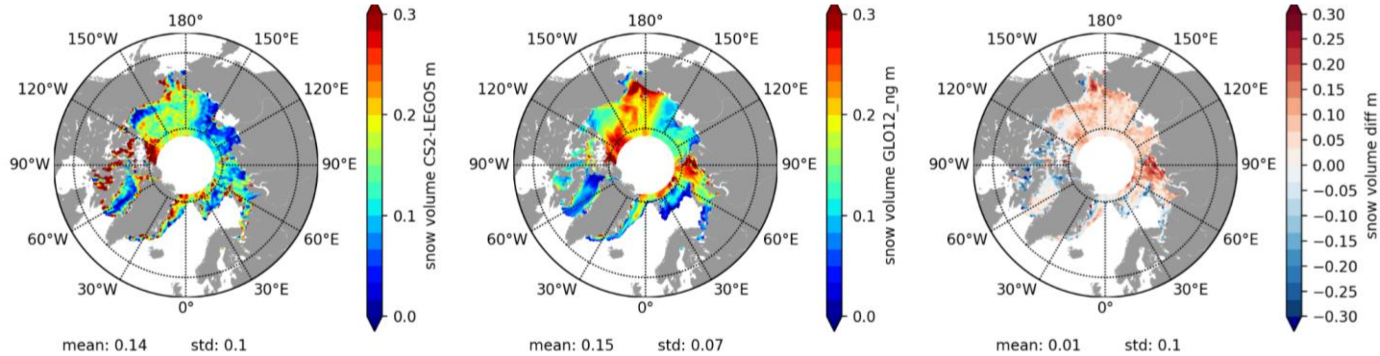


Figure 36. Monthly average for the month of March 2019 of sea ice snow volume. Observations (on the left) come from the Cryosat2 Ctoh/Legos (Centre de Topographie des Océans et de l'Hydrosphère) product. GLO12v4 system is in the middle and the difference between the GLO12v4 system and observations is on the right.

IV.5.1.5 Sea ice Temperature

On Figure 37, comparisons using sea ice temperature from Copernicus marine Service L4 product 011_008 tend to show that GLO12v4 strongly overestimates the sea ice temperature all year round (under investigation).

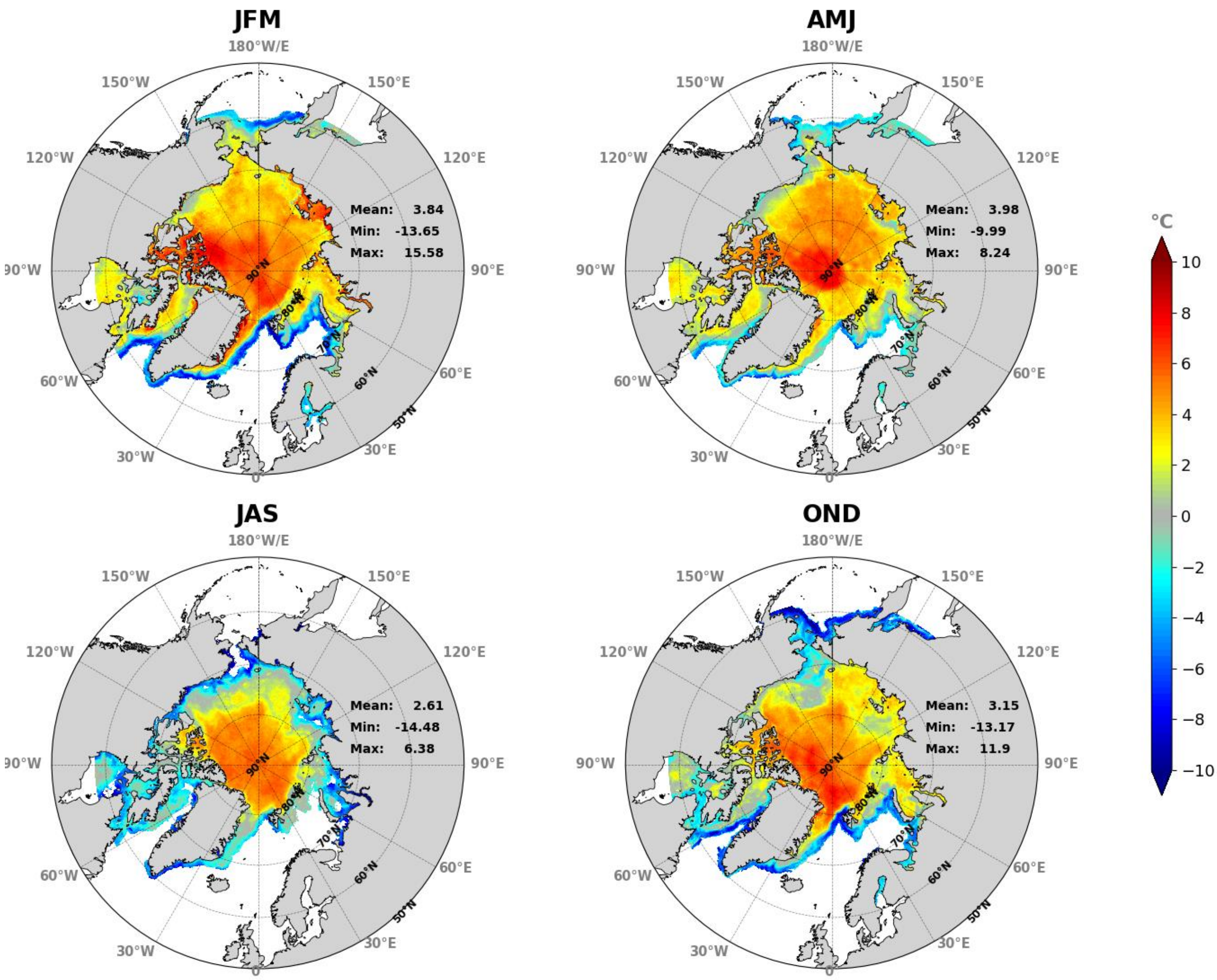


Figure 37. Seasonal mean difference between the GLO12v4 system and the observations (surface sea ice temperature from Danish Meteorological Institute) in the Arctic region for the year 2019.

IV.5.2 Validation in the Antarctic region

In the Antarctic, the time series of the PCT between 2017 and 2021 exhibits a good agreement between GLO12v4 and observations (Figure 38 and Figure 39). PCT values are close to one during the winter season and around 0.8 during the melting season in boreal summer. The main differences are in the marginal zones where the model may have a time lag with the observations (too much melting or too much ice formation) in the Baffin Bay, Weddell Sea and Ross Sea.

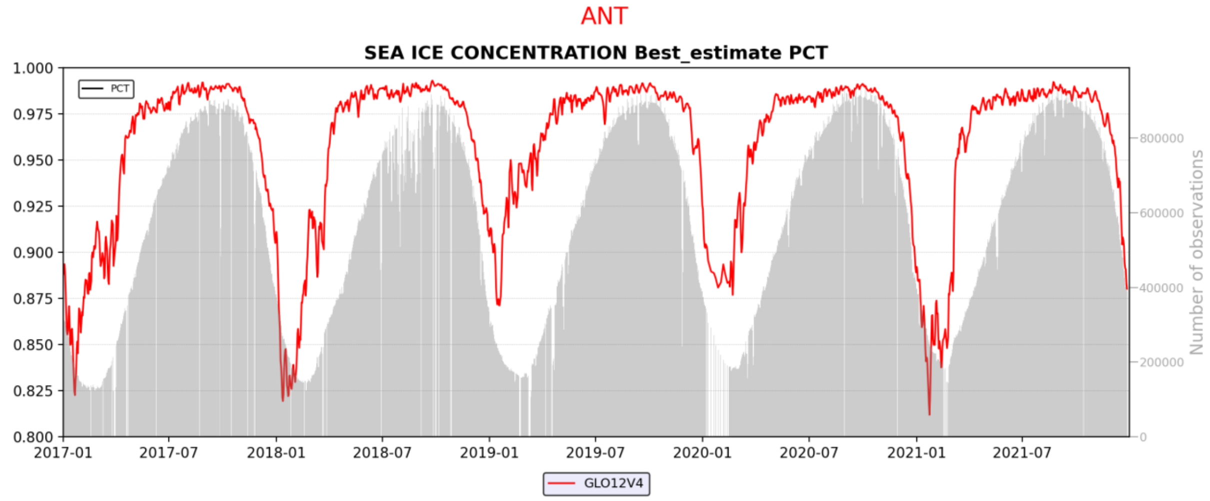


Figure 38. Time series of PCT scores using the GLO12v4 best analysis during the 2017-2021 period. The sea ice concentration product is computed with AMSR2 satellite data.

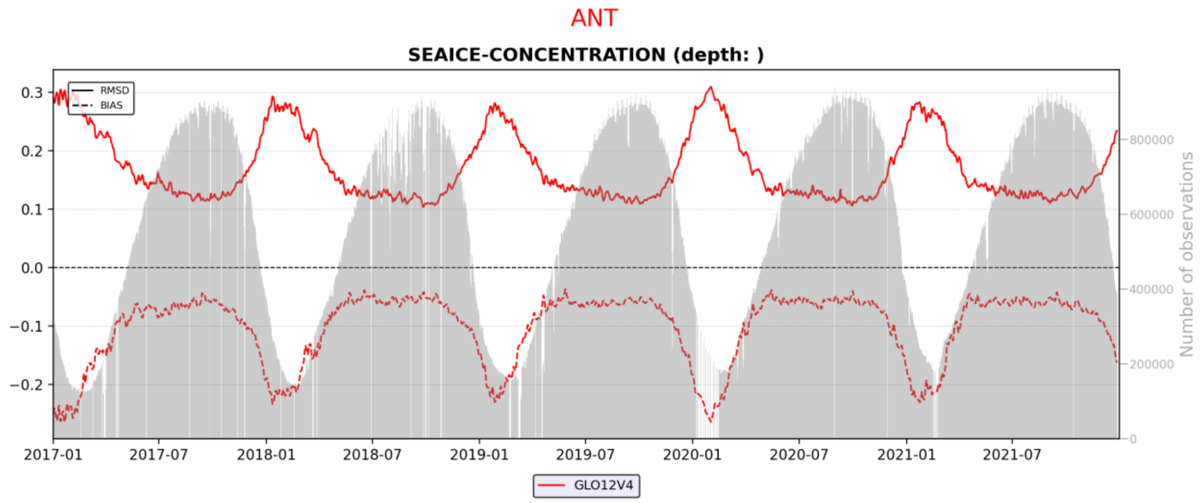


Figure 39. Time evolution of RMSD (top) and Mean difference using the GLO12v4 best analysis during the 2017-2021 period. The sea ice concentration product is computed with AMSR2 satellite data.

IV.6 Surface currents

IV.6.1 Large scale, Eulerian validation

Near surface currents (at 0.5 m depth) can be compared with velocities deduced from drifters without drogue (Copernicus Marice Service product) and ARGO floats (YoMaHa) surface trajectories (see Lellouche et al., 2013). We use these Eulerian comparisons to give an estimate of the surface current errors. A windage correction that minimizes the residual has been applied (maximum 5 % is imposed).

Figure 40a shows a scatterplot between the observed vs modelled zonal velocity component. The color indicates the density of measurement points. The general stretching of the cloud of points follows the 45° diagonal in a satisfactory way. Although the correlation is higher than 60 %, the regression slope is only about 0.5, i.e., the model underestimates zonal currents with respect to observations. Velocities lower than 20 cm/s have a strong weight in this 0.5 regression slope, as shown on Figure 40b where the grey envelope represents the uncertainty in the estimated slope. The slope is better (approximatively 70 %) for positive u velocities of about 50 cm/s. This gives an insight that the intensity of western boundary currents intensity is better reproduced. Note that the uncertainty about the slope grows with the current intensity, reflecting uncertainty linked with instability/turbulence regimes. The fact that low speeds are more underestimated than high speeds is particularly marked for the meridional velocity component v.

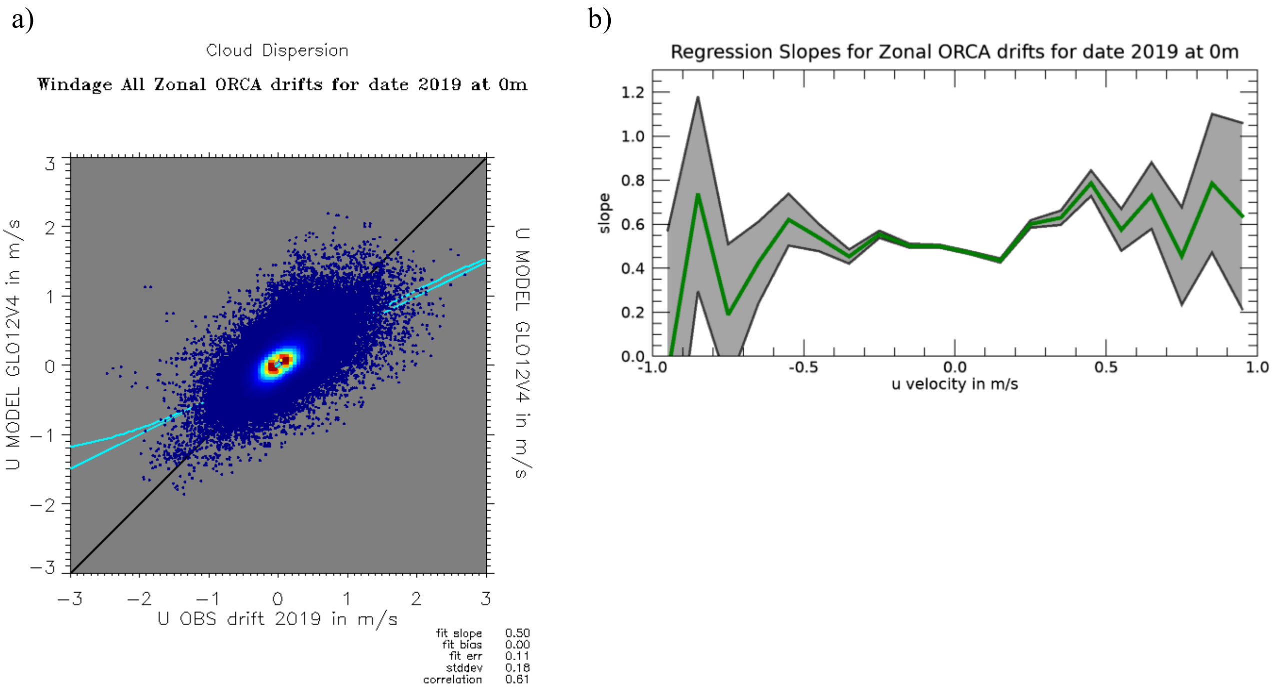


Figure 40. Scatter diagrams for GLO12v4 velocities near the surface with respect to drifters' velocities and ARGO floats velocities, for the calibration year 2019.

Figure 41 shows the differences in angle between the model and the observations, for all speeds (small and large). The median difference is 3°, so the model is very slightly to the right of the observations. The angle bias mainly concerns the southern hemisphere. This concerns mainly the small speeds, the speeds above 30cm/s being less affected.

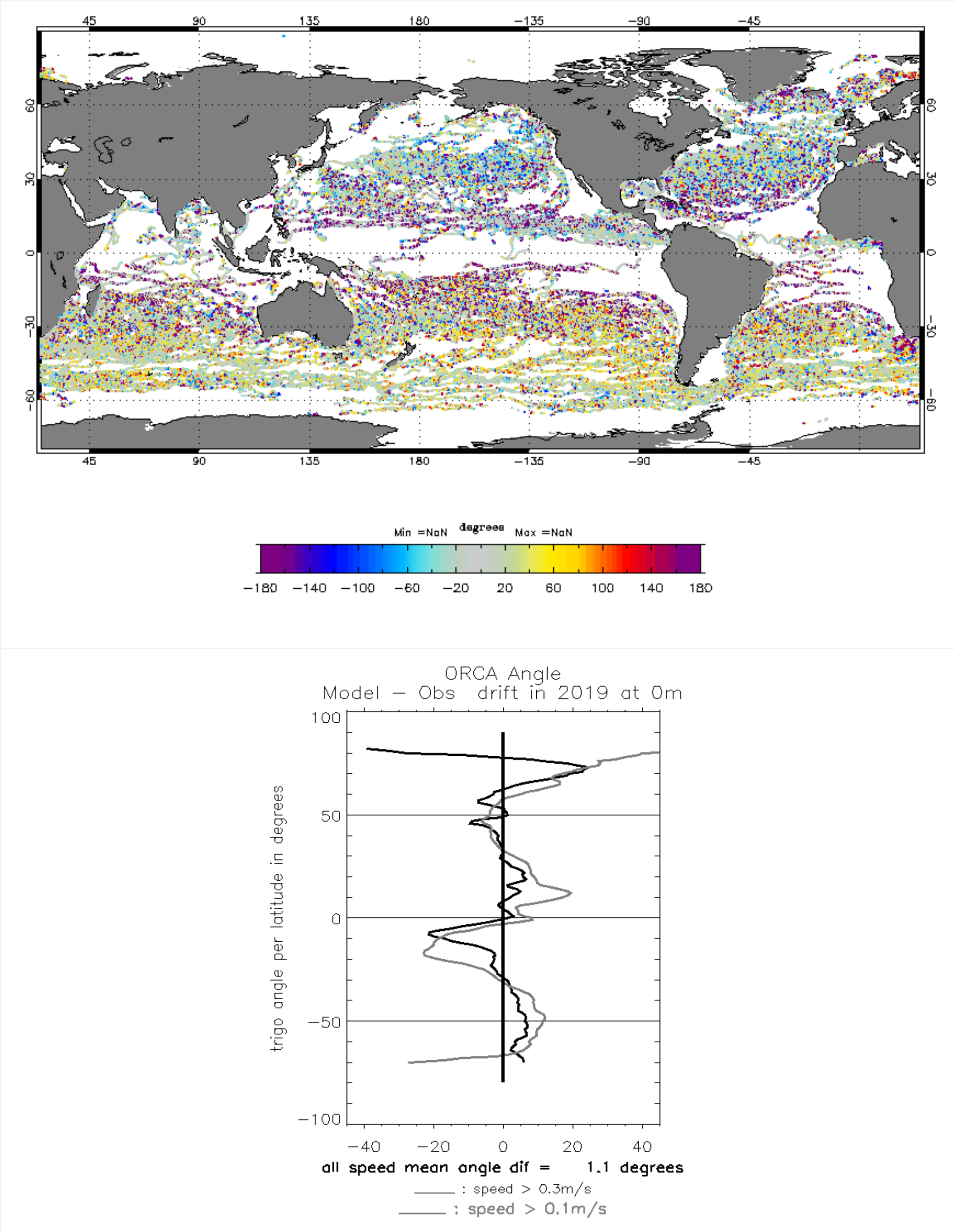


Figure 41. Zonal average trigonometric angle difference (°) between the GLO12v4 velocity vectors and surface drifter and ARGO float velocities, over the calibration year 2019 (top panel). Mean angle difference is on the bottom panel.

Figure 42 shows zonal and meridional currents for the observations and the model. While all structures are well represented, the currents would generally be underestimated. Misfits for u and v are also shown. The misfit in u clearly bears the signature of the wind, and this is also true in v in the eastern Pacific. The windage correction may be underestimated for light winds.

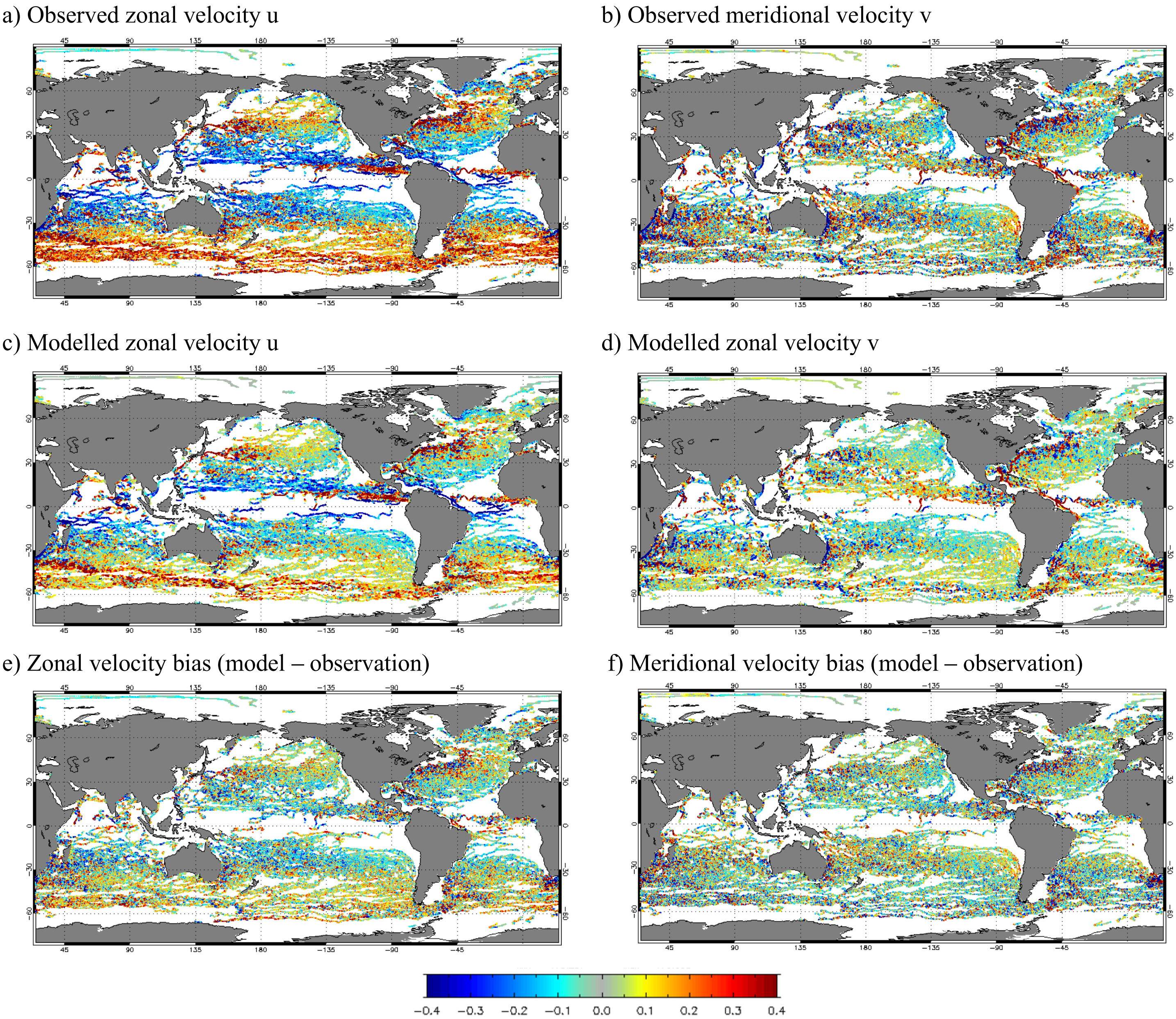


Figure 42. Maps of all observations gathered in 2019 for zonal u (a) and meridional v (b) Velocities from surface drifters and ARGO floats. c) and d) GLO12v4 analysis counterparts for a) and b), respectively. e) Zonal surface velocity bias and f) Meridional surface velocity bias deduced from the differences between GLO12v4 and the observations. Units are m/s.

IV.6.2 High frequency, nearshore validation

Figure 43 shows the location of the current meters used for the high frequency coastal validation. These current meters are located for the major part along the Spanish coasts and are distributed by the Copernicus Marine Service in the INSITU_GLO_NRT_OBSERVATIONS_013_033 product. The buoy network is operated by various met-ocean actors such as Météo-France, Puertos del Estado, etc. Whenever possible, the validation has been performed at hourly intervals and we tried to validate as much as we could over the 2019-2020 period for both GLO12v4 surface current and the utotal currents from cmems_mod_glo_phy_anfc_merged-uv_PT1H-I dataset (SMOC).

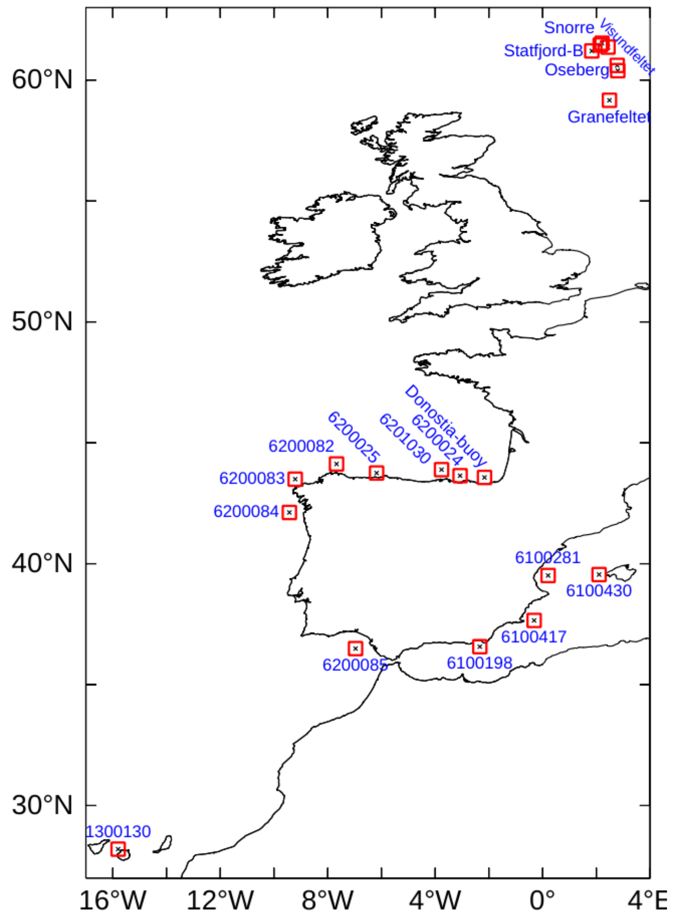


Figure 43. Location of the in situ buoys used for the coastal validation.

Comparing the time series of modelled vs measured currents at each mooring highlights how difficult it is to reproduce the complexity of a coastal circulation for a global-scale ocean model (not shown), notably in terms of variability and intensity. The hourly-recorded velocities sometimes exhibit peak values almost three times more energetic than their model equivalent, especially for the buoys situated along the north coast of Spain.

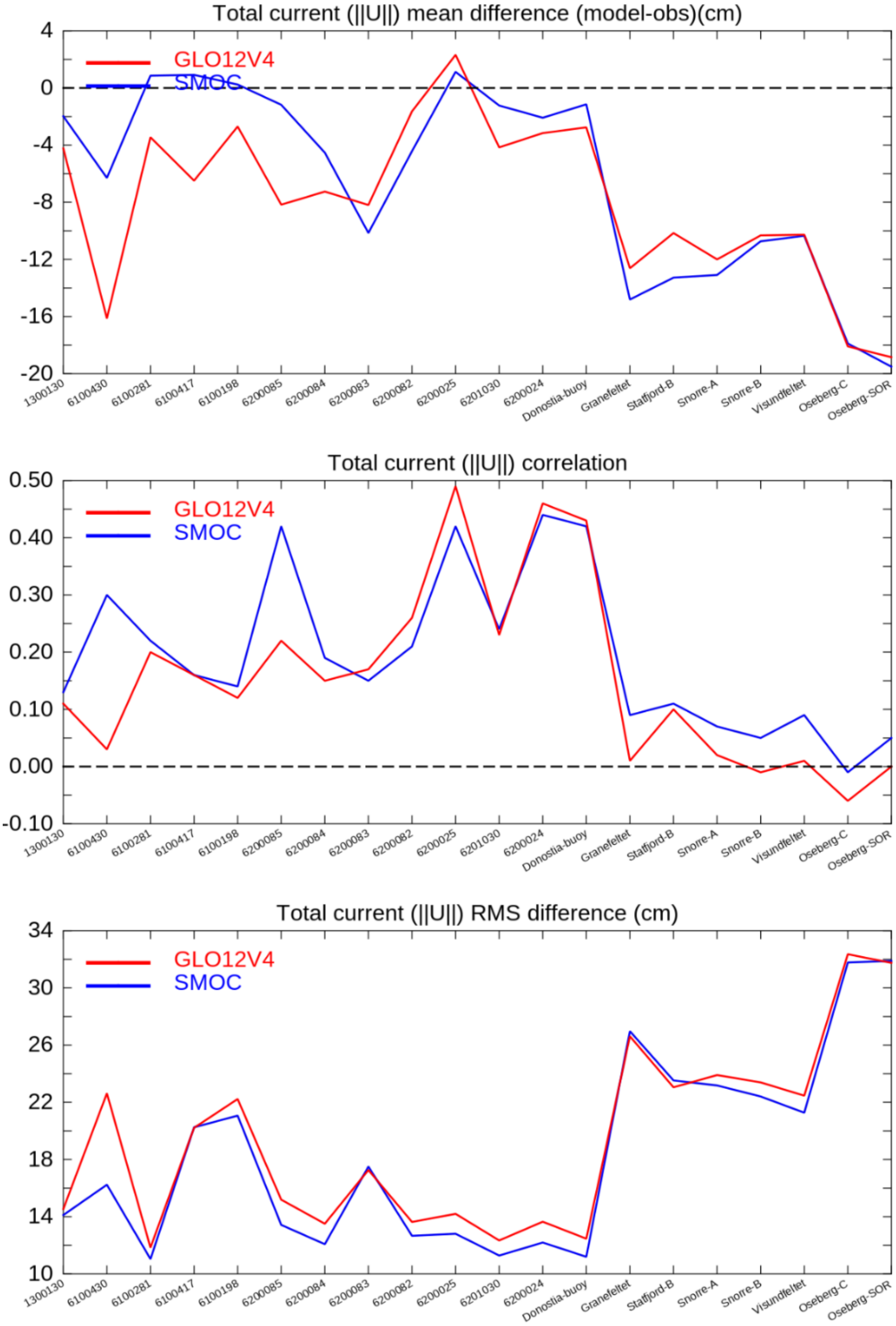


Figure 44. Statistical parameters for each mooring (mean difference, correlation and mean RMSD) derived from the comparison between observed and modelled time series of current velocity for both uo and utotal.

Figure 44 Figure 44summarizes the statistical results obtained for the **utotal variable** from cmems_mod_glo_phy_anfc_merged-uv_PT1H-i dataset (SMOC: surface current + waves + tides) versus the **uo variable** of the same dataset (GLO12v4: surface current only) at each current meters.
The statistical quantities retained are the correlation, the mean RMS and the mean errors, all computed over the entire validation period for each buoy.
In general terms, the buoys on the Catalan coast ([61289-61198]) are not visibly affected by the introduction of tidal and Stokes currents, while more differences are observed for the rest of the sites. Considering tide and waves always improves the low-biased current peaks in magnitude, which is not systematically the case when studying the (u, v) velocity components separately (not shown). Improving the current intensity but not its vectorial components suggest that introduced tidal oscillations may not be well phased with observation. However, the improvement in velocity mean difference is clear from 5 to 10 cm/s for most of the Atlantic sites. RMS mean differences and correlations are also clearly improved with Stokes and tide currents.
We have also performed some validation work using a collection of coastal radars including: the US high frequency radars from the Scripps Institution of Oceanography database (http://hfrnet-tds.ucsd.edu/thredds/catalog.html), the European HF radars from Puertos del Estado (http://opendap.puertos.es/thredds/catalog.html), and the BISC HF radar of the Basque government (http://www.emodnet-physics.eu). We have chosen to conserve hourly radar signals for this comparison, and we didn't filter out the tide signal.
Figure 45 Figure 45**Error! Reference source not found.**gives an example of a 2D correlation map obtained from comparing the USEG1 radar (located off the coasts of New Jersey, US) and the modeled currents. The addition of the tide and Stokes drift in utotal (labeled "SMOC_GLO12V4"**Error! Reference source not found.** on Figure 45) gives more correlated structures than was already obtained with uo (labeled "GLO12V4_RUNoper30" on Figure 45**Error! Reference source not found.**). These high correlation structures can be identified with high-frequency coastal processes that are related to waves or tide. The improvement can be quite strong, such as from 0.3 to 0.6 in correlation for the structure seen at (78°W, 33°N). New high correlation structures appear at 68°W, 42°N (e.g., the two appearing at the north-east side of the domain), proving that SMOC brings coastal variability not perceived by GLO12v4 alone.
Figure 46 Figure 46shows the complex correlation between observed surface currents and the total current of SMOC (utotal) at the several locations of HF radars.

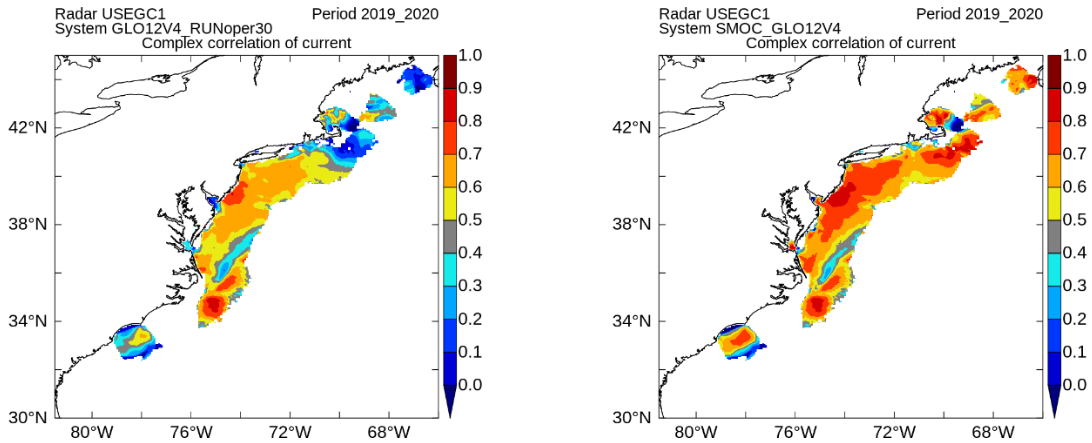


Figure 45. Complex correlation of surface currents between the general circulation current (GLO12v4 or uo), on the left, and the total current of SMOC (utotal), on the right.

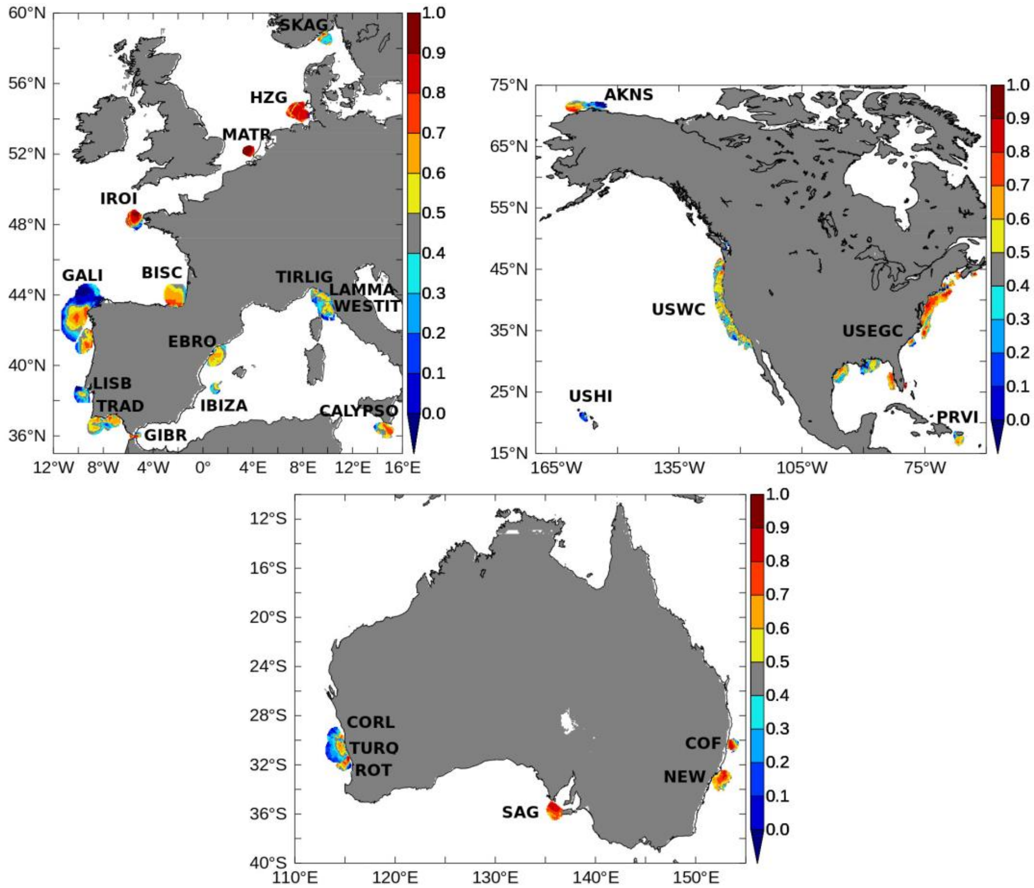


Figure 46. Complex correlation between observed surface currents and the total current of SMOC (utotal) at several locations of HF radars around Europe, United States and Australia..

Figure 47 sums up the results obtained for all the HF radar sites (see Figure 46 for the locations) through a Taylor diagram (graphical combination of the RMS error, standard deviation, and correlation). In general, the correlation for SMOC utotal is slightly better than for GLO12v4 uo (for the spatially averaged observed signals, as discussed above). Figure 47. shows especially that SMOC utotal displays a higher variability (ie normalized standard deviation) than GLO12v4 uo. To know if this an improvement or not, the symbol that refers to a buoy must get closer to the unity arc of the normalized standard deviation. This affords a clear benefit for some specific sites (FINN, GALI3, GIBR2, HZGZ, etc.) where the variability was underestimated with GLO12v4 uo. On the other hand, for some other locations like the AKNS radar and some USECG sites, the increased variability of SMOC utotal exceeds the observations' one, and therefore performs worse than GLO12v4 uo. The RMS error is also given by the red dashed arcs centered on "the observation point", i.e., the point where (correlation, normalized standard deviation) =(1,1). Considering the graphical criterion of the Taylor diagram where the distance between the symbols and the observation point should stay as low as possible, it is difficult to say that one model is better than the other in this coastal frame.

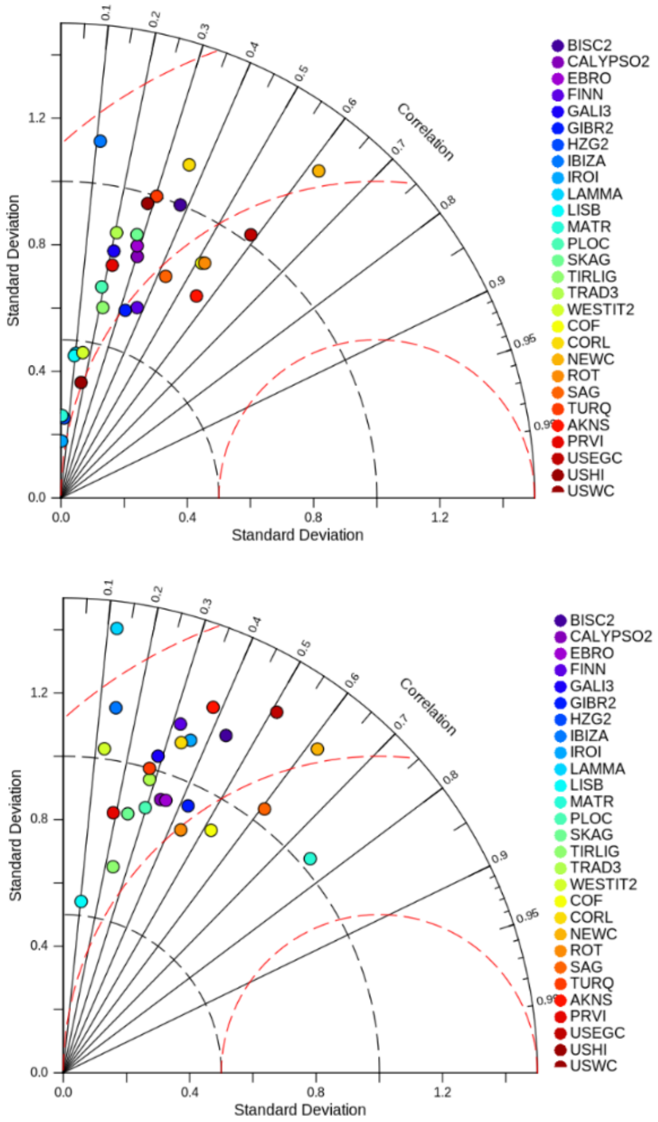


Figure 47. Taylor diagram summarizing the comparison of uo (top) and utotal (bottom) to HF radars. Locations of the different radars are displayed in Figure 46.

IV.6.3 Lagrangian validation

Our Lagrangian validation of (sub)surface currents consists in releasing particles within the ocean circulation model at location and time of real drifter observations in the ocean and compute their pathways using a particle advection scheme for a given drifting duration. The generated particle trajectories can then be compared to the ones of drifters in the real ocean, and through several metrics it enables to examine the model skill in reproducing the Lagrangian transport.

Here we use the particle tracking software Parcels (Lange and Sebille, 2017) which applies a Runge-Kutta 4 interpolation scheme to advect particles. The drifter trajectories observations come from the Global Drifter Program or GDP dataset at 6-h and 1-h resolution (Elipot et al., 2016). Drifters either have their drogue still attached meaning their pathways is representative of ocean circulation at 15 m depth or have lost it and their trajectories are therefore the result of surface processes.

The Lagrangian validation of GLOBAL_ANALYSISFORECAST_PHY_001_024 is carried out for both the surface and 15m depth covering the 2019 period. In a first instance at 15 m depth using the GLOBAL_ANALYSISFORECAST_PHY_001_024 daily outputs and then for the 0 m depth using the hourly output of cmems_mod_glo_phy_anfc_merged-uv_PT1H-I dataset (SMOC) – wherein uo is the component of GLOBAL_ANALYSISFORECAST_PHY_001_024.

IV.6.3.1 Validation of currents at 15 m depth

Lagrangian particle tracking experiments are carried out for the entire year of 2019. Every 5 days, virtual particles are deployed at locations of drogued SVP drifters and at depth of 15 m and then tracked for up to 30 days. With approximately 550 drogued SVP buoys drifting across the global ocean, a total of 28,708 particle trajectories are evaluated against their SVP equivalent across the year.

IV.6.3.1.1 General characteristics

The general Lagrangian characteristics are evaluated using the 30-day trajectory, a timescale chosen to span beyond the mesoscale –within a month, a sea water particle travels up to a few hundreds of kms a distance larger than the size of the dominant eddy locally. Particle displacement and direction (i.e straight distance and angle between seeding location and position after 30 days) are evaluated and confronted to results for SVP drifters (Figure 48). Values for displacement (in km) and direction (in degrees east) are given at the release location, and maps are generated by averaging results in 2x2 degree bins. Note that the number of observations vary greatly globally with highest number of observations in the subtropical gyres.

The large-scale characteristics of the Lagrangian transport are to first order well represented in the model. Largest displacements (larger than 600 km) are observed in regions of strongest basin scale currents such as the Gulf Stream, equatorial currents in all 3 basins, the Kuroshio, Agulhas current, and the Antarctic Circumpolar Current (ACC). Smallest displacements (approximately dozen of km) are observed on the eastern side of basin and within the subtropical gyres. Mean particle direction reproduces well the direction of the large-scale flow with the double gyre circulation in the north Pacific and Atlantic and Indian Ocean, and the eastward circumpolar movement of the ACC.

On the other hand, marked difference relative to displacement derived from the SVP dataset are present at a local/regional scale. Some differences can be noted in the vicinity of western boundary currents: larger displacement for the model is observed in the Kuroshio extension, Agulhas current and retroflexions, and the Atlantic East equatorial region.

Some patches of higher displacement are also present on the edges of the subtropical gyres in the North Atlantic (25°W-30°N) and the transition zone off the western North African and European coast (15°W, 35°N) and in the South Atlantic subtropical region (12°W-12°S).

Strong directional differences (larger than 90 degrees relative to the mean SVP direction) are sparse and localised. The most notable patches are in the north part of the North Pacific west of the Gulf of Alaska, where the trajectories move.

In the western part of the North Atlantic, to the west of the British Isles, pathways tend to be to the west instead of the northwest. West of the Iberian peninsula, trajectories point southward instead of south-westward.

In the southern hemisphere, directional errors on the northern boundary of the ACC suggest potential error in the position of the front. To the west of South Africa in the South Atlantic, pathways direct north westward instead of westward.

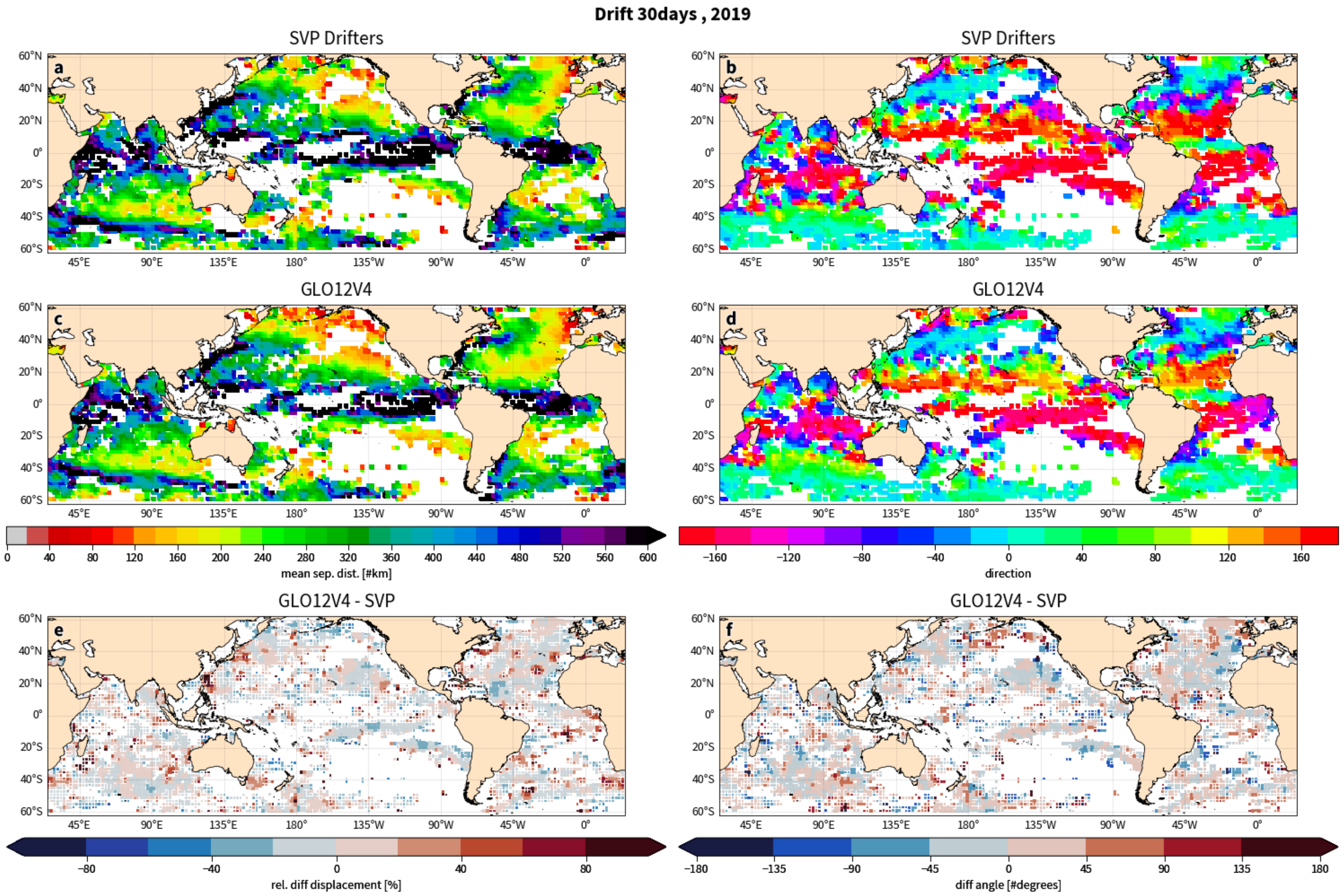


Figure 48. General Lagrangian characteristics for 30-day drift of the ocean dynamics at 15 m depth: drifter displacement and direction for SVP drifters (a,b) and GLO12v4 (c,d). Displacement corresponds to average straight-line distance travelled by all drifters departing from a given cell, and direction is the average direction between departure point and position after 30 days. Bottom panels show average relative difference in displacement between model and observations (e) and average difference in direction (f). The size of grid cells in the bottom panels is function of the number of observations in each 2°x2° bins. Largest are for bins with approximatively 50 observations and lowest for bins with less than 5 observations. Results were smoothed with a Gaussian window of size 3.

IV.6.3.1.2 Quantitative metrics: separation distance and Liu Index

The skill of the model is examined by comparing modelled and observed trajectories using the separation distance (*SD*) and the Liu index (*S*) (Liu and Weisberg, 2011). The former tells us how far apart the positions of the modelled particle and real drifter are after a given drifting time (here chosen to be 1,3 and 5 days), and the latter is a normalised cumulative separation distance.

The two metrics are defined as:

$$SD = d_N \tag{7}$$

$$\text{and } \; S = \frac{\sum_{i=1}^{N} d_i}{\sum_{i=1}^{N} l_{oi}} \tag{8}$$

where N is the drifting time (in days), d_i is the separation distance between the modelled and drifter positions after drifting time i (in days) after release, l_{oi} is length of the SVP trajectory.

a)

b)

c)

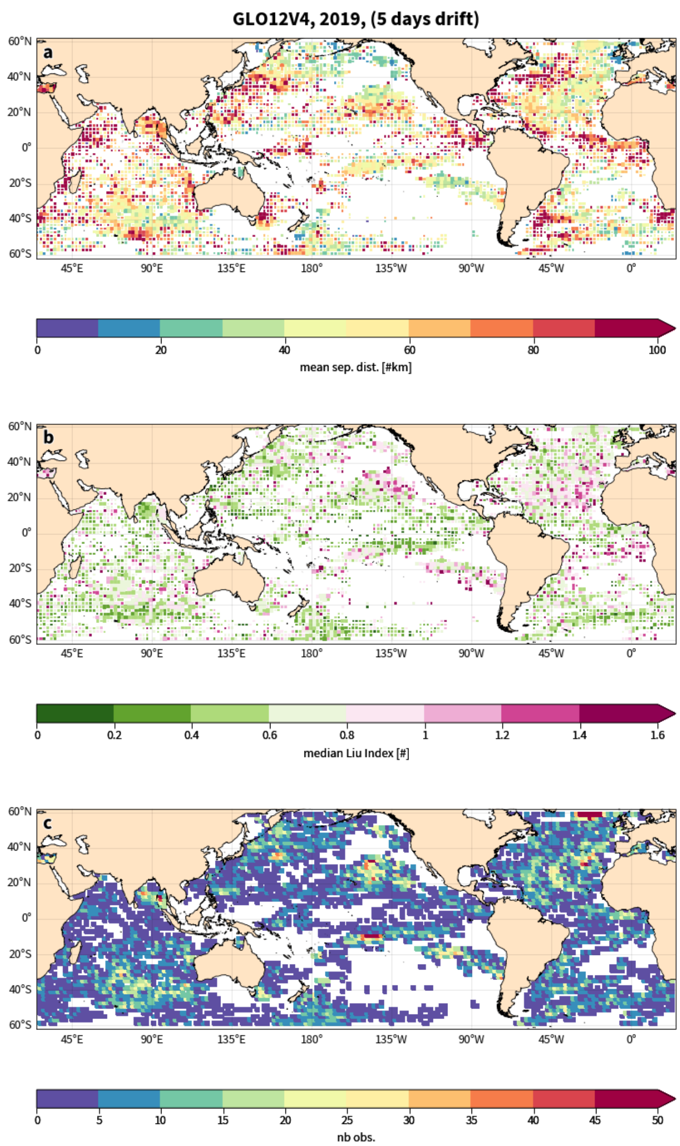


Figure 49. Mean separation distance (a) Liu Index (b) and number of obs. (c) after 5-day drift between SVP drogued drifters and their model equivalent (virtual particles seeded at same positions and time as SVP drifters) obtained with model velocities at 15 m, for the year 2019. Results are presented in 2x2 degree bins colored squares. Their size in panel a) and b) are relative to the number of observations (panel c) in each bin, with large square for large number of observations. Results were smoothed with a Gaussian filter.

The separation distance between modelled particle and SVP drifter positions are evaluated for a drifting time of 5 days (Figure 49). Longer distance consists in larger error in the representation of the currents by the model. This error is not geographically uniform and is larger for regions of strong and turbulent dynamics such as the west boundary currents, with errors of approximately 100 km up to 200 km found in these regions after 5 days. Quieter regions such as the subtropical gyres have lower errors (approximately 30km/5d). It is important to note that within these estimates a stochastics component exists; meaning that a separation distance of zero would never be reached due to the chaoticity of the currents which has as effect to disperse initially close particles according to an exponential or a power law (LaCasce, 2008).

An alternative and complementary approach to evaluate drifting skill is to evaluate not only the absolute distance between estimate and observation, but also a score that relates error as a function of distance travelled by the drifter, as for instance the Liu index (Liu and Weisberg, 2011). Indeed, in regions of slow dynamics, a modelled particle going in opposite direction to the true solution will present a separation distance that remains lower than errors observed in more turbulent regions; but a high Liu index as the cumulated separation distance across the 5 days will be larger than the distance travelled by the particle.

Values below 0.5 correspond to regions where the error across the 5 days relative to the distance travelled by the drifter is considered low; in contrast, values above 1 correspond to regions with large errors. Patterns of errors for the Liu index contrast with the separation distance (Figure 49) with large errors (values > 1) are not concentrated in western boundary currents but on the outskirts and the western side of the subtropical gyres such as in the North Pacific (30N, 135W), south Pacific (90W), and North Atlantic (22N, 25W). Focus on the equatorial Atlantic suggests that the model Lagrangian transport is better in the northern part than in the southern one, where high Liu index values are observed. In particular, poor errors in both metrics are present within the Gulf of Guinea.

Figure 50 shows the integrated distribution for each metric and their evolution over drifting time from 1 to 7 days. For the separation distance histogram, the values become more and more spread out as the forecast time scale increases (the distribution "flattens" as a function of the time), which illustrates the non-linear and chaotic aspect of oceanic processes in the framework of a Lagrangian integration.

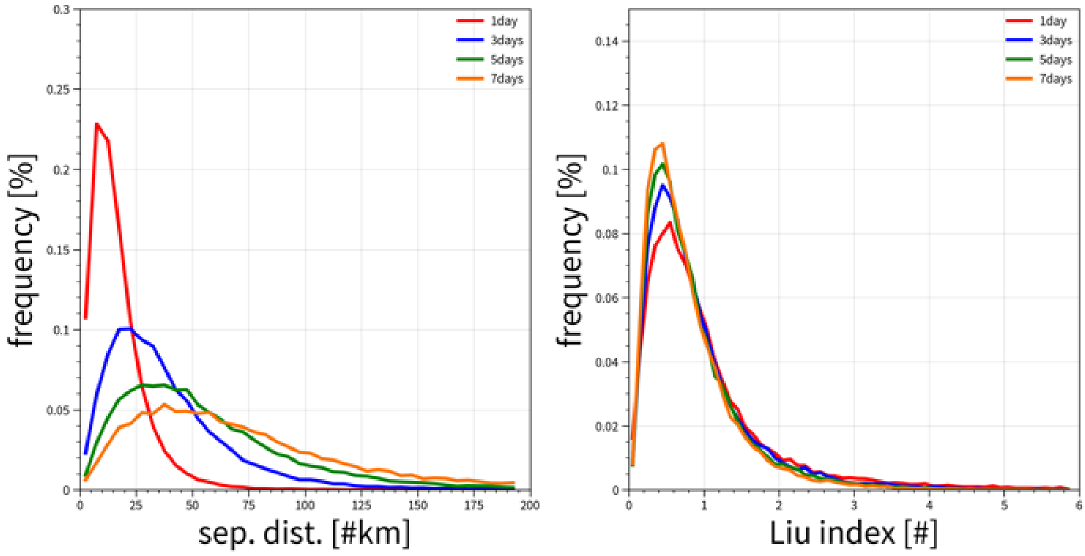


Figure 50. Separation distance and Liu index for 1-, 3-, 5- and 7-day drift for the global ocean and for the year 2019.

IV.6.3.2 Validation of surface currents

To validate the **cmems_mod_glo_phy_anfc_merged-uv_PT1H-i** dataset which comprises hourly surface currents from GLO12v4 and the SMOC currents (GLO12v4, Stokes component, and tidal component) over the 2019-2020 period, 171,484 Lagrangian simulations per velocity fields were launched with the PARCELS particle tracking software (Lange and Sebille, 2017).

These Lagrangian simulations aim to reproduce the trajectories from the PhoD hourly drifter database (Elipot et al., 2016). Only the PhoD drifters that have lost their drogue were used as we focus on surface displacement (possible windage on emerged parts are neglected). No filtering or special treatment were applied on the buoy data which were used as such. The period of validation (2019-2020) represents the use of 2683 buoys from the database.

Each buoy has been separated into 3-day trajectory pieces that are treated independently for each Lagrangian simulation. A Lagrangian simulation consists of seeding a virtual drifter at the beginning of a 3-day observed trajectory and advecting it for 3 days with a particular current field that is assessed.

Current field data or models that are evaluated in this document will bear the following names on the figures:

- GLO12V3_SMOC_uo: previous system version for uo (surface current alone)
- GLO12V3_SMOC_utotal: previous system version for utotal (surface current + wave + tides)
- GLO12V4_SMOC_uo: current system version for uo
- GLO12V4_SMOC_utotal: current system version for utotal

The validation involves comparing statistically the virtual trajectories to the observed one under various Lagrangian criterions such as separation distance, velocity, and direction.

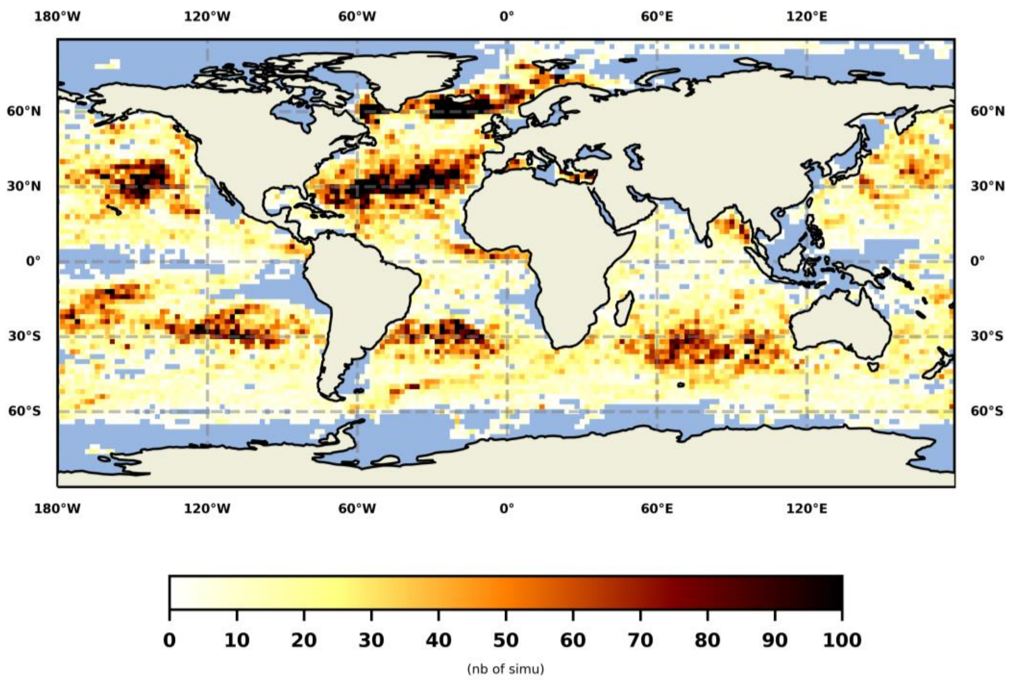


Figure 51. Number of Lagrangian simulations launched by 2°x2° bins for 2019-2020 per dataset.

Figure 51 shows the geographical distribution of the number of observed positions retained as seeding points for 2019-2020. It must be understood that most observed areas have more statistical weight in the global averages, such as the North Atlantic, the convergence zone of the subtropical gyres, where the number of simulations is greater than 100 per 2°x2° block. It can be noted that the equatorial divergence zone and the latitudes above 60° are practically not observed. **Error! Reference source not found.** gives more precision about the number of Lagrangian simulations used in some regions of interest that will be focused on later. For instance, the inshore regions (bathymetry < 200 m) only have 4000 simulations in total, which can be seen unsignificant compared to other basin-scale regions. In the same way, Antarctic, Arctic and Guinea Gulf region results shouldn't be given too much credit. It should also be mentioned that the number of simulations retained for each model very slightly differs because some beached virtual drifters may be discarded from the validation (results of Stokes drift directed towards the coasts).

Table 10. Number of Lagrangian simulations used for 2019-2020 by regions of interest.

Zone	Global	Inshore	Mid latitude	Pac. Tropics	Atlantic Tropic	Equator	India	Atl North	Atl South
Nb of simulations	1.5E5	4.0E3	1.3E5	4.E5	1.5E5	2.5E3	2.E4	3.E4	1.5E4
Zone	Pac South	Pac North	Antarctic	Arctic	Gulf Stream	Kuroshio	Guinea Gulf		
Nb of simulations	3.5E4	3.5E4	1.5E3	1.8E3	7.E3	5.E3	0.3E3		

Figure 52 shows the Lagrangian forecasts made for buoy n° 653 514 620 which travels in the subtropical Indian Ocean. The buoy's trajectory reflects interesting processes, such as eddies and direction changes, both of various scales. A quality criterion of the forecasts is that the predicted trajectories remain geographically close to the observed trajectory; which is most of the time the case for all models. The reproduction of small-scale motions (probably inertial type oscillations) at the beginning of the trajectory is on the other hand difficult; oscillations are reproduced, but the mean motion goes in the wrong direction most of the time. This is especially the case for the two meso-scale loops in the very beginning, showing the difficulty to phase meso-scale eddies for drift applications. Nevertheless, we can clearly see the positive contribution of using utotal compared to uo chronologically at 73°E, 76°E and 66°E, when the buoy changes smoothly its direction with loops of about 100 km. These movements would be related to a change in the wind regime and associated waves. It is difficult to qualify the contribution of the new system (GLO12v4) compared to the old one (GLO12v3) on these visual criteria for this buoy.
The buoy n° 66 312 850 is released along the Florida coast and is rapidly captured by the Gulf Stream flow. The current GLO12v4 system improvement over the previous GLO12v3 version is clearer for this example. Forecasted undulations of trajectories are more in agreement with those observed; thanks to a better current dynamics and more realistic thermal fronts, as seen at 72°W, 65°W and 68°W. The part of the drift east of 55°W is subject to the turbulent activity of the Gulf Stream and it is difficult for all the models to produce realism in these conditions. GLO12v4 does nevertheless a better job for the meanders between 50 and 45°W. The contribution of the Stokes velocities present in utotal is negligible compared to uo for this western boundary current case.

The separation distance or Lagrangian distance error is a classical metric that consists in scoring the evolution of the distance between the forecast and the observation. It gives a practical measure of the error and distance to be covered in the event of a search and rescue operation at sea if a deterministic forecast is used. The smaller it is, the better is the forecast. Typically, the average separation distance monotonically increases with time when averaged over enough members.

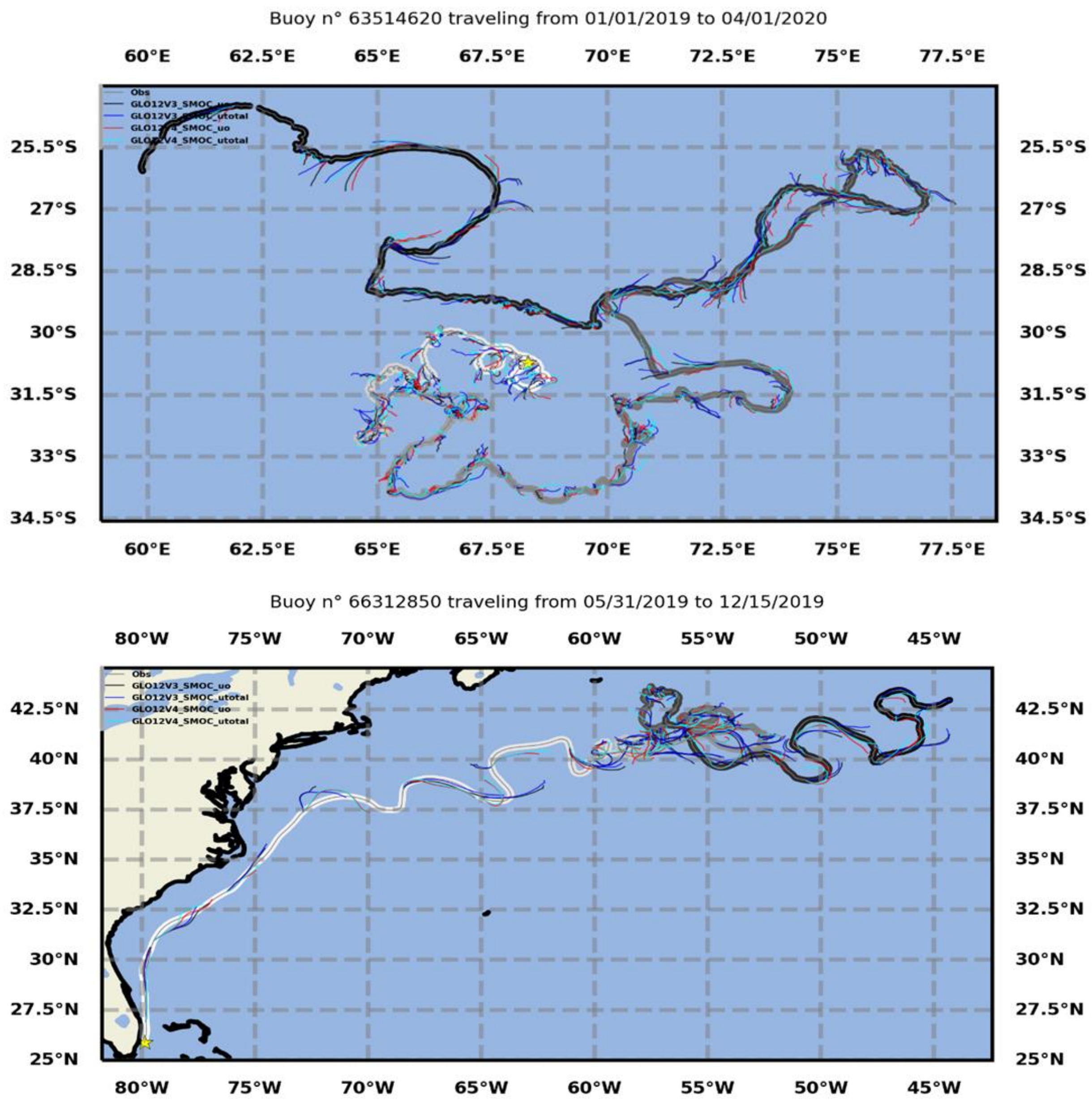


Figure 52. Example of Lagrangian forecasts launched from buoy n° 653 514 620 (Indian Ocean) and n°66 312 850 (Gulf Stream). Each coloured segment corresponds to a 3-day forecast calculated from a given current field: GLO12v3 is the previous system version, GLO12v4 the current version, uo is the surface current only of GLO system, utotal is the surface current added with tide and wave effects. The forecasts are launched from the position of the real buoy which starts from the yellow star and is coloured from white to black according to the time.

Figure 53 gives the values of the average separation distance for the last forecast period (72h) and various zones of interest. For the global area, the new system GLO12v4 added with Stokes and tide currents (GLO12v4 utotal) obtains an average Lagrangian performance of 40 km/3d, which represents an improvement of 7.5 % compared to GLO12v4 uo. Although these areas are poorly observed, the most problematic areas are at the equator and western boundaries current like the Gulf Stream with errors of the order of more than 60 km/3d. Nevertheless, for western boundaries current, the current GLO12v4 system significantly improves the results compared to previous GLO12v3 version in western boundaries currents, in agreement with what is explained for Figure 52. We can note a significant improvement in the south Pacific using utotal over uo; thanks to the large Stokes drift contribution caused by waves born from high winds and large fetch.

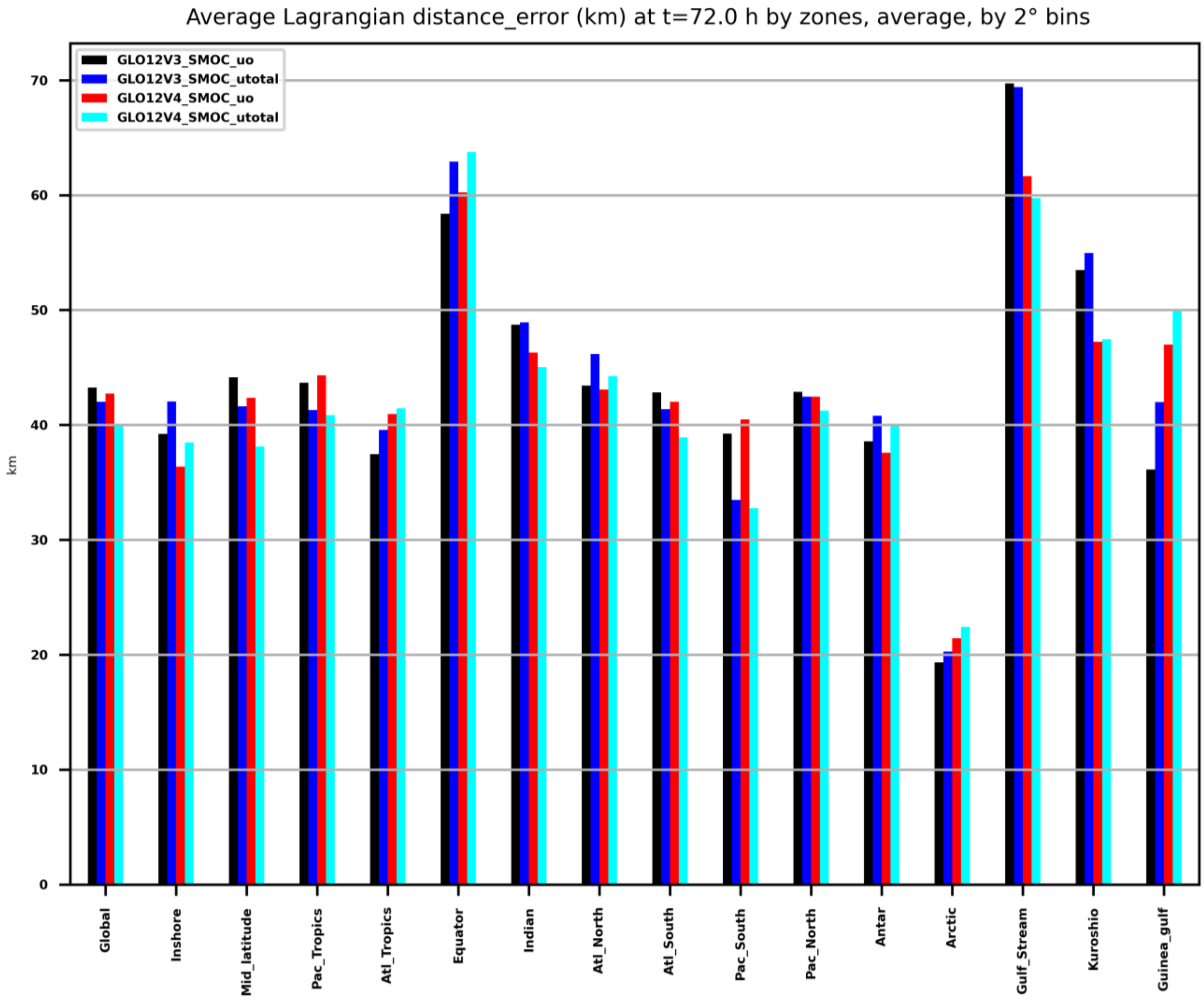


Figure 53. Average separation distance (km) for 3 days of drift for zones of interest for the 171,484 simulations of the 2019-2020 period.

Figure 54 is a global mapping of 2°x2° average bins of the separation distance for 3 days of Lagrangian forecasts for the GLO12v4 utotal velocity field. The most important errors (reaching up to 80 km/3d) are located in the large unstable currents such as the Gulf Stream, the Brazilian Current, and the Agulhas

Current. The smallest errors are found in the subtropical gyres where the large-scale circulation is weak (error of about 10 km/3d). Figure 55 measures the percentage improvement of the current system for surface drift compared to the previous version. The current system significantly improves the previous one in the mid and high latitudes by about ten percent. However, it degrades the forecasts in the Pacific and the tropical Atlantic. Figure 56 shows the improvement of utotal compared to uo for the current system. It expresses the contribution of Stokes velocities (and of the tide near the coast) for surface drift applications. We can see very significant improvements in relation to synoptic winds, such as improvements greater than 50% in the Southern Ocean. Therefore, the Lagrangian performance is significantly improved (20-30%) in mid-latitudes by this link to westerly winds and drift due to induced waves. Note that the local results are quite different from the average results per box in Figure 53 since we see local degradations, which can be as important as the nearby improvements. Degradations are strong in the North Sea, in the Sea of Japan, and in the North Indian.

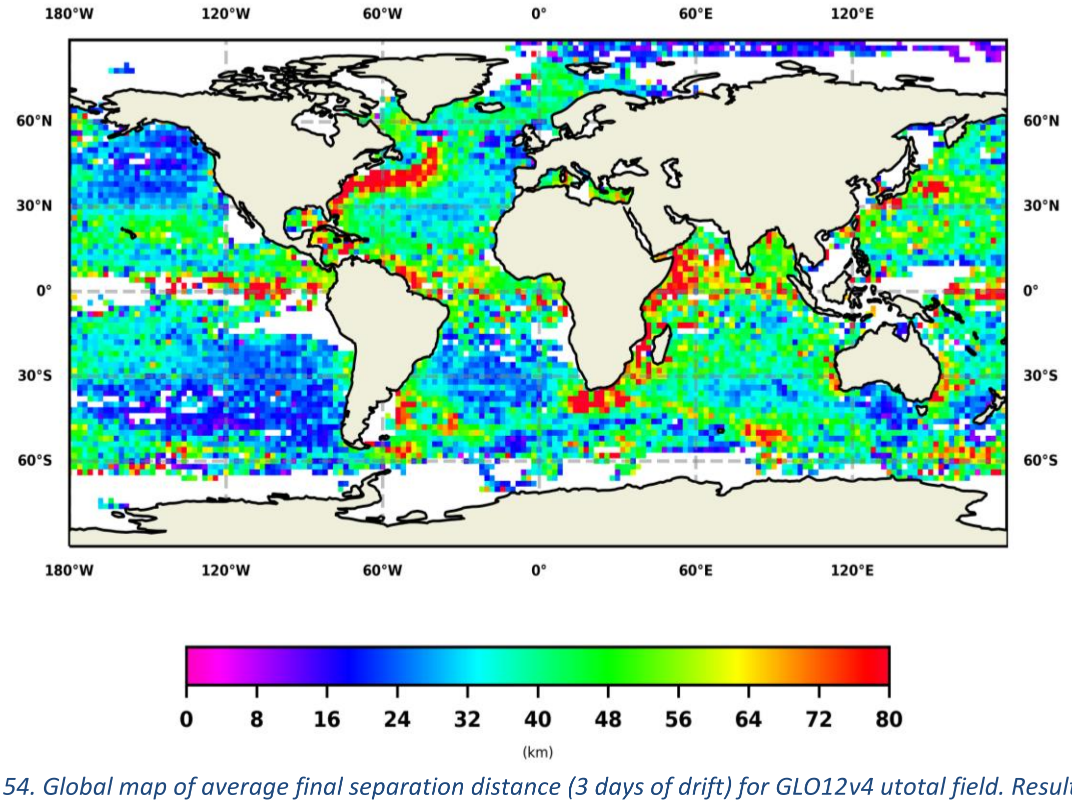


Figure 54. Global map of average final separation distance (3 days of drift) for GLO12v4 utotal field. Results were binned and average over 2°x2° boxes (date 01/01/2019-01/01/2021).

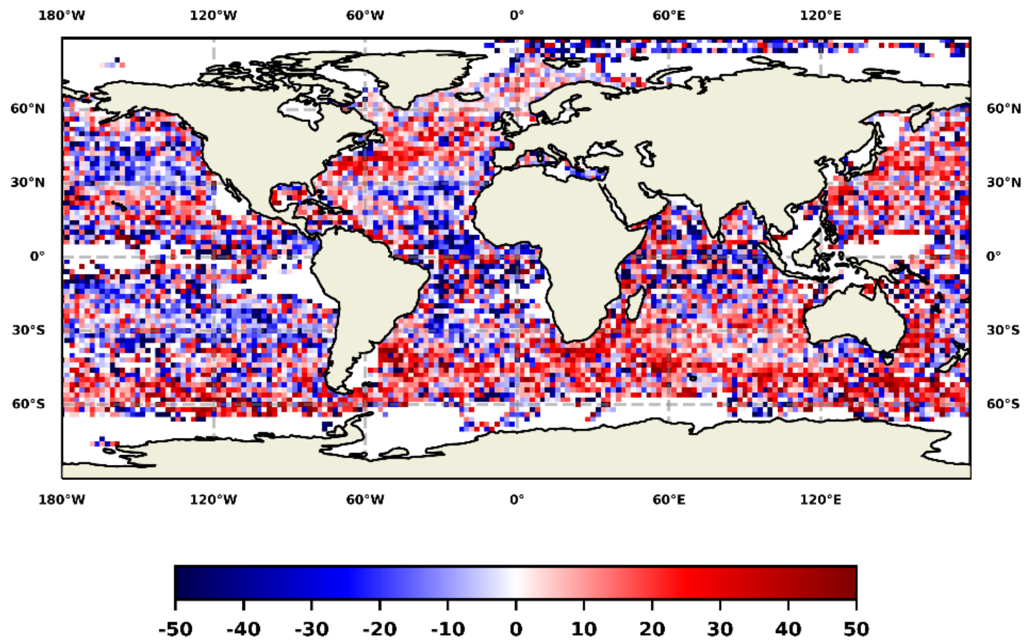


Figure 55. Improvement (%) of GLO12v4 utotal over GLO12v3 utotal for the final separation distance (after 72 h), binned over 2°x2°boxes. This map shows where the current GLO12v4 system is better than the previous GLO12v3 version for surface drift purposes.

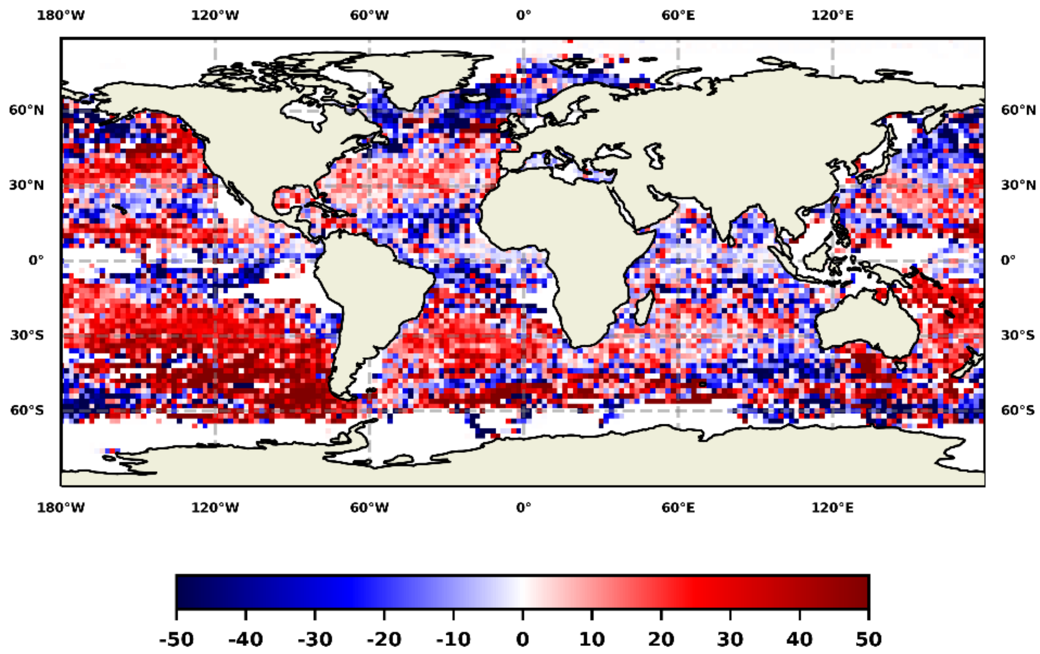


Figure 56. Improvement (%) of GLO12v4 utotal over GLO12v4 uo for the final separation distance (after 72 h), binned over 2°x2° boxes. This map shows were Stokes and tide currents improve the surface drift.

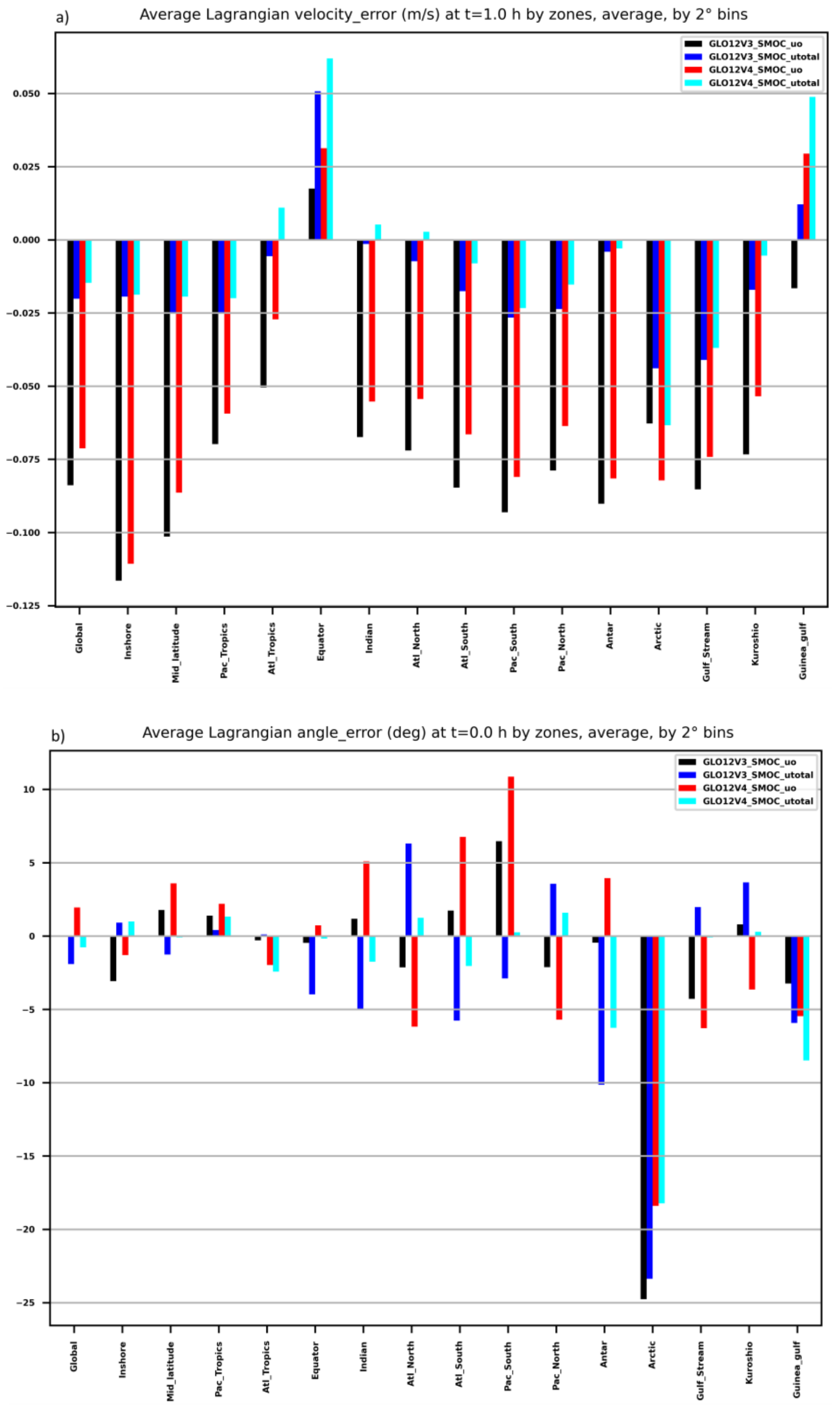
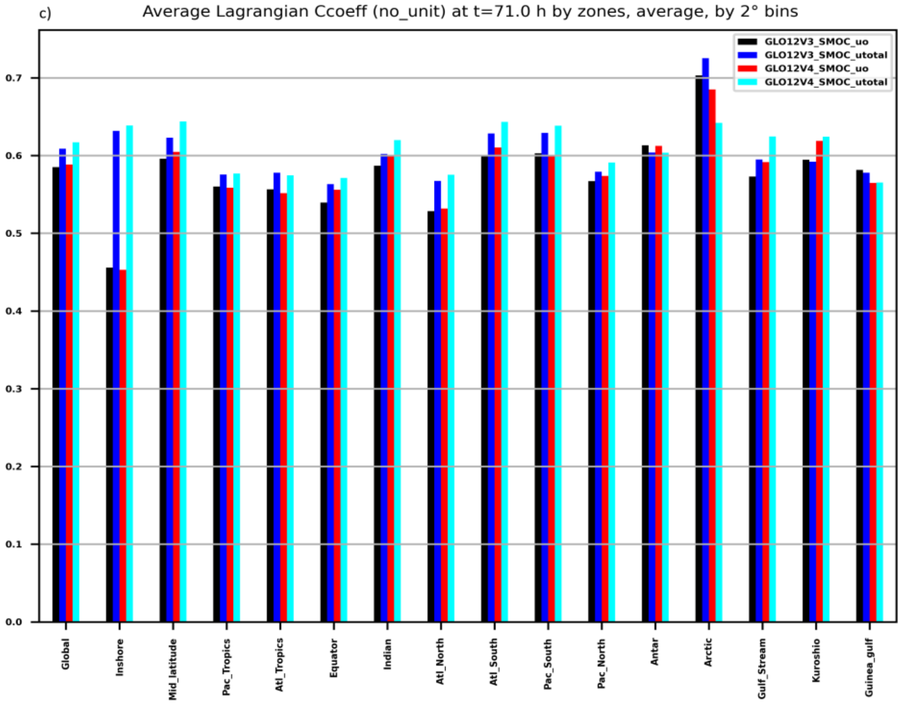


Figure 57. Lagrangian statistics by zones of interests (2019-2020): a) Velocity error (m/s) at initialization, b) Direction error (°) at initialization, c) Vector correlation coefficient (on velocity component times series) between virtual and observed for 3 day long, d) Best rotation angle (°) of virtual trajectory to fit observed one (based on time series of velocity component).



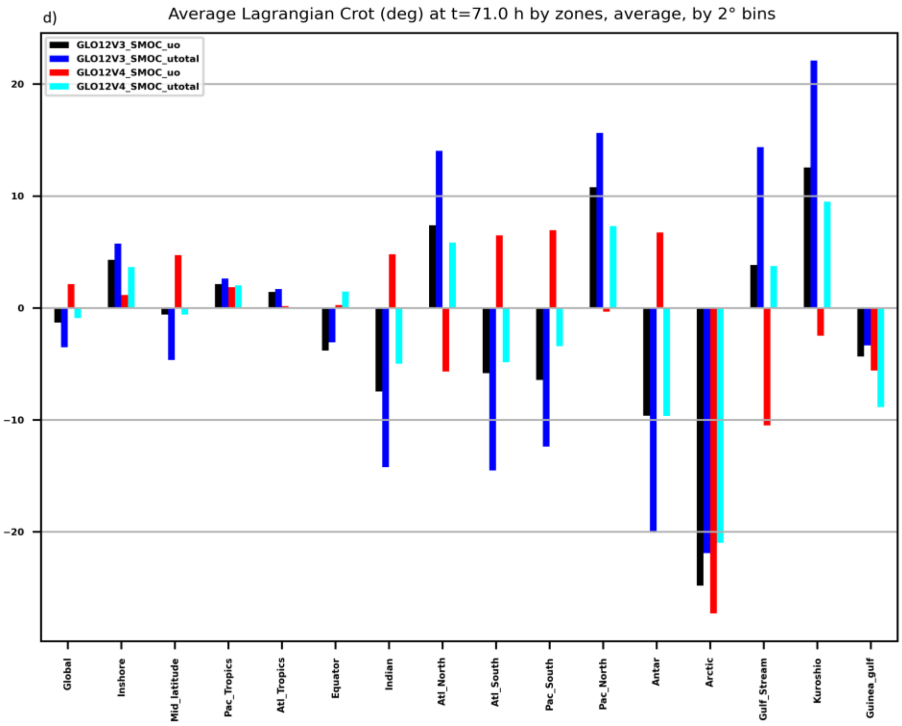
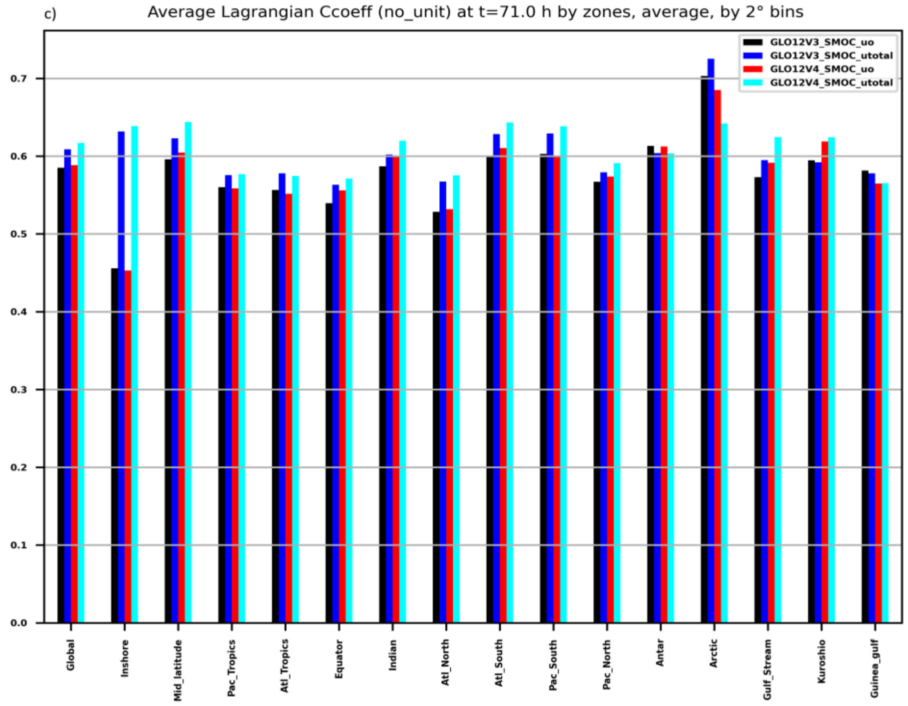


Figure 57 (continued). Lagrangian statistics by zones of interests (2019-2020): a) Velocity error (m/s) at initialization, b) Direction error (°) at initialization, c) Vector correlation coefficient (on velocity component times series) between virtual and observed for 3 day long, d) Best rotation angle (°) of virtual trajectory to fit observed one (based on time series of velocity component).

Figure 57. Lagrangian statistics by zones of interests (2019-2020): a) Velocity error (m/s) at initialization, b) Direction error (°) at initialization, c) Vector correlation coefficient (on velocity component times series) between virtual and observed for 3 day long, d) Best rotation angle (°) of virtual trajectory to fit observed one (based on time series of velocity component).



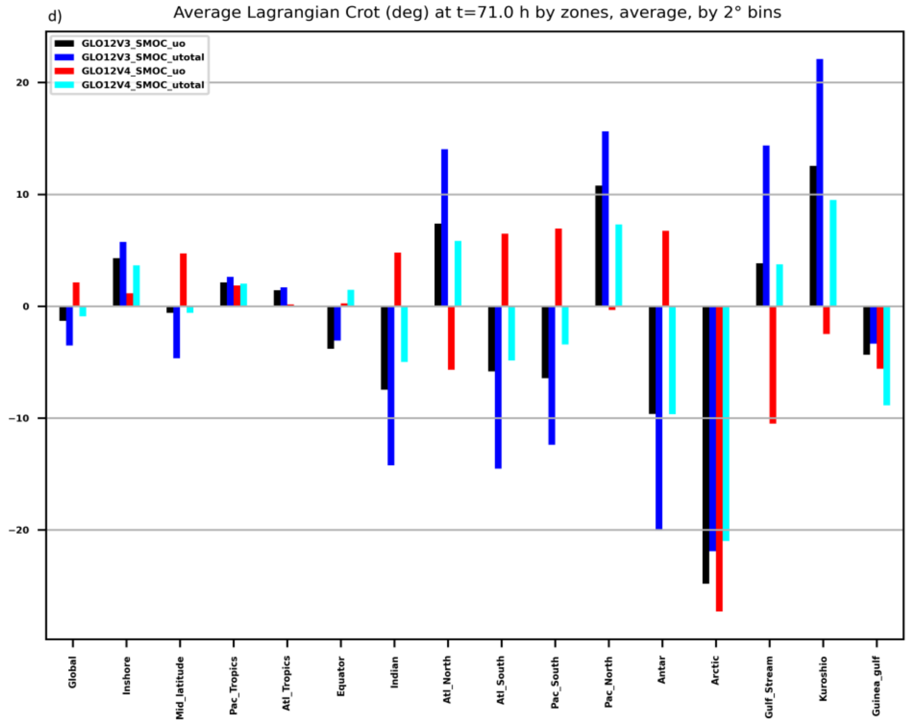


Figure is a collection of figures similar to Figure 53 for several interesting Lagrangian statistics such as initial velocity error, initial direction error, and vector correlation for the complete forecasted trajectory with the observed one. Concerning the error in initial velocity (a), we see that the predictions made with uo velocities underestimate the observed velocities by a little less than 10 cm/s. In comparison, adding Stokes drift reduces this negative bias to a few cm/s. We notice that in equatorial zone, the uo velocities tend to be too strong of a few cm/s, and that adding the Stokes drift positively biases the velocity even more. Concerning the initial direction error (b), the errors are on average about 5°. A positive (respectively negative) error corresponds to a forecast "too far to the right" (respectively "too far to the left"). Adding the Stokes velocities with utotal allows again to decrease these errors with for example a cancellation of the angle biases in the South Pacific. We can see a general trend where the direction biases in the northern hemisphere are too far to the left, and conversely in the southern hemisphere. This result suggests errors in the directional distribution of wind energy on the first vertical levels of the model. In the Atlantic, adding Stokes drift effectively corrects the velocity angle, but we observe a change of sign for the angle bias, which shows that the correction is slightly too strong. The vector correlations (c) were calculated on all the time scales (from 0 to 72 h), and they naturally decrease with the time scale. The vector correlation coefficient for 3 days of drift remains higher than 0.6 on average, which shows that a 1/12° model and hourly outputs can represent the velocity variations that the buoys undergo up to 3 days. We can note that on the shelves ("inshore" on the figure), the correlation of uo with observation is 0.45 and utotal improves this value to 0.65. The shelves are in fact isolated from the large-scale circulation and the tide and local wind play therefore a major role. It is also possible to find the best rotation to apply to the modeled time series to match the observed one (d). The results obtained are practically equivalent to the initial angle biases (c) when looking at basin-scale areas. It could be more practical at a local scale, as in for example in Kuroshio or the Gulf stream, where a rotation between 5 and 10° clockwise would improve the velocity time series correlation.

IV.7 Current at 1000 m depth

To validate the currents at 1000 m, we use Yomaha drifts at 1000 m and we compared them to the background model first trajectory (observations in the assimilation window are not yet assimilated). However, the Yomaha currents correspond in fact to a drift over 10 days at the depth of the profilers; thus we use a 10-day time average to make the comparison. But the correspondence will not be good if the profiler has travelled a long distance (possibly in circles). This validation, which can therefore be very approximate locally, is not a substitute for a true Lagrangian validation.Figure 58 shows that zonal current large-scale structures at 1000 m are well represented, and the currents do not seem to be underestimated. The agreement is less good in the tropics. The model appears noisier despite the 10-day average; this is probably the signature of locally important currents, but which only affect the profiler for a shorter period (passage over a threshold, or between two opposing eddies).

The differences in angle between the model and the observations are modest, whatever the velocities (small or large). The median difference is less than 1° (not shown). In general, the correlation is good where currents are intense (ACC, Gulf Stream, and Kuroshio). In global, the slope of the scatterplot exceeds 60% in u and v. The difference between the modulus of observed and model velocities shows a slight general underestimation in the subtropical gyres.

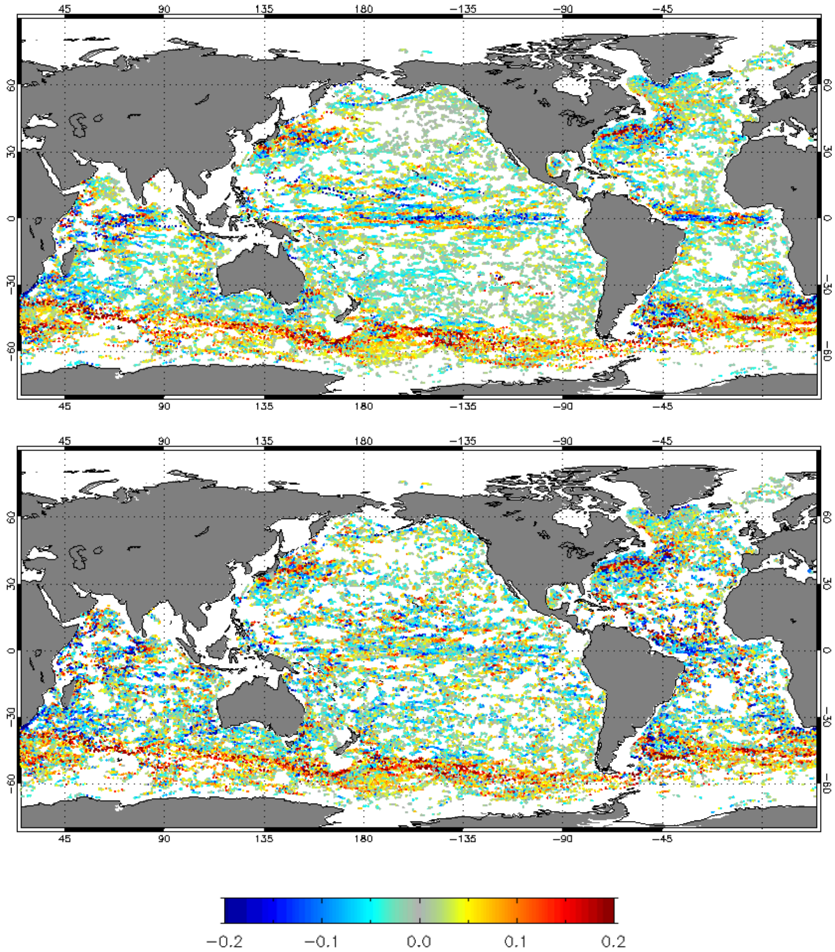


Figure 58. Zonal current for the Yomaha observations (top panel) and the GLO12v4 model (bottom panel) at 1000 m and for the year 2019. Units are m/s.

IV.8 Mixed Layer Depth

Large scale mixed layer depth features are captured by GLO12v4, as shown by Figure 59. The depth of the deep convection is overestimated, and the thermocline is too deep in the upwelling areas, consistently with temperature diagnostics shown earlier in the document. Part of the differences with the climatological mixed layer depth deduced from observations may be explained by the resolution of GLO12v4 (higher than the observational products) and the interannual variability. The current version of the system (GLO12v4) performs slightly better than the previous one (GLO12v3) with respect to observations.

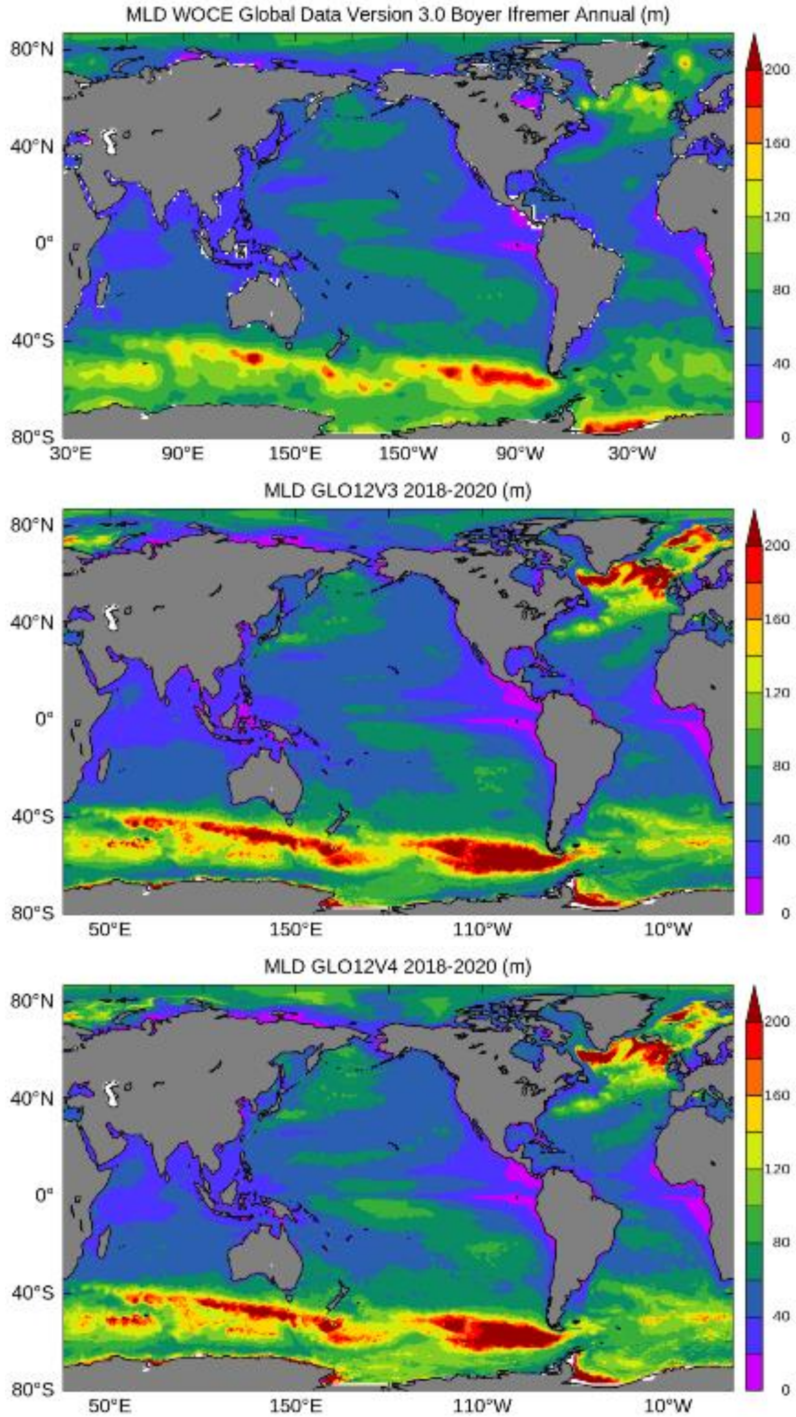


Figure 59. Climatological annual mean mixed layer depth (m). De Boyer Montegut climatology (top), GLO12v3 previous system (middle) (2018-2020 average) and GLO12v4 current system (bottom) (2018-2020 average).

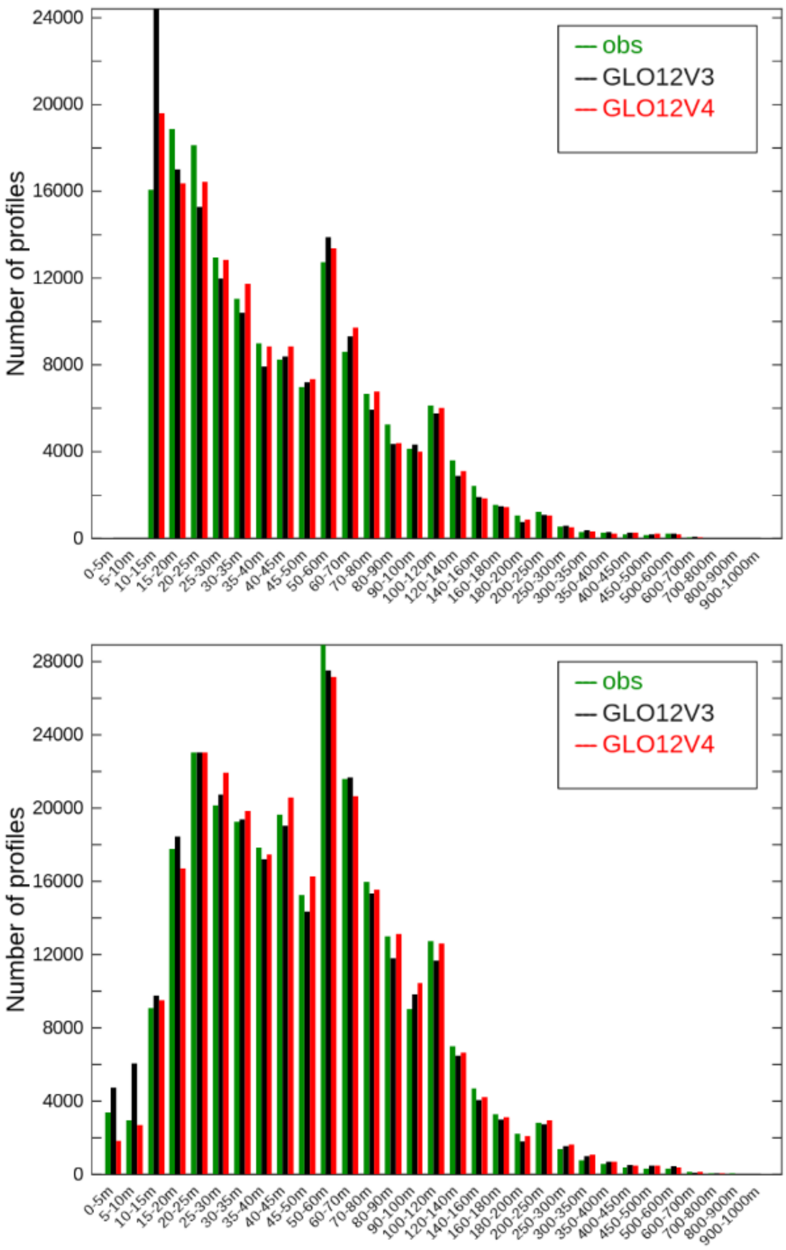


Figure 60. 2019 distributions, for the global ocean, of Mixed Layer Depth, calculated from profiles with the density criteria (MLD defined as the depth at which density difference from the surface reaches 0.03 kg/m3) (top) and temperature criteria (MLD defined as the depth at the temperature difference from the surface reaches 0.2°C) (bottom); GLO12v3 is in black, GLO12v4 is in red and the in situ observations in green.

Comparing the distributions of mixed layer depth at global scale in 2019 in Figure 60 and the seasonal cycle of differences in Figure 61, we note that GLO12v4 is closer to observations than the previous version GLO12v3. Larger discrepancies with observations occur during winter convection.

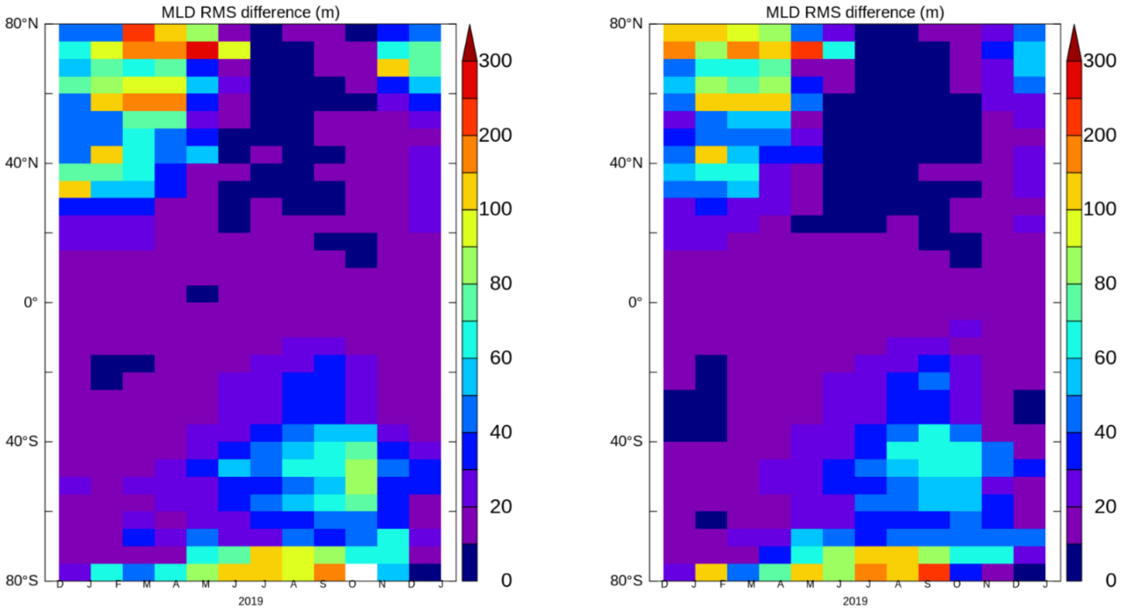


Figure 61. monthly MLD RMSD (CORA dataset) (m) in function of latitude in 2019, (left) GLO12v3, (right) GLO12v4. The profiles are bin averaged by month, latitudes and for the whole longitudes.

IV.9 Validation of high frequencies at tide-gauges

GLO12 hourly outputs at the surface can be compared with tide gauges measurements, from which we filter out the inverse barometer effect and the tides.

IV.9.1.1 <u>Sea surface Height</u>

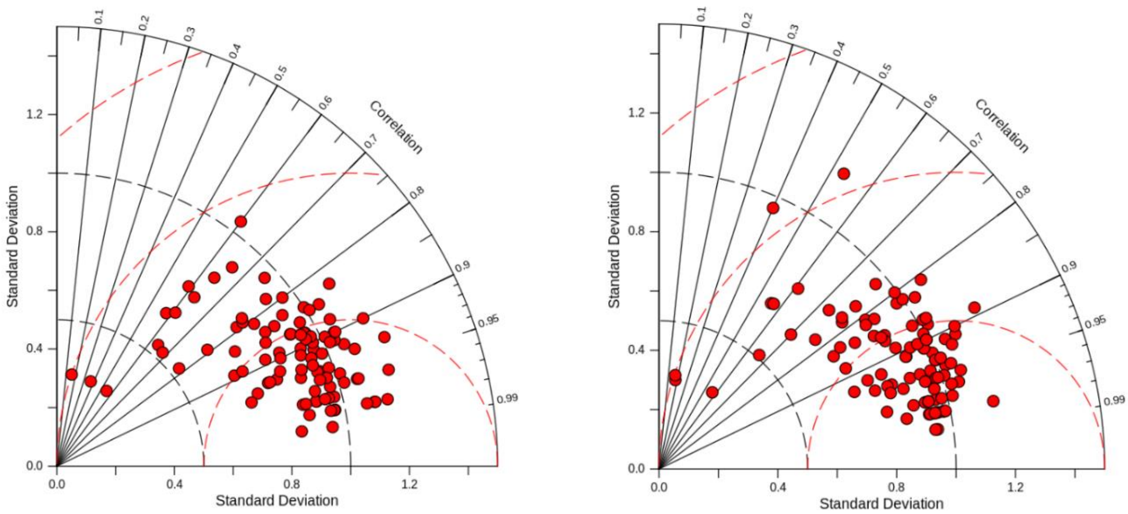


Figure 62. Taylor diagram of the sea surface height errors (2019-2020). Comparisons between observations at tide gauges and GLO12 systems. (left): GLO12v3. (right): GLO12v4.

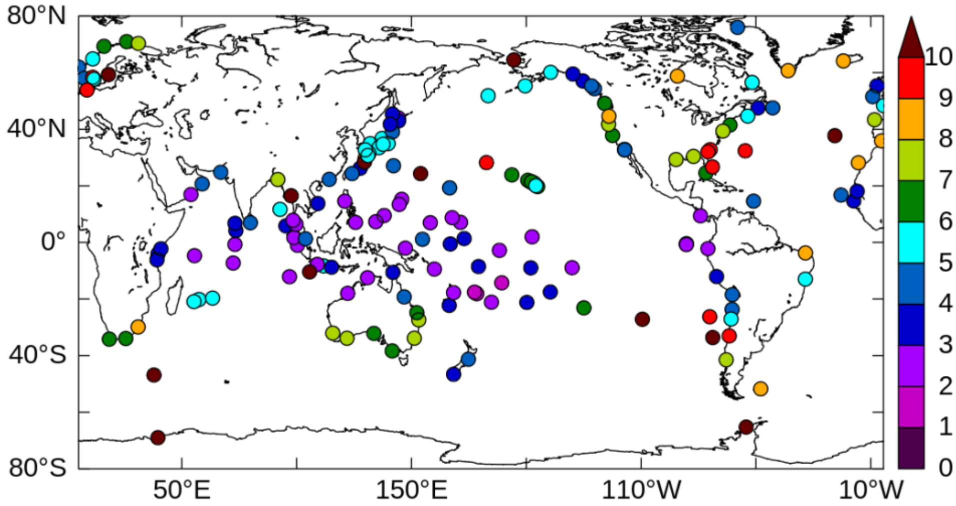


Figure 63. Sea surface height RMSD (cm) between observations at tide gauges and GLO12v4 for the years 2019 to 2020.

Although tides and inverse barometer are filtered out from this comparison, large uncertainties are expected in the high frequency SSH comparisons due to processes which are not (fully) resolved by the system GLO12v4, and due to the location of tide gauges in coastal areas.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES.

V.1 Product corrections in July 2023

Following user feedback, the GLO MFC production unit has worked on some technical fixes:

- to correct for anomalous values that are erroneously generated by the interpolation process (from native to standard grid) in some coastal regions, close to bathy areas. The anomalous values are only in the 3D datasets.
- to fix an issue with bottom pressure at 73°E.

The proposed corrections allow these problems to be fully corrected in real time.
A system update was performed on Wednesday 26th of July 2023, i.e., users saw patched data from day 20230719 onward (being delivered from Thursday 27th am).
The past timeseries will be delivered in June 2024 to provide users with a fully consistent timeseries.

Moreover, some spurious (negative) values have been identified in several variables, e.g., MLD.
For this last problem, we have redelivered the past timeseries and corrected the real time in June 2024.

V.2 Product corrections in November 2024

Many interactions with the Hydrography and Oceanography Service of the French Navy (SHOM) have helped to identify a few problems with the GLO12v4 system:

- Comparisons with in-situ salinity and temperature profiles show a degradation in some areas, in particular in the tropical zone, and especially below 300 m. This degradation is present in the English Channel and confirms the suspicious profiles noted by the SHOM. This degradation does not affect the performance of GLO12v4 in terms of mixing layer or surface channel thickness, GLO12v4 being closer to observations than GLO12v3 for overall statistics in the 0-300 m layer.
- GLO12v4 appears to have unrealistically high variability below 300 m, particularly in tropical areas.
- Fronts (SST gradients) have a more diffuse dynamic in GLO12v4, very different from GLO12v3.

Actions have therefore been taken to resolve the problems identified in GLO12v4. The changes made in GLO12v4 concern the ocean model, the SEEK assimilation and the 3D-Var temperature and salinity bias correction. Our conjecture is that in GLO12v4, the model is over-corrected both through SEEK and bias correction, which introduces some of the observational noise into the model. As the model is a little too turbulent, this noise is not diffused. In addition, variability is much greater at depth (50% more at around 2000 m) in GLO12v4. While GLO12v3 variability is probably underestimated, the GLO12v4 one is likely overestimated. This explains the degradation in temperature RMS below 300 m: the profiles are well assimilated, but advected anarchically by an under diffusive model.

Here are the changes made to the GLO12v4 system:

- The viscosity on dynamics has been increased in the ocean model. Indeed, the change from NEMO3.1 to NEMO3.6 (new advection scheme and time-splitting for SSH) and the higher resolution and higher frequency atmospheric forcing (1/15° - 1 h) used in GLO12v4 (versus 1/8° - 3 h for GLO12v3) results in too much energy appearing in the model.

- The intensity of the SEEK correction has been reduced over the entire water column, in particular the velocity corrections in the tropical band which become very weak below 2500 m.
- The 3D-Var bias correction for temperature and salinity has been significantly revised:
  - 1/2° horizontal grid instead of 1° previously,
  - More accurate land-sea mask with at least 2 points in the channels,
  - Replacement of the 1-point periodicity by a wide overlap band (10° from 360°E to 370°E),
  - Pseudo observations at the bottom to limit increments in the bottom boundary layer,
  - Addition of a Quality Control for observations to reject innovations (differences between observations and model equivalents) that exceed the 95% quantile,
  - Increase of the size of the scales corrected by this bias correction.

VI QUALITY CHANGES SINCE PREVIOUS VERSION

VI.1 Product corrections in July 2024

The product corrections made in July 2023 and described in section V.1 do not have any impact on the behaviour and the quality of the system.

VI.2 Product corrections in November 2024

The product corrections made in November 2024 are described in section V.2.

The new GLO12v4 system benefited from a series of improvements compared with the previous GLO12v3 system. GLO12v4 is more energetic and has more variability, closer in this respect to the real ocean. Comparisons with in situ salinity and temperature profiles showed a significant overall improvement in biases and absolute deviations from observations. However, there was a deterioration in some regions at depths below 300 m, particularly in the tropical zone, where RMSD can be locally up to 2 times higher in GLO12v4 than in GLO12v3. The proposed corrections allow these problems to be fully corrected in real time.

VI.2.1 Impact on vertical velocities

To quantify the impact of the changes, a corrected version of the GLO12v4 system, taking into account all the changes described in section V.2, has been used. This corrected version (hereafter referred to as “GLO12v4corrected”) has been in operation since January 2024 and covers an entire annual cycle.

The vertical velocities of the GLO12v4 system prior to November 2024 have been identified as excessively high. Figure 64 compares the monthly averages of vertical velocities at 1565 metres in November 2024 for the different systems. The GLO12v4 system shows significantly higher velocities than the GLO12v4corrected system. Figure 65, which presents the same averages for February 2025, shows a comparable order of magnitude. This is corroborated by Figure 66. In November 2024, the vertical velocities of the GLO12v4 system (blue curve) are greater than those of the GLO12v3 and GLO12v4 corrected systems, whilst in February 2025, the probability densities of the three systems appear very similar.

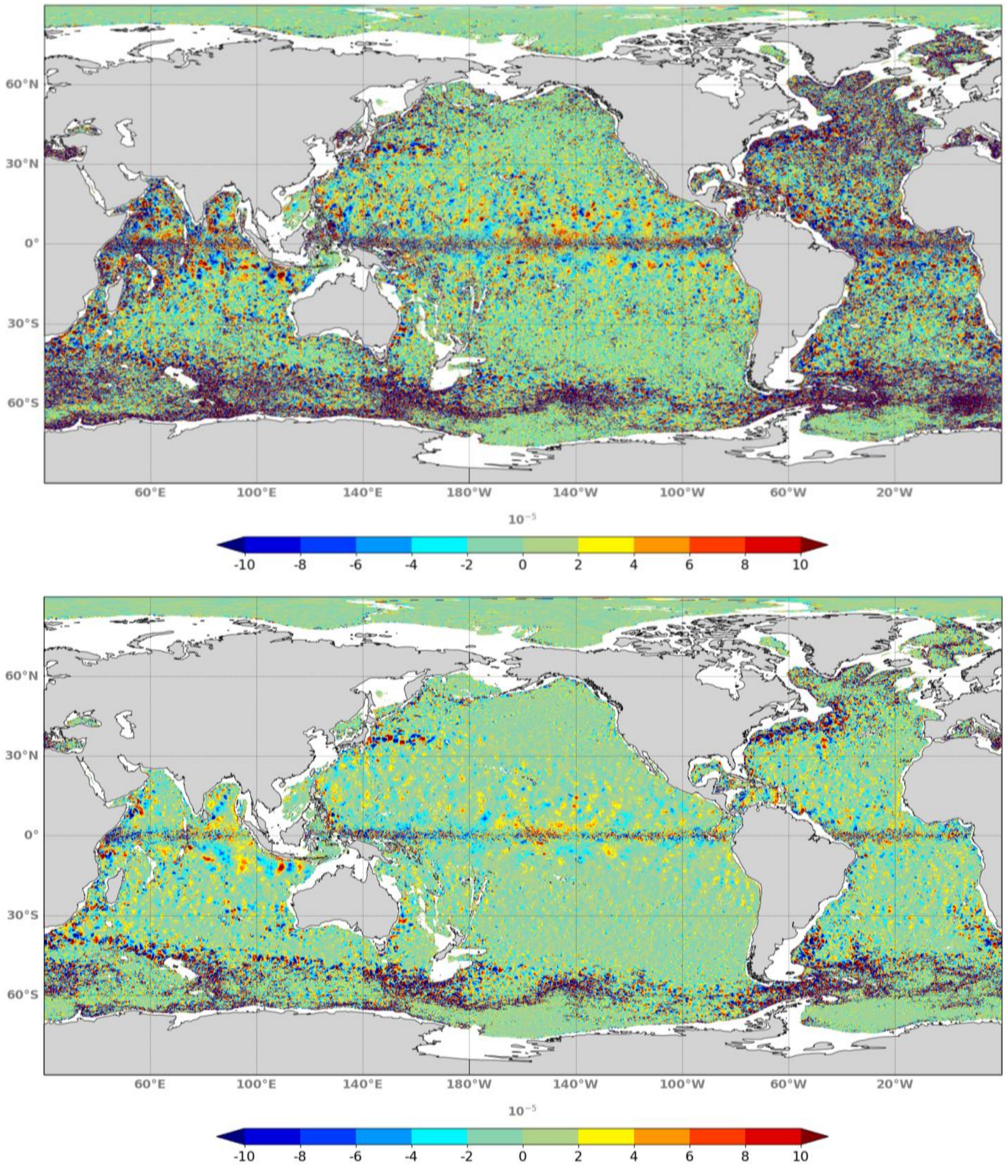


Figure 64. Monthly mean vertical velocity in November 2024 at 1565 m for GLO12v4 (top) and GLO12v4corrected (bottom).

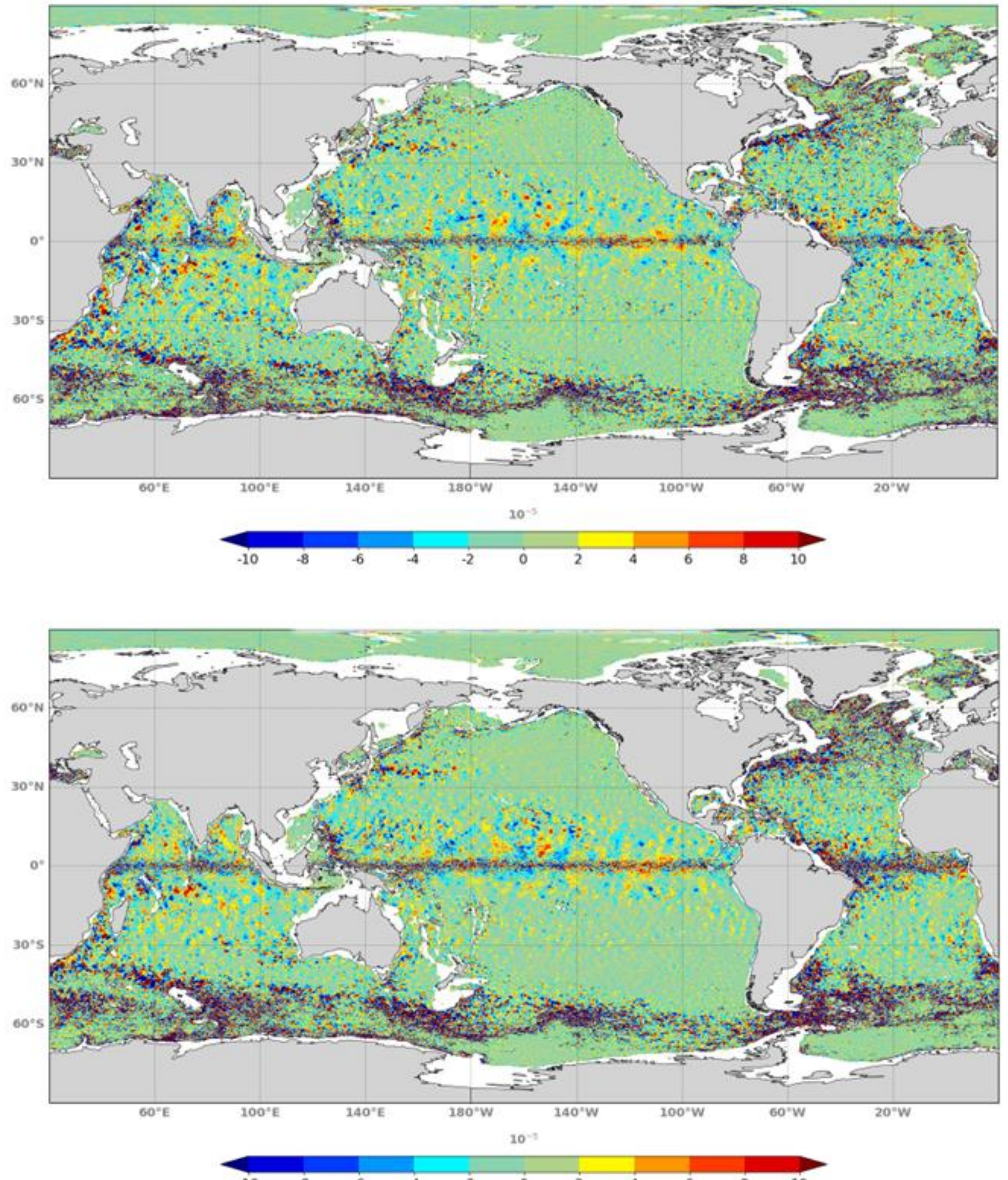


Figure 65. Monthly mean vertical velocity in February 2025 at 1565 m for GLO12v4 (top) and GLO12v4corrected (bottom).

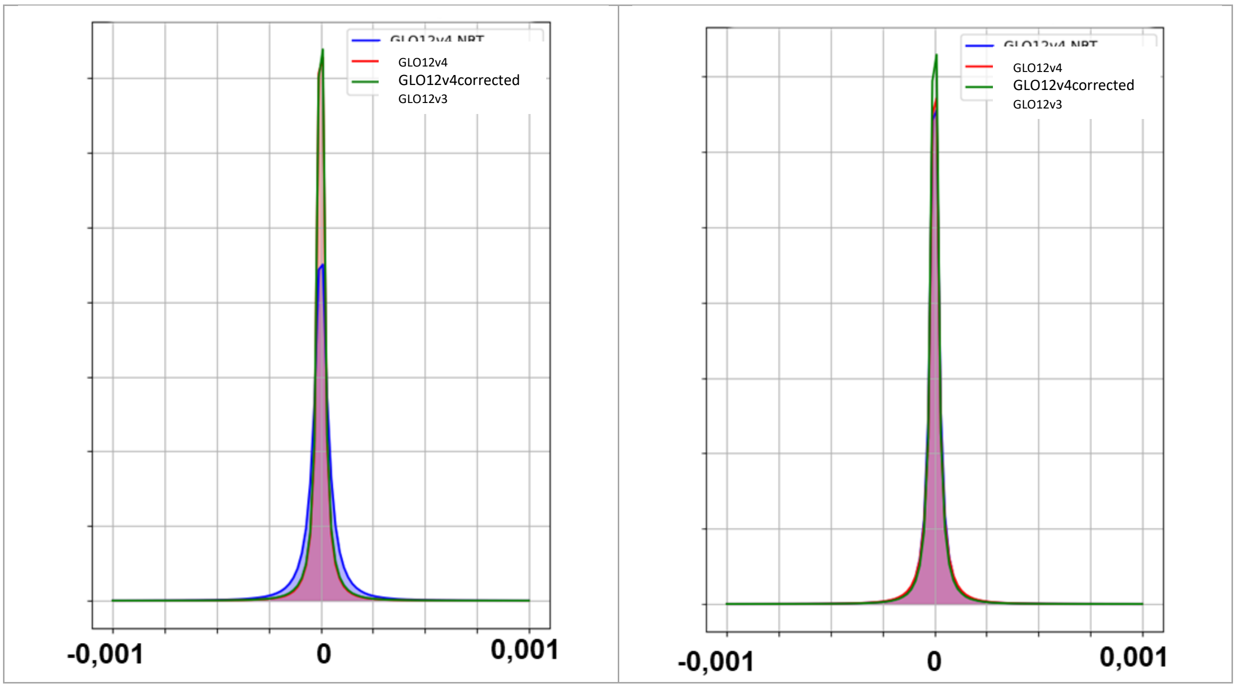


Figure 66: Distribution of the monthly mean vertical velocity in November 2024 (left) and February 2025 (right) at 1565 m.These first results enabled us to measure the impact of the corrections made on the GLO12v4 system after 5 months.

To complete this validation exercise, the vertical velocities of the other Copernicus Marine Service MFCs (IBI, MED and ARC) have been compared with those of the GLO12v4 system (GLO-MFC).

Figure 67 shows a comparison between the GLO12v4 system over the IBI region and the IBI-MFC near-real-time system at 1565 m. The GLO12v4 velocities in November 2024 are higher than those of the regional system. In February 2025, the opposite is seen, which is more consistent as the regional system has a higher resolution.

Figure 68 shows a comparison between the GLO12v4 system over the Mediterranean region and the MED-MFC near-real-time system at 1565 m. The GLO12v4 velocities in November 2024 are significantly higher than those of the regional system. In February 2025, the probability densities are much closer.

Figure 69 shows a comparison between the GLO12v4 system over the Arctic region and the ARC-MFC near-real-time system at 1565 m. The GLO12v4 velocities in November 2024 are significantly higher than those of the regional system. In February 2025, the probability densities are much closer.

Whatever the regional product with which we compare the GLO12v4 system in February 2025, we are in a comparable order of magnitude. Users can therefore start again using the vertical velocities of this system.

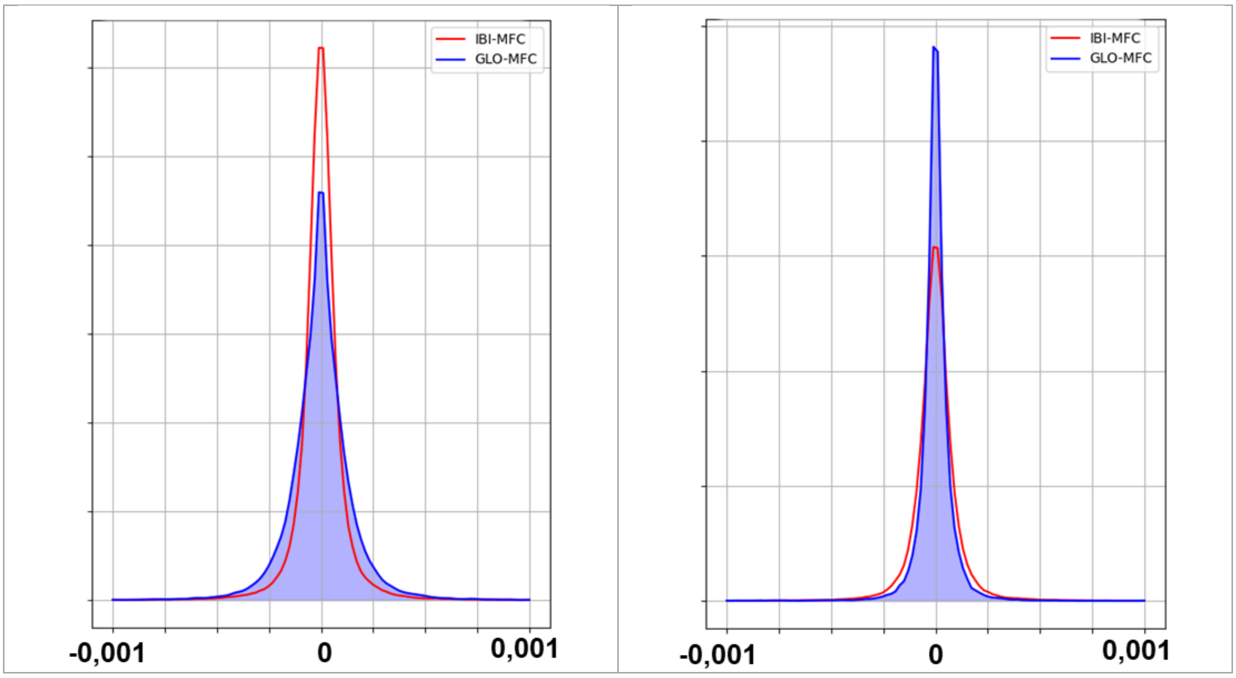


Figure 67: Distribution of monthly mean vertical velocity in November 2024 (left) and February 2025 (right) at 1565 m on IBI region.

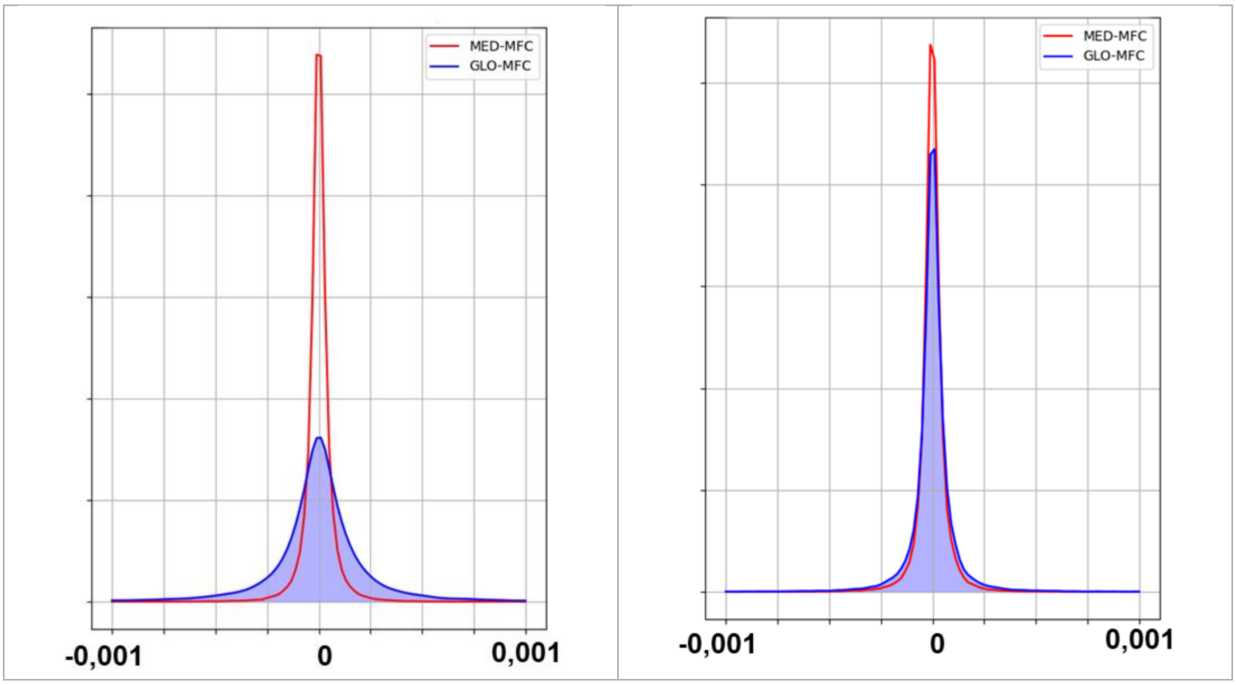


Figure 68: Distribution of monthly mean vertical velocity in November 2024 (left) and February 2025 (right) at 1565 m in MED region.

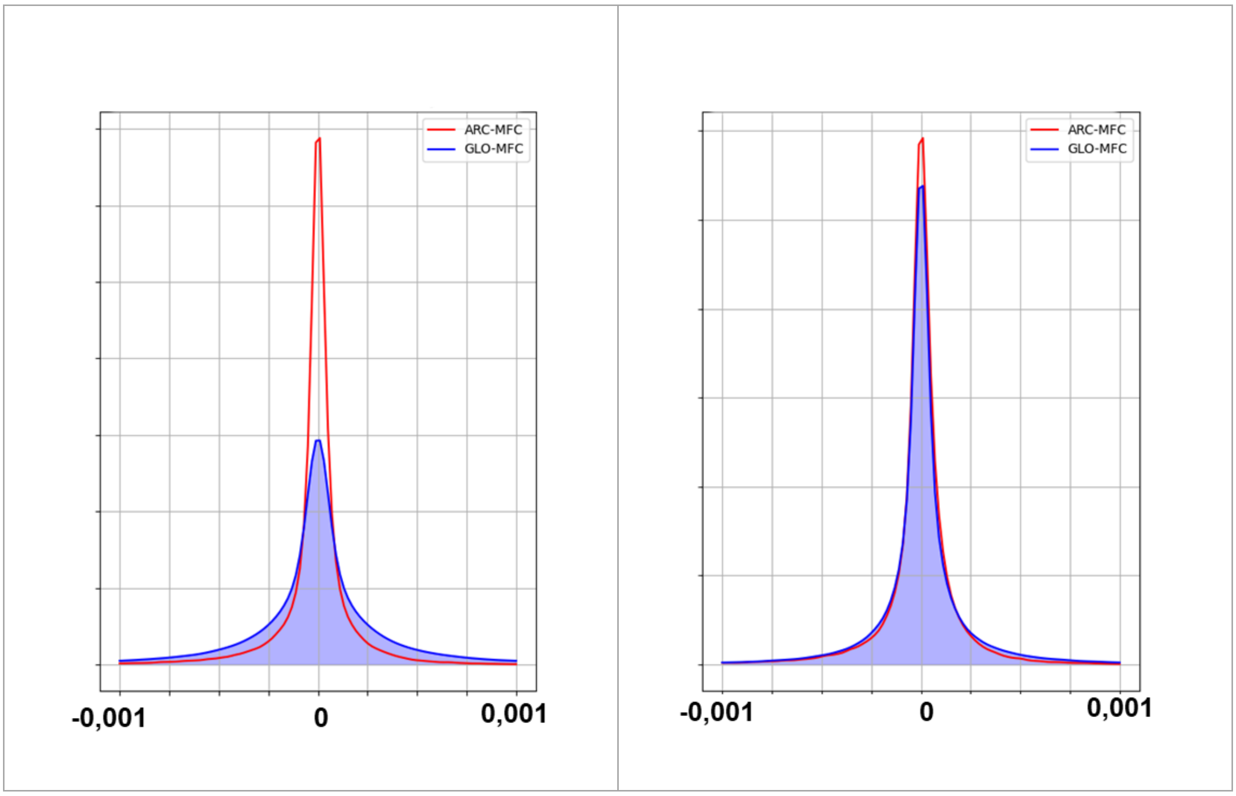


Figure 69: Distribution of monthly mean vertical velocity in November 2024 (left) and February 2025 (right) at 1565 m in ARC region.

VI.2.2 Comparison of EAN scores on the Feb 2024 – Feb 2025 period

For the previous EANs computed on 2019 (see section I.3), CORA5.2 in-situ delayed time dataset (INSITU_GLO_PHY_TS_DISCRETE_MY_013_001 product) was used. To validate the corrections performed on the GLO12v4 system, new EANs have been calculated over a one-year period (between February 2024 and February 2025) for the best analyses of GLO12v4 and GLO12v4corrected, this time using the dataset INSITU_GLO_PHY_TSASSIM_DISCRETE_NRT_013_047 based on the Coriolis database available close to real time. Even if the two datasets are different, all the types of platforms are used: Argos profiles, Vessels, Moorings, Drifters, Sea mammals, etc.

Table 11 shows the RMSD and the mean bias of 2D variables with respect to sea ice cover from AMSR2 satellite, Sea level anomaly (SLA) from SEALEVEL_GLO_PHY_L3_NRT_008_044 Copernicus Marine L3 product, sea surface temperature (SST) from in situ drifting buoys “TS_DB” Coriolis files and 3-day filtered zonal and meridional currents from INSITU_GLO_PHY_UVASSIM_DISCRETE_NRT_013_054 Copernicus Marine product (surface drifting buoys at 15 m).

Table 12 and Table 13 show the RMSD and the mean bias of 3D variables with respect to in situ temperature and salinity vertical profiles for all available platforms.

For 2D variables, RMSD and mean bias scores for both systems are similar. GLO12v4corrected shows slight improvements in SST and surface currents. Corrections made on GLO12v4 have enhanced the surface ocean circulation without degrading global scores.

For 3D variables, RMSD and mean bias scores are similar for both systems across the entire water column. The GLO12v4corrected system shows slightly better results in the first 800 meters.

Table 11. Global RMSD and mean bias for 2D variables for the one-year period (from February 2024 to February 2025) for GLO12v4 and GLO12v4corrected systems

2D observations	**GLO12V4**		**GLO12v4corrected**	
	Mean bias	**RMSD**	**Mean bias**	**RMSD**
Sea Ice Cover (%)	-0.6	5.4	-0.6	5.3
SLA (cm)	0.1	5.4	-0.8	5.4
SST (K)	-0.06	0.62	-0.07	0.61
Ucurr (cm/s)	0.9	15.3	1.0	14.4
Vcurr (cm/s)	0.1	14.0	0.1	12.9

Table 12. Global RMSD and mean bias for 3D Temperature for the one-year period (from February 2024 to February 2025) for GLO12v4 and GLO12v4corrected systems.

3D T (K) in situ observations	**GLO12V4**		**GLO12v4corrected**	
	Mean bias	**RMSD**	**Mean bias**	**RMSD**
0-5 m	-0.047	0.925	-0.051	0.894
5-100 m	0.002	0.747	0.004	0.705
100-300 m	0.008	0.698	0.018	0.631
300-800 m	-0.005	0.441	-0.006	0.379
800-2000 m	-0.010	0.181	-0.011	0.153
2000-5000 m	-0.016	0.131	-0.008	0.134
0-5000 m	-0.008	0.470	-0.008	0.423

Table 13. Global RMSD and mean bias for 3D Salinity for the one-year period (from February 2024 to February 2025) for GLO12v4 and GLO12v4corrected systems.

3D S (PSU) in situ observations	GLO12V4		GLO12v4corrected	
	Mean bias	RMSD	Mean bias	RMSD
0-5 m	-0.0111	0.3111	0.0003	0.2925
5-100 m	-0.0398	0.2824	-0.0386	0.2752
100-300 m	-0.0191	0.1843	-0.0172	0.1711
300-800 m	absolute value < 0.0001	0.06	absolute value < 0.0001	0.0541
800-2000 m	-0.0004	0.0278	-0.0005	0.0218
2000-5000 m	0.0018	0.0166	0.0033	0.0201
0-5000 m	-0.0083	0.1274	-0.00799	0.1212

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VIII APPENDIX

VIII.1 Defining ocean basins/areas for regional EANs

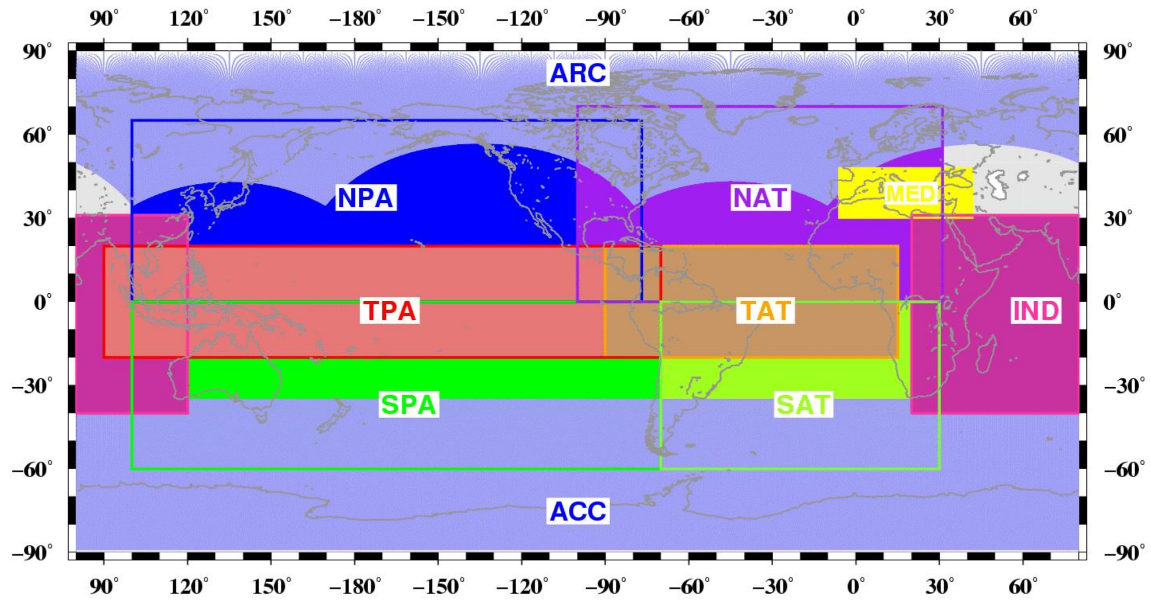

Figure 70. Colours show the geographical extent of ocean basins for which estimated accuracy numbers (EANs) are computed: Antarctic Circumpolar (ACC), Arctic (ARC), North Pacific (NPA), North Atlantic (NAT), Mediterranean (MED), Tropical Pacific (TPA), Tropical Atlantic (TAT), South Pacific (SPA), South Atlantic (SAT), and Indian (IND).

VIII.2 Data assimilation glossary

misfit	Difference between the observation and its model counterpart at the time and location of the observation. The model counterpart is possibly transformed to be compared to the observation (filtered in space and time, for instance).
innovation	Same as misfit.
forecast	In the data assimilation context, it is the name given to the model counterpart for the misfit computation (model value before any correction).
background	Same as forecast.
guess	Same as forecast.
analysis	Model forecast (or guess) corrected with the increment (or correction).
residual	Difference between the observation and its analysis counterpart at the time and location of the observation.
hindcast	Numerical experiment describing a period in the past.